



US012414858B2

(12) **United States Patent**
Huh et al.

(10) **Patent No.:** **US 12,414,858 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **CURVED EXPANDABLE INTERBODY DEVICES AND DEPLOYMENT TOOLS**

(71) Applicant: **Spinal Elements, Inc.**, Carlsbad, CA (US)

(72) Inventors: **Edwin Huh**, Irvine, CA (US); **Jason Blain**, Encinitas, CA (US); **Taylor Semington**, San Diego, CA (US); **Nicholas Chen**, Carlsbad, CA (US); **Ketchen Smith**, Escondido, CA (US); **David Ortiz**, Carlsbad, CA (US); **Geoffrey Toon**, Vista, CA (US); **Morton Albert**, Carlsbad, CA (US)

(73) Assignee: **Spinal Elements, Inc.**, Carlsbad, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/417,624**

(22) Filed: **Jan. 19, 2024**

(65) **Prior Publication Data**

US 2024/0225851 A1 Jul. 11, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/523,620, filed on Nov. 10, 2021, now Pat. No. 11,911,284.

(Continued)

(51) **Int. Cl.**

A61F 2/44 (2006.01)

A61F 2/30 (2006.01)

A61F 2/46 (2006.01)

(52) **U.S. Cl.**

CPC **A61F 2/4425** (2013.01); **A61F 2/30771** (2013.01); **A61F 2/4465** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC **A61F 2/44-447**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,845,428 A 2/1932 Llewellyn

2,440,123 A 4/1948 Smith

(Continued)

FOREIGN PATENT DOCUMENTS

CA 1 329 525 5/1994

CA 2 521 526 11/2004

(Continued)

OTHER PUBLICATIONS

Official Communication in Australian Application No. 2006227755, dated Dec. 8, 2010.

(Continued)

Primary Examiner — Eduardo C Robert

Assistant Examiner — Steven J Cotroneo

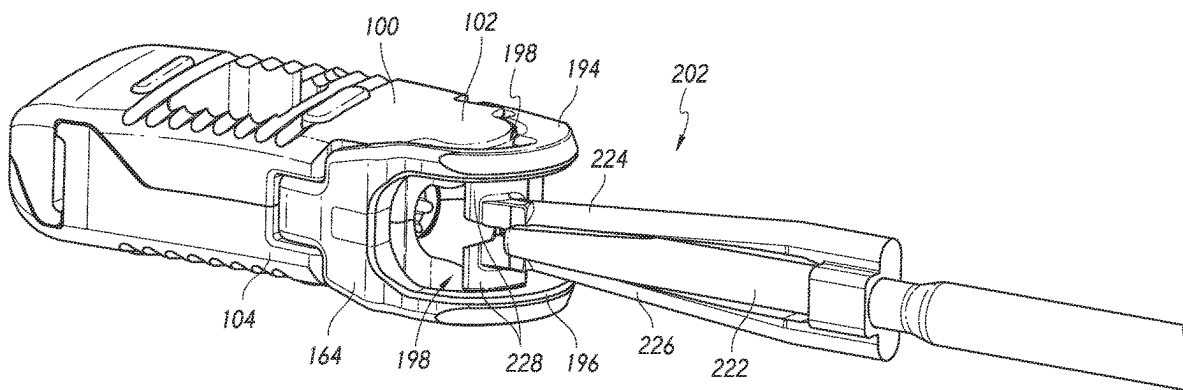
(74) *Attorney, Agent, or Firm* — Knobbe, Martens, Olson & Bear, LLP

(57)

ABSTRACT

A curved expandable interbody device for placement between vertebrae having an upper structure, a lower structure, and a screw mechanism, wherein actuation of the screw mechanism moves the upper and lower structures between a collapsed configuration and an expanded configuration. A deployment tool couples to the curved expandable interbody device for positioning the device between adjacent vertebrae and actuating the screw mechanism, wherein the deployment tool can lock to the curved expandable interbody device and pivot relative to the curved expandable interbody device.

19 Claims, 33 Drawing Sheets



--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

(56)

References Cited

U.S. PATENT DOCUMENTS

6,656,181	B2	12/2003	Dixon et al.	7,718,109	B2	5/2010	Robb et al.
6,656,224	B2	12/2003	Middleton	7,806,932	B2	10/2010	Webb et al.
6,660,038	B2	12/2003	Boyer, II et al.	7,815,682	B1	10/2010	Peterson
6,669,700	B1	12/2003	Farris et al.	7,819,903	B2	10/2010	Fraser et al.
6,676,892	B2	1/2004	Das et al.	7,837,734	B2	11/2010	Zucherman et al.
6,682,561	B2	1/2004	Songer et al.	7,857,839	B2	12/2010	Duong et al.
6,682,563	B2	1/2004	Scharf	7,892,239	B2	2/2011	Warnick et al.
6,695,845	B2	2/2004	Dixon et al.	7,914,561	B2	3/2011	Konieczynski et al.
6,695,846	B2	2/2004	Richelsoph et al.	7,931,840	B2	4/2011	Michelson
6,702,817	B2	3/2004	Beger et al.	7,935,137	B2	5/2011	Gorhan et al.
6,719,794	B2	4/2004	Gerber et al.	7,959,675	B2	6/2011	Gately
6,730,127	B2	5/2004	Michelson	7,963,981	B2	6/2011	Binder et al.
6,740,087	B2	5/2004	Knox	7,972,366	B2	7/2011	Richelsoph et al.
6,743,255	B2	6/2004	Ferree	7,985,255	B2	7/2011	Bray et al.
6,746,450	B1	6/2004	Wall et al.	7,998,212	B2	8/2011	Schwab et al.
6,755,833	B1	6/2004	Paul et al.	8,025,697	B2	9/2011	McClellan, III et al.
6,758,849	B1	7/2004	Michelson	8,043,293	B2	10/2011	Warnick
6,767,369	B2	7/2004	Boyer, II et al.	8,062,375	B2	11/2011	Glerum et al.
6,776,798	B2	8/2004	Camino et al.	8,088,163	B1	1/2012	Kleiner
6,780,186	B2	8/2004	Errico et al.	8,100,955	B2	1/2012	Blain et al.
6,808,537	B2	10/2004	Michelson	8,142,886	B2	3/2012	Noble et al.
6,830,570	B1	12/2004	Frey et al.	8,147,554	B2	4/2012	Hansell et al.
6,833,006	B2	12/2004	Foley et al.	8,157,845	B2	4/2012	Warnick et al.
6,835,208	B2	12/2004	Marchosky	8,231,661	B2	7/2012	Carls
6,884,242	B2	4/2005	LeHuec et al.	8,252,060	B2	8/2012	Hansell et al.
6,890,334	B2	5/2005	Brace et al.	8,267,966	B2	9/2012	McCormack et al.
6,890,335	B2	5/2005	Grabowski et al.	8,268,001	B2	9/2012	Butler et al.
6,890,355	B2	5/2005	Michelson	8,277,510	B2	10/2012	Kleiner
6,932,610	B2	8/2005	Ono et al.	8,282,675	B2	10/2012	Maguire et al.
6,962,606	B2	11/2005	Michelson	8,282,682	B2	10/2012	Kirschman
6,964,664	B2	11/2005	Freid et al.	8,292,960	B2	10/2012	Kleiner
6,974,480	B2	12/2005	Messerli et al.	8,343,224	B2	1/2013	Lynn et al.
6,984,234	B2	1/2006	Bray	8,366,777	B2	2/2013	Matthis et al.
7,001,389	B1	2/2006	Navarro et al.	8,382,842	B2	2/2013	Greenhalgh et al.
7,001,672	B2	2/2006	Justin et al.	8,394,125	B2	3/2013	Assell
7,025,769	B1	4/2006	Ferree	8,409,290	B2	4/2013	Zamani et al.
7,048,739	B2	5/2006	Konieczynski et al.	8,414,590	B2	4/2013	Oh et al.
7,063,701	B2	6/2006	Michelson	8,425,529	B2	4/2013	Milz et al.
7,077,844	B2	7/2006	Michelson	8,430,930	B2	4/2013	Hunt
7,112,222	B2	9/2006	Fraser et al.	8,435,298	B2 *	5/2013	Weiman A61F 2/44 623/17.11
7,128,761	B2 *	10/2006	Kuras A61F 2/4425 606/247	8,444,650	B2	5/2013	Warnick et al.
7,135,024	B2	11/2006	Cook et al.	8,470,039	B2	6/2013	Blain
7,141,068	B2	11/2006	Ross et al.	8,480,745	B2	7/2013	Liu et al.
7,172,627	B2	2/2007	Fiere et al.	8,491,658	B1	7/2013	Etminan
7,182,782	B2	2/2007	Kirschman	8,496,691	B2	7/2013	Blain
7,186,254	B2	3/2007	Dinh et al.	8,496,708	B2	7/2013	Blain
7,217,291	B2	5/2007	Zucherman et al.	8,506,636	B2	8/2013	Dye
7,220,263	B2	5/2007	Cordaro	8,518,120	B2	8/2013	Glerum et al.
7,232,463	B2	6/2007	Falahee	8,545,566	B2	10/2013	Niemiec et al.
7,232,464	B2	6/2007	Mathieu et al.	8,556,979	B2	10/2013	Glerum et al.
7,235,082	B2	6/2007	Bartish et al.	8,579,904	B2	11/2013	Siccardi et al.
7,273,481	B2	9/2007	Lombardo et al.	8,603,175	B2	12/2013	Thibodeau
7,306,605	B2	12/2007	Ross	8,617,246	B2	12/2013	Malone
7,309,340	B2	12/2007	Fallin et al.	8,628,578	B2	1/2014	Miller et al.
7,473,277	B2	1/2009	Boyer, II et al.	8,632,595	B2	1/2014	Weiman
7,481,829	B2	1/2009	Baynham et al.	8,652,137	B2	2/2014	Blain et al.
7,481,830	B2	1/2009	Wall et al.	8,652,143	B2	2/2014	McClellan, III et al.
7,503,933	B2	3/2009	Michelson	8,679,183	B2	3/2014	Glerum et al.
7,521,017	B2	4/2009	Kunze et al.	8,685,095	B2	4/2014	Miller et al.
7,524,325	B2	4/2009	Khalili	8,685,098	B2	4/2014	Glerum et al.
7,537,664	B2	5/2009	O'Neill et al.	8,690,949	B2	4/2014	Messerli et al.
7,540,882	B2	6/2009	Michelson	8,696,721	B2	4/2014	Blain
7,547,324	B2	6/2009	Cragg et al.	8,702,798	B2	4/2014	Matthis et al.
7,621,943	B2	11/2009	Michelson	8,709,086	B2	4/2014	Glerum
7,628,816	B2	12/2009	Magerl et al.	8,715,355	B2	5/2014	Kleiner
7,641,665	B2	1/2010	Zubok et al.	8,728,387	B2	5/2014	Jones et al.
7,641,690	B2	1/2010	Abdou	8,740,942	B2	6/2014	Blain
7,651,517	B2	1/2010	Konieczynski et al.	8,758,443	B2	6/2014	Ullrich, Jr. et al.
7,658,766	B2	2/2010	Melkent et al.	8,795,366	B2	8/2014	Varela
7,674,294	B2	3/2010	Karahalios et al.	8,795,370	B2	8/2014	Kirschman
7,674,297	B2	3/2010	Falahee	8,801,785	B2	8/2014	Brittan et al.
7,693,981	B2	4/2010	Clubb et al.	8,801,791	B2	8/2014	Soo et al.
7,708,778	B2	5/2010	Gordon et al.	8,801,793	B2	8/2014	Mckay
7,708,779	B2	5/2010	Edie et al.	8,801,794	B2	8/2014	Blain
				8,840,668	B1	9/2014	Donahoe et al.
				8,843,229	B2	9/2014	Vanasse et al.
				8,845,731	B2	9/2014	Weiman
				8,852,279	B2	10/2014	Weiman

(56)

References Cited

U.S. PATENT DOCUMENTS

8,888,853 B2	11/2014	Glerum et al.	10,512,545 B2	12/2019	Arnone
8,888,854 B2	11/2014	Glerum et al.	10,517,740 B2	12/2019	Ball et al.
8,926,704 B2	1/2015	Glerum et al.	10,524,929 B2	1/2020	Shoshtaev
8,986,307 B2	3/2015	Kirschman	10,555,819 B2	2/2020	Miccio
8,986,383 B2	3/2015	Castro	10,568,664 B2	2/2020	Blain et al.
8,992,703 B2	3/2015	O'Neill et al.	10,610,373 B2	4/2020	Jang et al.
8,998,924 B2	4/2015	Simpson et al.	10,624,757 B2	4/2020	Bost
9,060,873 B2	6/2015	Abdou	10,660,763 B2	5/2020	Wilson et al.
9,078,706 B2	7/2015	Kirschman	10,667,927 B2	6/2020	Lamborne et al.
9,095,385 B2	8/2015	Wallenstein et al.	10,687,876 B2	6/2020	Vrionis et al.
9,101,410 B1	8/2015	Urrea	10,702,397 B2	7/2020	Simpson et al.
9,138,327 B1	9/2015	McClellan, III	10,709,575 B2	7/2020	Robinson
9,138,330 B2	9/2015	Hansell et al.	10,758,361 B2	9/2020	Blain
9,198,772 B2	12/2015	Weiman	10,765,525 B2	9/2020	Sansur et al.
9,204,974 B2	12/2015	Glerum et al.	10,905,567 B2	2/2021	Kuyler et al.
9,220,542 B2	12/2015	Kerboul et al.	10,993,810 B2	5/2021	Magagnoli
9,220,547 B2	12/2015	Blain et al.	11,026,801 B2	6/2021	Suh et al.
9,226,836 B2	1/2016	Glerum	11,147,682 B2	10/2021	Trudeau et al.
9,237,957 B2	1/2016	Klimek et al.	11,147,687 B2	10/2021	Hyeon et al.
9,259,327 B2	2/2016	Niemiec et al.	11,173,047 B2 *	11/2021	Milz A61F 2/4465
9,265,546 B2	2/2016	Blain	11,179,247 B2	11/2021	Jebesen et al.
9,271,765 B2	3/2016	Blain	11,213,404 B2	1/2022	Foley et al.
9,271,845 B2	3/2016	Hunt et al.	11,285,014 B1 *	3/2022	Josse A61F 2/4455
9,320,610 B2	4/2016	Alheid et al.	11,364,057 B2	6/2022	Blain et al.
9,358,126 B2	6/2016	Glerum et al.	11,382,769 B2	7/2022	Blain et al.
9,399,086 B2	7/2016	Melkent et al.	11,654,030 B2	5/2023	Valkoun et al.
9,402,736 B2	8/2016	Etminan	11,696,837 B2	7/2023	Messerli
9,415,137 B2	8/2016	Meridew et al.	11,911,284 B2	2/2024	Huh et al.
9,421,108 B2	8/2016	Hunt	2001/0007941 A1	7/2001	Steiner et al.
9,427,328 B2	8/2016	Drochner et al.	2001/0014807 A1	8/2001	Wagner et al.
9,433,707 B2	9/2016	Swords et al.	2001/0047207 A1	11/2001	Michelson
9,439,778 B2	9/2016	Biedermann et al.	2002/0004683 A1	1/2002	Michelson
9,456,901 B2	10/2016	Jones et al.	2002/0016595 A1	2/2002	Michelson
9,463,099 B2	10/2016	Levy et al.	2002/0022843 A1	2/2002	Michelson
9,545,317 B2	1/2017	Hunt	2002/0045898 A1	4/2002	Freid et al.
9,549,823 B2	1/2017	Hunt et al.	2002/0116064 A1	8/2002	Middleton
9,554,918 B2	1/2017	Weiman	2002/0147450 A1	10/2002	LeHuec et al.
9,572,669 B2	2/2017	Hunt et al.	2002/0151895 A1	10/2002	Soboleski et al.
9,585,707 B2	3/2017	Blain	2002/0169507 A1	11/2002	Malone
9,615,934 B2	4/2017	Khurana	2002/0169508 A1	11/2002	Songer et al.
9,636,226 B2	5/2017	Hunt	2003/0040798 A1	2/2003	Michelson
9,662,223 B2	5/2017	Matthis et al.	2003/0060828 A1	3/2003	Michelson
9,662,226 B2	5/2017	Wickham	2003/0093082 A1	5/2003	Campbell et al.
9,668,876 B2	6/2017	Blain et al.	2003/0135279 A1	7/2003	Michelson
9,700,428 B2	7/2017	Niemiec et al.	2003/0139812 A1	7/2003	Garcia et al.
9,707,317 B2	7/2017	Hunter et al.	2003/0167091 A1	9/2003	Scharf
9,757,235 B2	9/2017	Hunt et al.	2003/0171753 A1	9/2003	Collins et al.
9,757,247 B2	9/2017	Mantri	2003/0187442 A1	10/2003	Richelsoph et al.
9,770,343 B2	9/2017	Weiman	2003/0212399 A1	11/2003	Dinh et al.
9,782,269 B2	10/2017	Hansell et al.	2004/0010254 A1	1/2004	Cook et al.
RE46,647 E *	12/2017	Messerli A61B 17/1659	2004/0068319 A1	4/2004	Cordaro
9,867,713 B2	1/2018	Milz et al.	2004/0087951 A1	5/2004	Khalili
9,889,020 B2	2/2018	Baynham	2004/0097935 A1	5/2004	Richelsoph et al.
9,936,984 B2	4/2018	Blain	2004/0127900 A1	7/2004	Konieczynski et al.
9,949,841 B2	4/2018	Glerum et al.	2004/0127904 A1	7/2004	Konieczynski et al.
9,962,271 B2	5/2018	Glerum	2004/0176778 A1	9/2004	Zubok et al.
9,968,462 B2	5/2018	Weiman	2004/0181227 A1	9/2004	Khalili
9,980,823 B2	5/2018	Matthis et al.	2004/0210219 A1	10/2004	Bray
9,980,825 B2	5/2018	Nichols et al.	2004/0220570 A1	11/2004	Frigg et al.
9,987,051 B2	6/2018	Nunley et al.	2004/0230192 A1	11/2004	Graf
9,987,149 B2	6/2018	Simpson et al.	2004/0260306 A1	12/2004	Fallin et al.
10,004,607 B2	6/2018	Weiman et al.	2005/0027296 A1	2/2005	Thramann et al.
10,022,245 B2	7/2018	Frasier et al.	2005/0027360 A1	2/2005	Webb et al.
10,028,841 B2	7/2018	Moore et al.	2005/0033433 A1	2/2005	Michelson
10,034,770 B2	7/2018	Etminan	2005/0049593 A1	3/2005	Duong et al.
10,064,737 B2	9/2018	Tsai et al.	2005/0085913 A1	4/2005	Fraser et al.
10,092,412 B2	10/2018	Drochner et al.	2005/0101960 A1	5/2005	Fiere et al.
10,130,490 B2	11/2018	Hansell et al.	2005/0149191 A1	7/2005	Cragg et al.
10,154,912 B2	12/2018	Glerum	2005/0159746 A1	7/2005	Grab et al.
10,245,152 B2	4/2019	Kloss	2005/0177237 A1	8/2005	Shapple et al.
10,271,957 B2	4/2019	Niemiec et al.	2005/0177245 A1	8/2005	Leatherbury et al.
10,299,938 B1	5/2019	Ehteshami	2005/0192576 A1	9/2005	Michelson
10,369,009 B2	8/2019	Joly et al.	2005/0277933 A1	12/2005	Wall et al.
10,470,892 B2	11/2019	Abdou	2006/0074488 A1	4/2006	Abdou
10,478,313 B1 *	11/2019	Sweeney, III A61F 2/4611	2006/0122604 A1	6/2006	Gorhan et al.
			2006/0200147 A1	9/2006	Ensign et al.
			2006/0224241 A1	10/2006	Butler et al.
			2006/0235403 A1	10/2006	Blain
			2006/0235409 A1	10/2006	Blain

(56)

References Cited

U.S. PATENT DOCUMENTS

2006/0235418	A1	10/2006	Gil et al.
2006/0241597	A1	10/2006	Mitchell et al.
2006/0241616	A1	10/2006	Konieczynski et al.
2006/0247650	A1	11/2006	Yerby et al.
2007/0055252	A1	3/2007	Blain et al.
2007/0055373	A1	3/2007	Hudgins et al.
2007/0093897	A1	4/2007	Gerbac et al.
2007/0123863	A1	5/2007	Winslow et al.
2007/0135814	A1	6/2007	Farris
2007/0213820	A1	9/2007	Magerl et al.
2007/0250166	A1	10/2007	McKay
2007/0255414	A1	11/2007	Melkent et al.
2007/0255416	A1	11/2007	Melkent et al.
2007/0282441	A1	12/2007	Stream
2007/0282446	A1	12/2007	Li
2008/0103598	A1	5/2008	Trudeau et al.
2008/0140207	A1	6/2008	Olmos
2008/0161925	A1	7/2008	Brittan et al.
2008/0167686	A1	7/2008	Trieu et al.
2008/0177390	A1	7/2008	Mitchell et al.
2008/0249569	A1	10/2008	Waugh
2008/0294260	A1	11/2008	Gray
2009/0072006	A1	3/2009	Clauson et al.
2009/0082819	A1	3/2009	Blain et al.
2009/0182430	A1	7/2009	Tyber et al.
2009/0198339	A1	8/2009	Kleiner et al.
2009/0270929	A1	10/2009	Suddaby
2010/0069960	A1	3/2010	Chaput
2010/0070037	A1	3/2010	Parry et al.
2010/0145459	A1	6/2010	McDonough et al.
2010/0211176	A1	8/2010	Greenhalgh
2011/0015744	A1	1/2011	Squires et al.
2011/0015745	A1	1/2011	Bucci
2011/0040384	A1	2/2011	Junn et al.
2011/0106259	A1	5/2011	Lindenmann et al.
2011/0153020	A1	6/2011	Abdelgany et al.
2011/0172775	A1	7/2011	Flickinger et al.
2011/0184522	A1	7/2011	Melkent et al.
2011/0224796	A1	9/2011	Weiland et al.
2011/0264218	A1	10/2011	Asaad
2011/0282453	A1	11/2011	Greenhalgh et al.
2012/0016480	A1	1/2012	Gerber et al.
2012/0029636	A1	2/2012	Ragab et al.
2012/0046749	A1	2/2012	Tatsumi
2012/0197402	A1	8/2012	Blackwell et al.
2012/0277801	A1	11/2012	Marik et al.
2013/0006357	A1	1/2013	Kruger
2013/0030529	A1	1/2013	Hunt
2013/0110238	A1	5/2013	Lindemann et al.
2013/0110248	A1	5/2013	Zipnick
2013/0123923	A1	5/2013	Pavlov et al.
2013/0158667	A1	6/2013	Tabor et al.
2013/0158669	A1	6/2013	Sungarian et al.
2013/0173003	A1*	7/2013	Matthis A61F 2/4465 623/17.16
2013/0181015	A1	7/2013	Cason
2013/0197643	A1	8/2013	Greenberg et al.
2013/0218276	A1	8/2013	Fiechter
2013/0268078	A1	10/2013	Richelsoph
2013/0297024	A1	11/2013	Khurana
2013/0325129	A1	12/2013	Huang
2014/0012318	A1	1/2014	Goel
2014/0066758	A1	3/2014	Marik et al.
2014/0094922	A1	4/2014	Abdou
2014/0180422	A1	6/2014	Klimek
2014/0236296	A1	8/2014	Wagner et al.
2014/0309741	A1	10/2014	Ganter et al.
2014/0324173	A1	10/2014	Kirschman
2015/0173917	A1	6/2015	Radcliffe et al.
2015/0190242	A1	7/2015	Blain et al.
2015/0320568	A1	11/2015	Ameil et al.
2015/0342648	A1	12/2015	McCormack et al.
2016/0000576	A1	1/2016	Kirschman
2016/0022438	A1	1/2016	Lamborne et al.
2016/0120660	A1	5/2016	Melkent et al.

2016/0151171	A1	6/2016	Mozelski et al.
2016/0213485	A1	7/2016	Schauler et al.
2016/0296338	A1	10/2016	Kim et al.
2017/0056201	A1	3/2017	Liang et al.
2017/0095342	A9	4/2017	Waugh
2017/0105844	A1	4/2017	Kuyler et al.
2017/0119439	A1	5/2017	Ozdil
2017/0119538	A1	5/2017	Baynham
2017/0231782	A1	8/2017	Perez-Cruet et al.
2017/0333205	A1	11/2017	Joly et al.
2018/0014947	A1	1/2018	Baynham
2018/0042732	A1*	2/2018	Seifert A61F 2/4611
2018/0104063	A1	4/2018	Asaad
2018/0214279	A1	8/2018	Etminan et al.
2018/0235769	A1	8/2018	Levy et al.
2018/0250051	A1	9/2018	Vrionis et al.
2018/0256336	A1	9/2018	Mueller
2018/0289508	A1	10/2018	Glerum
2018/0326323	A1	10/2018	Shoshtaev
2018/0325693	A1*	11/2018	Weiman A61F 2/4465
2019/0046333	A1*	2/2019	Hansell A61F 2/4465
2019/0053910	A1	2/2019	Sansur et al.
2019/0091036	A1	3/2019	Levy et al.
2019/0133785	A1	5/2019	Georges
2019/0175357	A1	6/2019	Sharabani
2019/0254840	A1	8/2019	Gray et al.
2020/0085586	A1	3/2020	Ludwig et al.
2020/0093612	A1	3/2020	Blain et al.
2020/0146729	A1	5/2020	Blain et al.
2020/0197149	A1	6/2020	Folger et al.
2020/0229943	A1	7/2020	Abdou
2020/0246157	A1	8/2020	Berry
2020/0315679	A1	10/2020	Vrionis et al.
2020/0345503	A1	11/2020	Sansur et al.
2020/0345505	A1	11/2020	Etminan et al.
2021/0000608	A1	1/2021	Blain et al.
2021/0137702	A1	5/2021	Neubardt
2021/0145600	A1	5/2021	Sharifi-Mehr et al.
2021/0145607	A1	5/2021	Kuyler et al.
2021/0154021	A1	5/2021	Bae et al.
2022/0015919	A1	1/2022	Reah et al.
2022/0031469	A1	2/2022	Suh et al.
2022/0192841	A1	6/2022	Blain et al.
2022/0241089	A1	8/2022	Clauss et al.
2022/0346843	A1	11/2022	Blain et al.
2022/0387189	A1	12/2022	Blain et al.
2023/0165690	A1	6/2023	Blain et al.

FOREIGN PATENT DOCUMENTS

DE	30 27 138	12/1981
DE	30 27 148	12/1981
DE	297 01 099	4/1997
DE	197 02 201	8/1998
DE	20 2004 015 912	12/2004
EP	0 242 842	10/1987
EP	0 974 319	1/2000
EP	1 029 510	8/2000
EP	1 346 697	9/2003
EP	1 470 803	10/2004
EP	2 719 360	5/2016
EP	2 967 904	5/2019
FR	2 766 353	1/1999
FR	2 813 519	3/2002
FR	2 859 904	3/2005
FR	2 923 155	5/2009
JP	2002-515287	5/2002
JP	2003-518977	6/2003
JP	2004-500156	1/2004
JP	2008-522787	7/2008
JP	2008-537498	9/2008
JP	2010-510852	4/2010
JP	2014-523751	9/2014
JP	2015-500701	1/2015
WO	WO 88/003781	6/1988
WO	WO 89/004150	5/1989
WO	WO 93/010725	6/1993
WO	WO 94/000066	1/1994
WO	WO 95/035067	12/1995

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 00/024343	5/2000
WO	WO 01/003570	1/2001
WO	WO 01/049191	7/2001
WO	WO 01/078615	10/2001
WO	WO 01/089428	11/2001
WO	WO 03/017856	3/2003
WO	WO 03/071966	9/2003
WO	WO 2004/006792	1/2004
WO	WO 2005/027760	3/2005
WO	WO 2006/020464	2/2006
WO	WO 2013/184946	12/2013
WO	WO 2017/075079	5/2017
WO	WO 2020/219789	10/2020
WO	WO 2021/055363	3/2021

OTHER PUBLICATIONS

Official Communication in Australian Application No. 2012211502, dated Jul. 17, 2013.

Notice of Acceptance in Australian Application No. 2012211502, dated Sep. 10, 2014.

Official Communication in Australian Application No. 2014274519, dated Sep. 17, 2015.

Official Communication in Australian Application No. 2014274519, dated Jun. 17, 2016.

Official Communication in Australian Application No. 2014274519, dated Aug. 26, 2016.

Notice of Acceptance in Australian Application No. 2014274519, dated Sep. 22, 2016.

Official Communication in Australian Application No. 2016277588, dated Sep. 7, 2017.

Official Communication in Australian Application No. 2018271345, dated Jul. 31, 2019.

Official Communication in European Application No. 06738204.4, dated Mar. 26, 2009.

Official Communication in European Application No. 06738204.4, dated Apr. 6, 2010.

Official Communication in European Application No. 06738204.4, dated Apr. 5, 2011.

Official Communication in European Application No. 06738204.4, dated Oct. 28, 2011.

Official Communication in European Application No. 06738204.4, dated Jul. 18, 2012.

Official Communication in European Application No. 06738204.4, dated Oct. 14, 2013.

Extended European Search Report for European Application No. 11160061.5, dated Nov. 2, 2011.

Official Communication in European Application No. 11160061.5, dated Jul. 9, 2012.

Extended European Search Report for European Application No. 11160063.1, dated Nov. 2, 2011.

Official Communication in European Application No. 11160063.1, dated Jul. 12, 2012.

Official Communication in European Application No. 11160063.1, dated Nov. 27, 2012.

Official Communication in European Application No. 11160063.1, dated Oct. 14, 2013.

Official Communication in European Application No. 14190344.3, dated Feb. 10, 2015.

Official Communication in European Application No. 14190344.3, dated Jan. 4, 2016.

Official Communication in European Application No. 14190344.3, dated Sep. 8, 2016.

Official Communication in European Application No. 18150661.9, dated May 25, 2018.

Official Communication in European Application No. 18150661.9, dated Aug. 23, 2019.

Official Communication in European Application No. 18150661.9, dated Jan. 18, 2023.

Official Communication in Japanese Application No. 2008-501962, dated May 10, 2011.

Official Communication in Japanese Application No. 2008-501962, dated Nov. 13, 2012.

Official Communication in Japanese Application No. 2011-210533, dated Mar. 5, 2013.

Official Communication in Japanese Application No. 2011-210533, dated Dec. 3, 2013.

Notice of Allowance in Japanese Application No. 2011-210533, dated May 7, 2014.

Notice of Allowance in Japanese Application No. 2013-117602, dated May 7, 2014.

Notice of Allowance in Japanese Application No. 2013-117602, dated Feb. 2, 2015.

International Search Report and Written Opinion in International Application No. PCT/US2006/009120, dated Oct. 20, 2006.

International Preliminary Report on Patentability and Written Opinion in International Application No. PCT/US2006/009120, dated Sep. 18, 2007.

Official Communication in Australian Application No. 2016212009, dated Sep. 6, 2019.

Official Communication in Australian Application No. 2016212009, dated May 26, 2020.

Official Communication in Australian Application No. 2016212009, dated Jul. 14, 2020.

Official Communication in Australian Application No. 2020281016, dated Nov. 24, 2021.

Official Communication in Australian Application No. 2020281016, dated Aug. 26, 2022.

Official Communication in Australian Application No. 2020281016, dated Oct. 7, 2022.

Official Communication in Australian Application No. 2020281016, dated Nov. 16, 2022.

Official Communication in Australian Application No. 2020281016, dated Nov. 23, 2022.

Official Communication in Canadian Application No. 2,972,788, dated Jan. 31, 2022.

Official Communication in Canadian Application No. 2,972,788, dated Oct. 31, 2022.

Official Communication in European Application No. 16743832.4, dated Jul. 24, 2018.

Official Communication in European Application No. 16743832.4, dated Jan. 26, 2023.

Official Communication in Japanese Application No. 2017-557269, dated Oct. 21, 2019.

Official Communication in Japanese Application No. 2017-557269, dated Jul. 13, 2020.

Official Communication in Japanese Application No. 2017-557269, dated Nov. 1, 2021.

Official Communication in Japanese Application No. 2020-181320, Sep. 21, 2021.

Official Communication in Japanese Application No. 2020-181320, Feb. 13, 2023.

International Search Report and Written Opinion in International Application No. PCT/US2016/013062, dated Mar. 16, 2016.

International Preliminary Report on Patentability and Written Opinion in International Application No. PCT/US2016/013062, dated Aug. 10, 2017.

Official Communication in European Application No. 19862906.5, dated Sep. 23, 2022.

Invitation to Pay Additional Search Fees in International Application No. PCT/US2019/052211, dated Nov. 14, 2019.

International Search Report and Written Opinion in International Application No. PCT/US2019/052211, dated Feb. 3, 2020.

International Preliminary Report on Patentability and Written Opinion in International Application No. PCT/US2019/052211, dated Apr. 1, 2021.

International Search Report and Written Opinion in International Application No. PCT/US2021/072934, dated Feb. 24, 2022.

Invitation to Pay Additional Search Fees in International Application No. PCT/US2021/072934, dated Jan. 13, 2022.

International Search Report and Written Opinion in International Application No. PCT/US2021/072934, dated Mar. 18, 2022.

(56)

References Cited

OTHER PUBLICATIONS

International Search Report and Written Opinion in International Application No. PCT/US2022/050050, dated Feb. 22, 2023.

Official Communication in Australian Application No. 2020257035, dated Jun. 24, 2021.

Official Communication in Australian Application No. 2022201902, dated May 10, 2023.

Official Communication in Brazilian Application No. BR112021005206-2, dated Jun. 5, 2023.

International Preliminary Report on Patentability and Written Opinion in International Application No. PCT/US2021/072934, dated Jun. 29, 2023.

International Preliminary Report on Patentability in International Application No. PCT/US2021/072334, dated Jun. 1, 2023.

International Preliminary Report on Patentability and Written Opinion in International Application No. PCT/US2022/050050, dated Jun. 13, 2024.

* cited by examiner

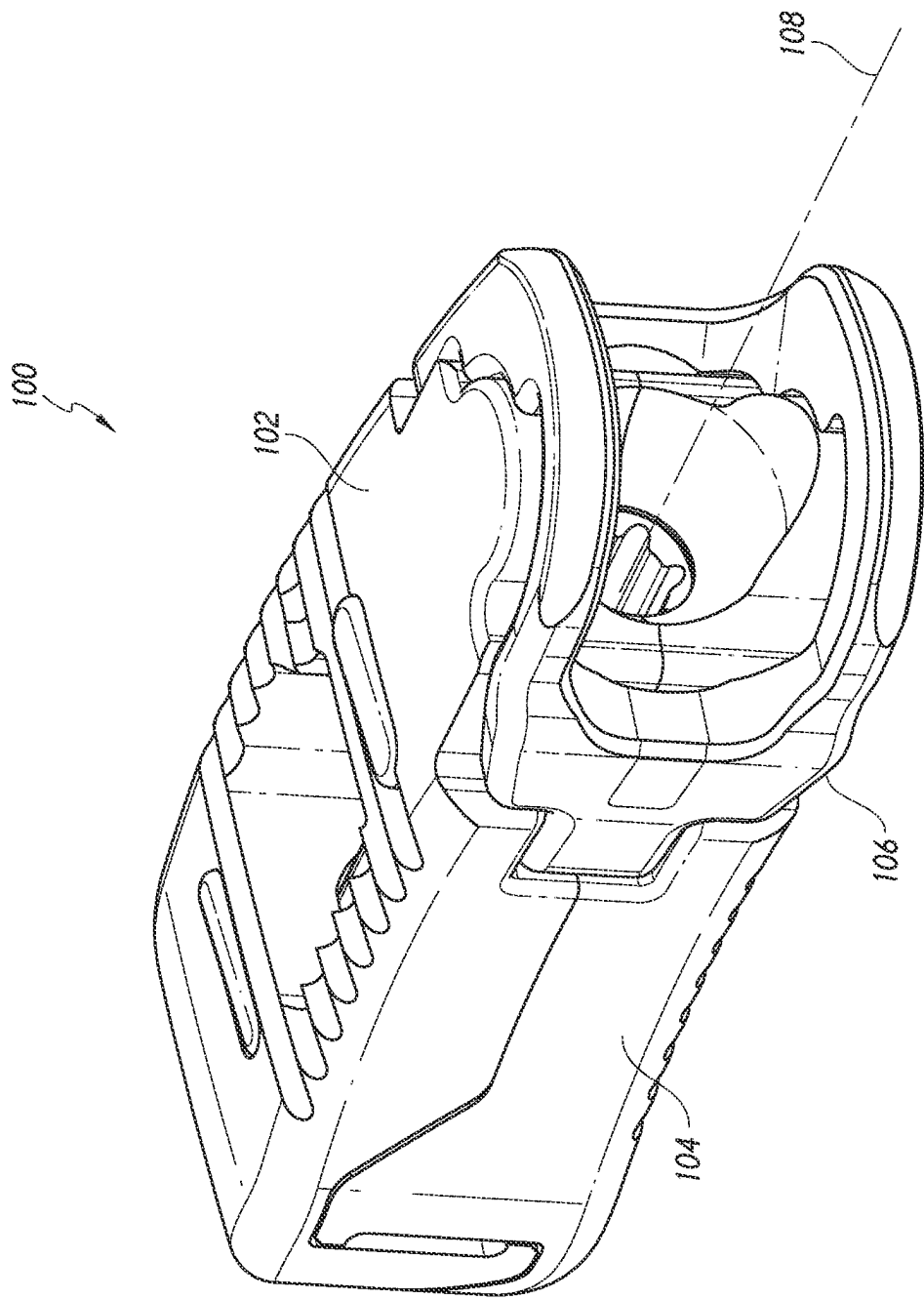


FIG. 1

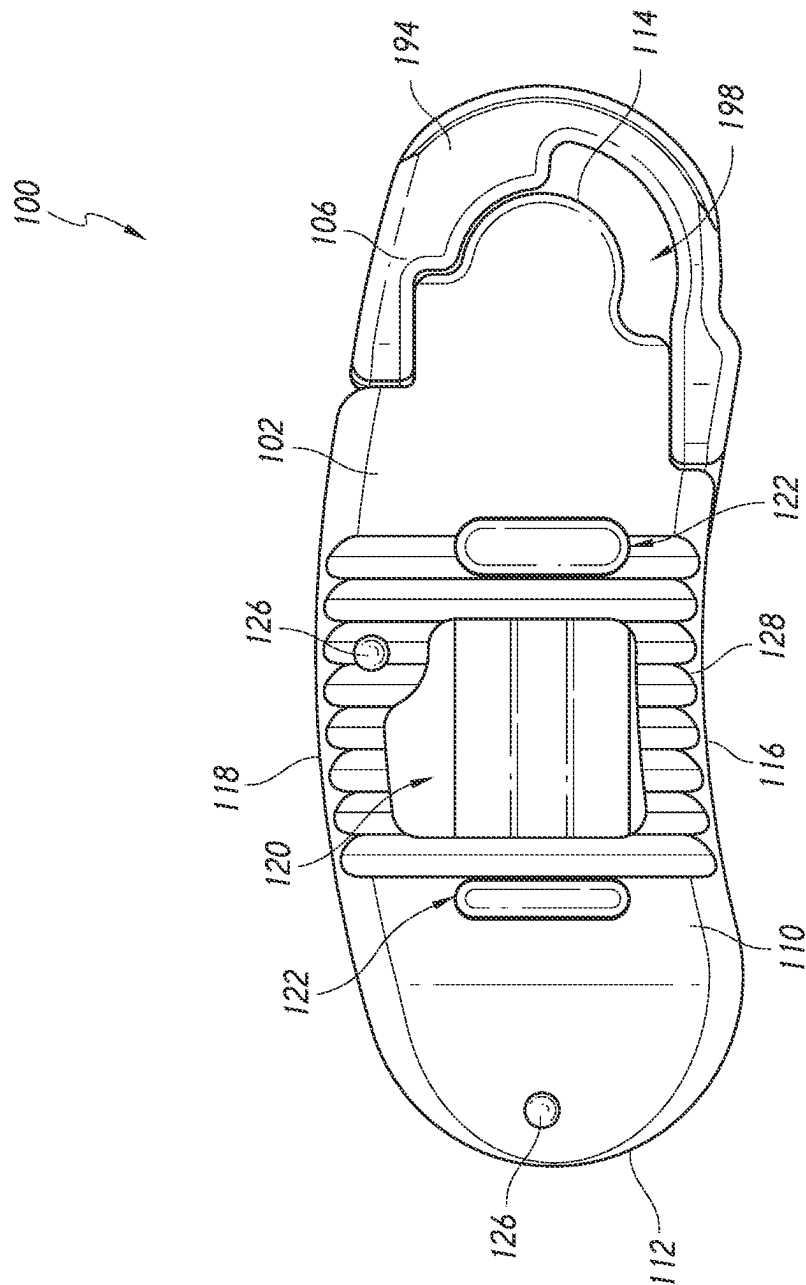
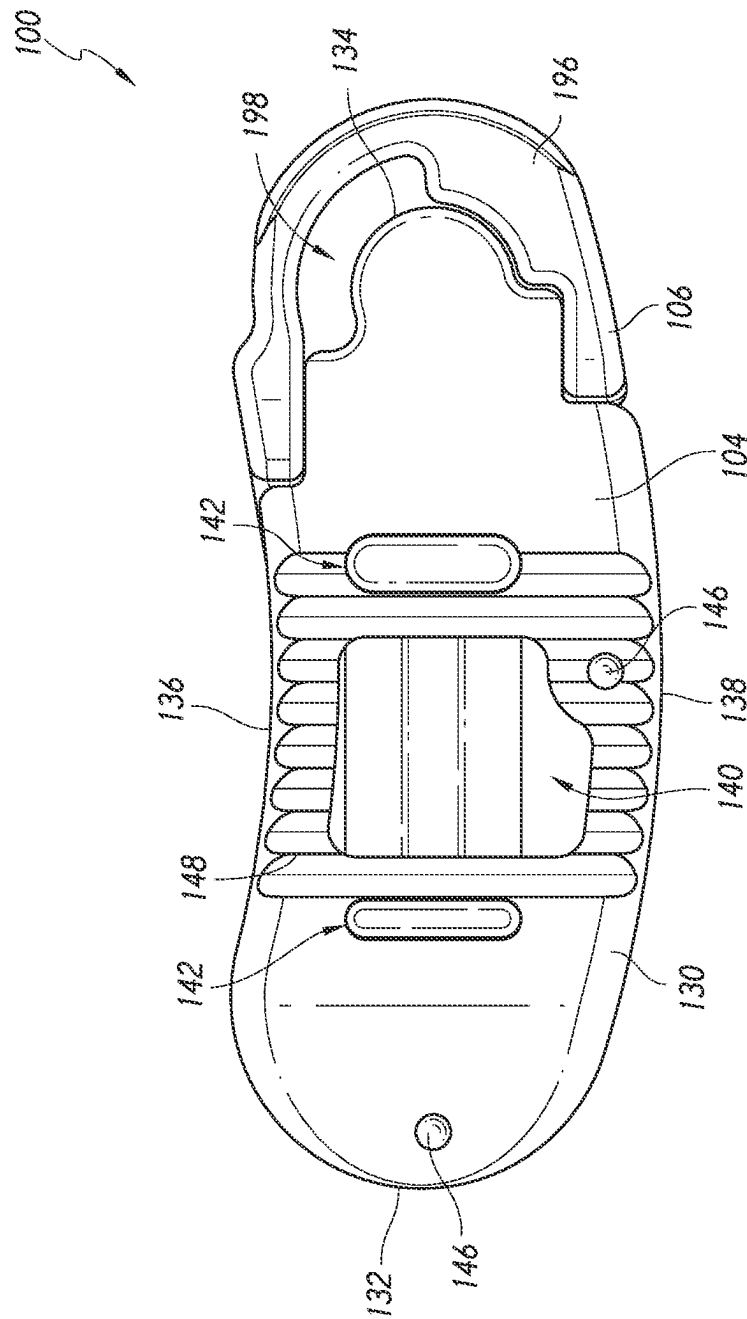


FIG. 2



354

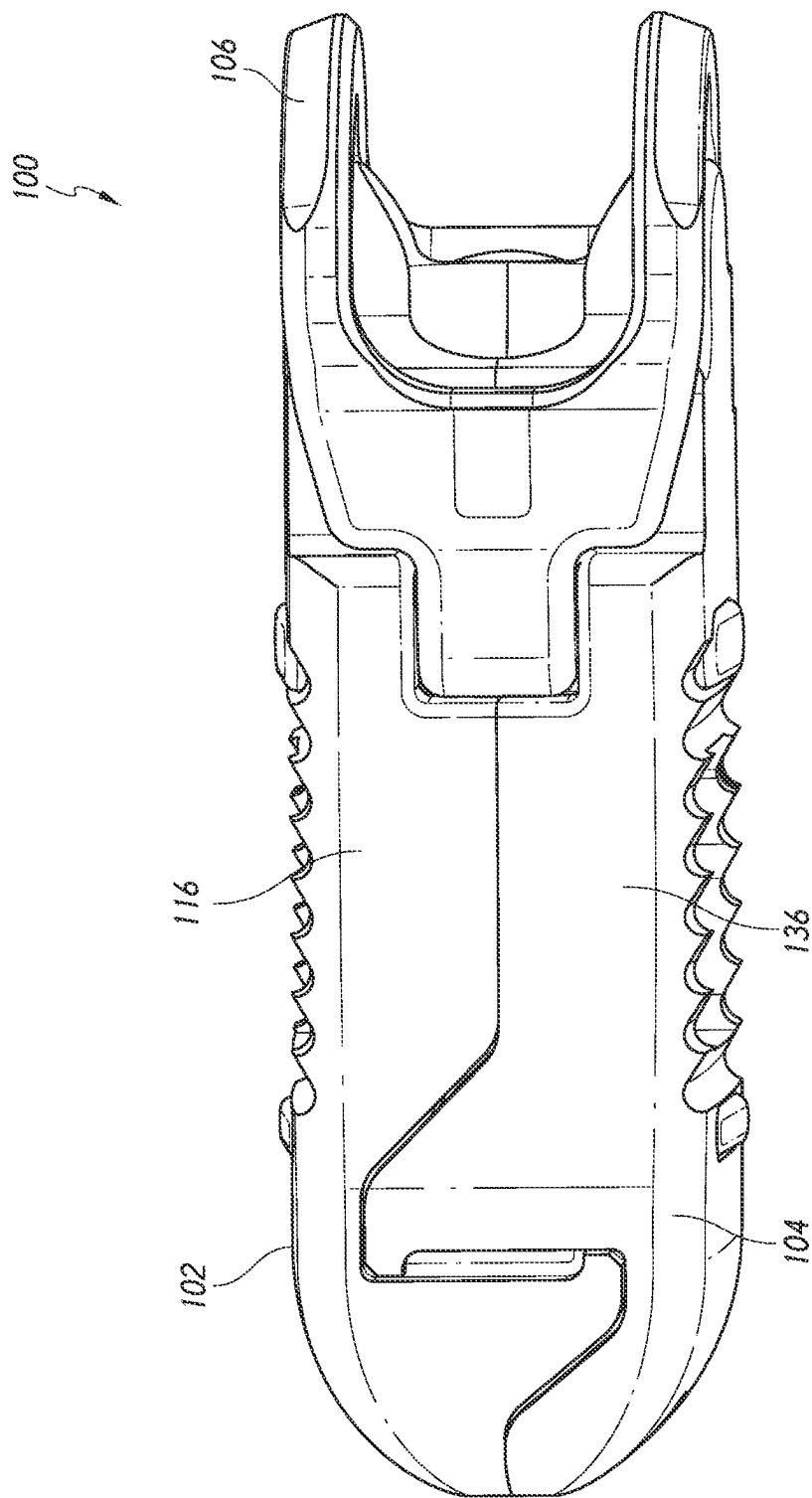


FIG. 4

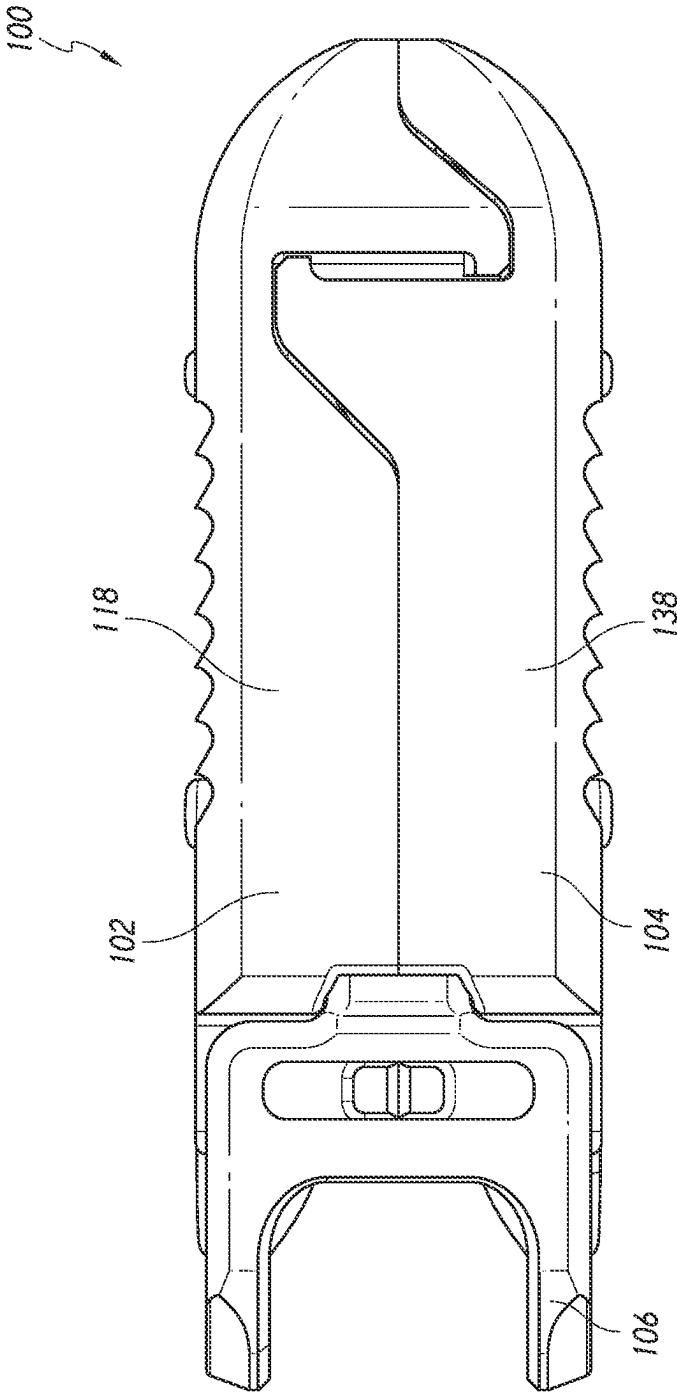


FIG. 5

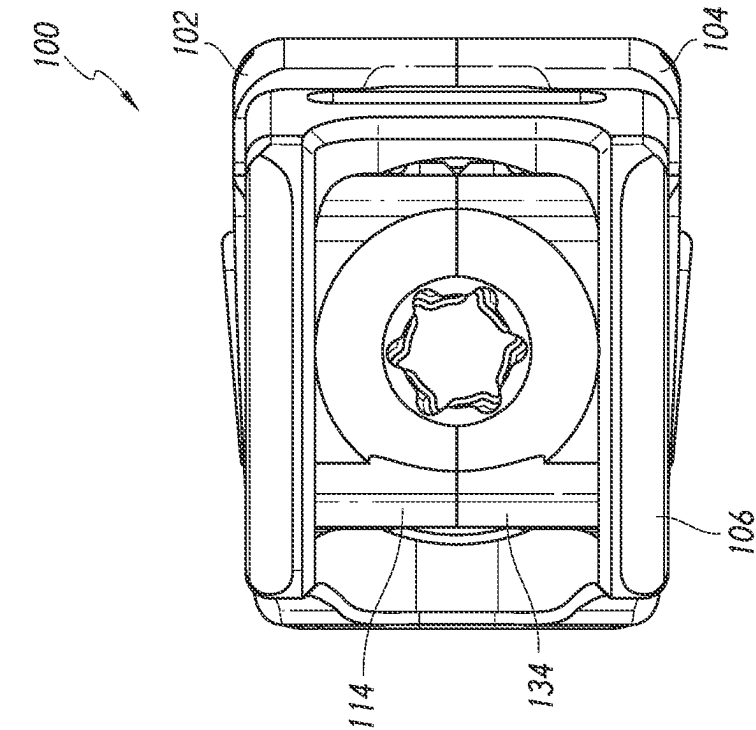


FIG. 6

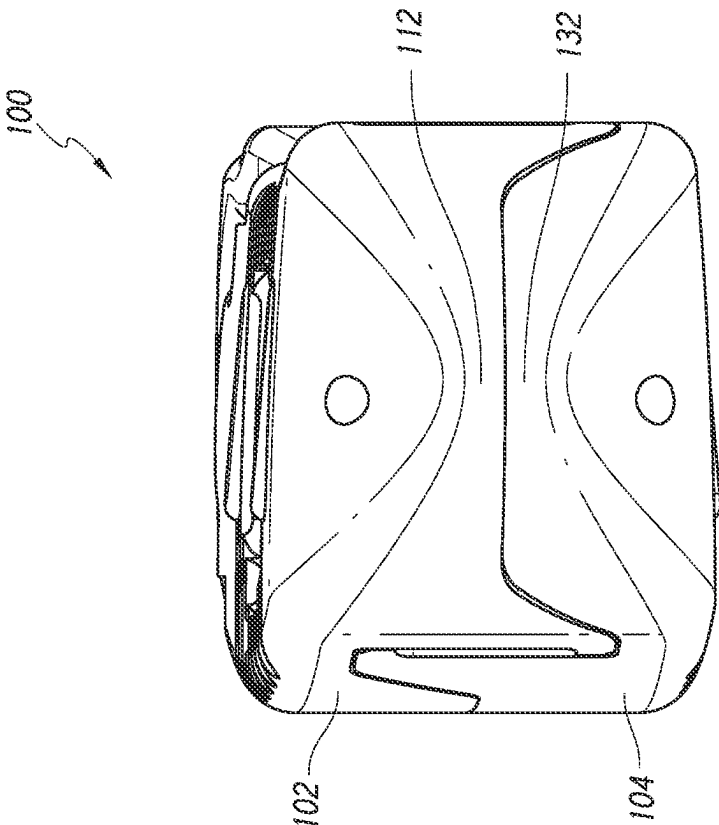


FIG. 7

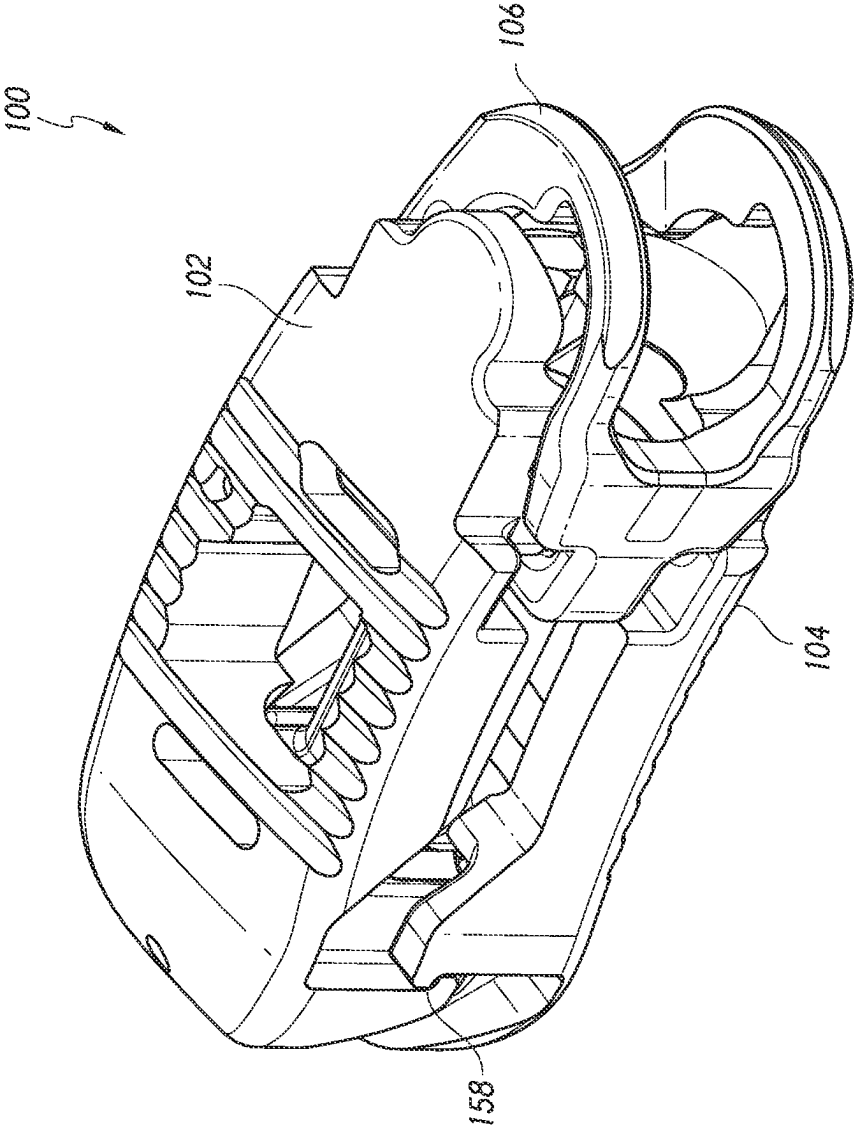


FIG. 8

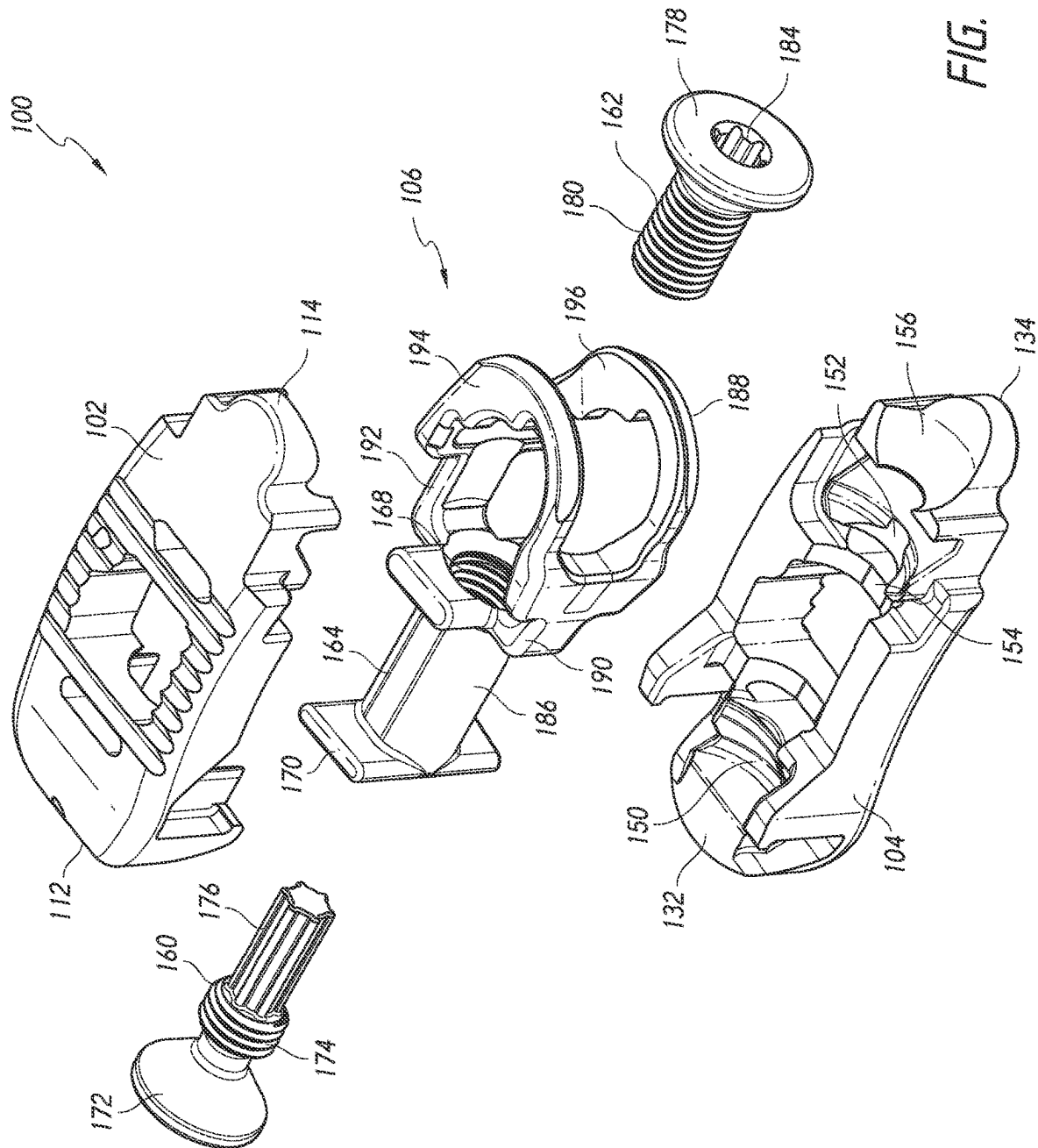


Fig. 9

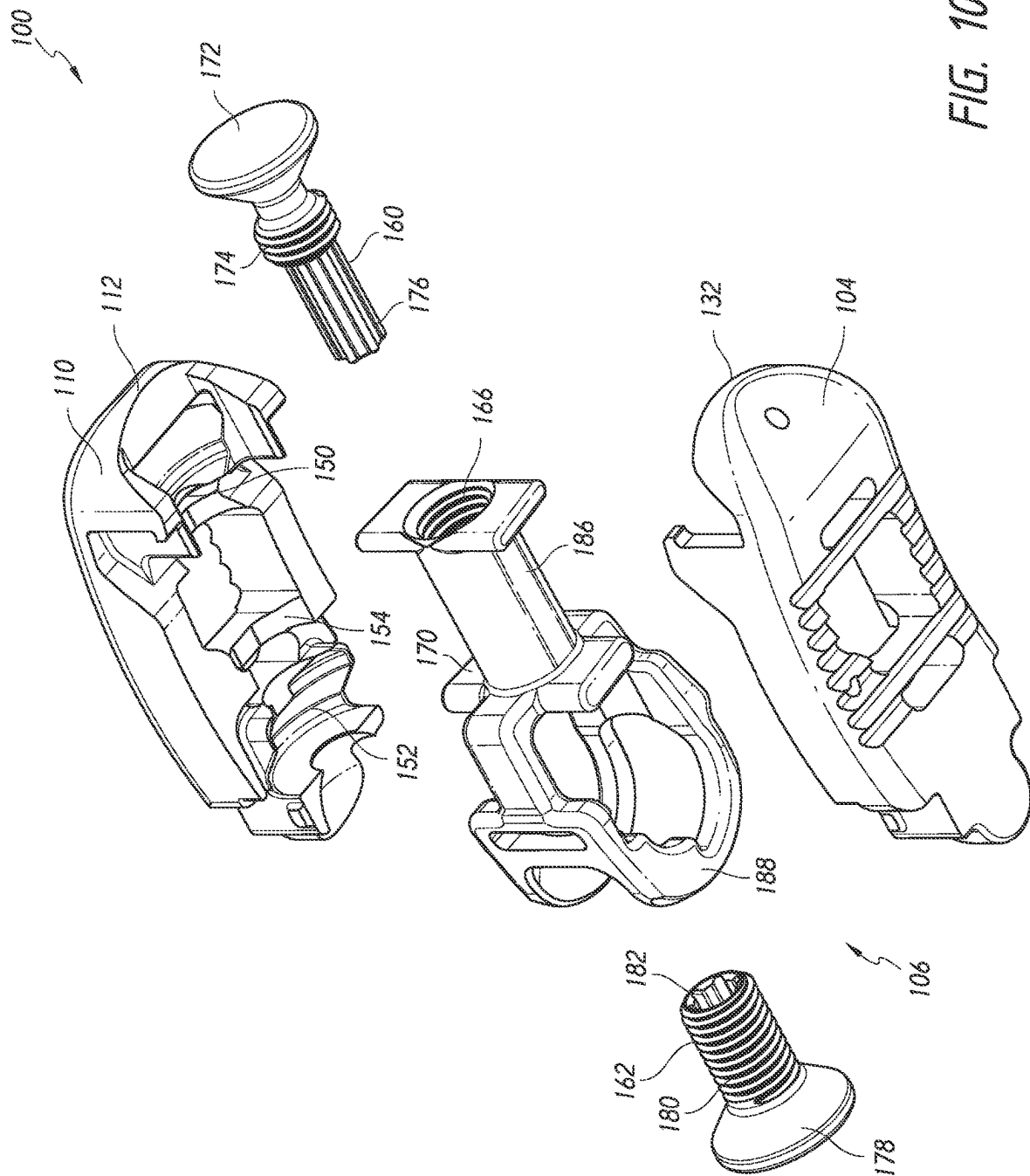


FIG. 10

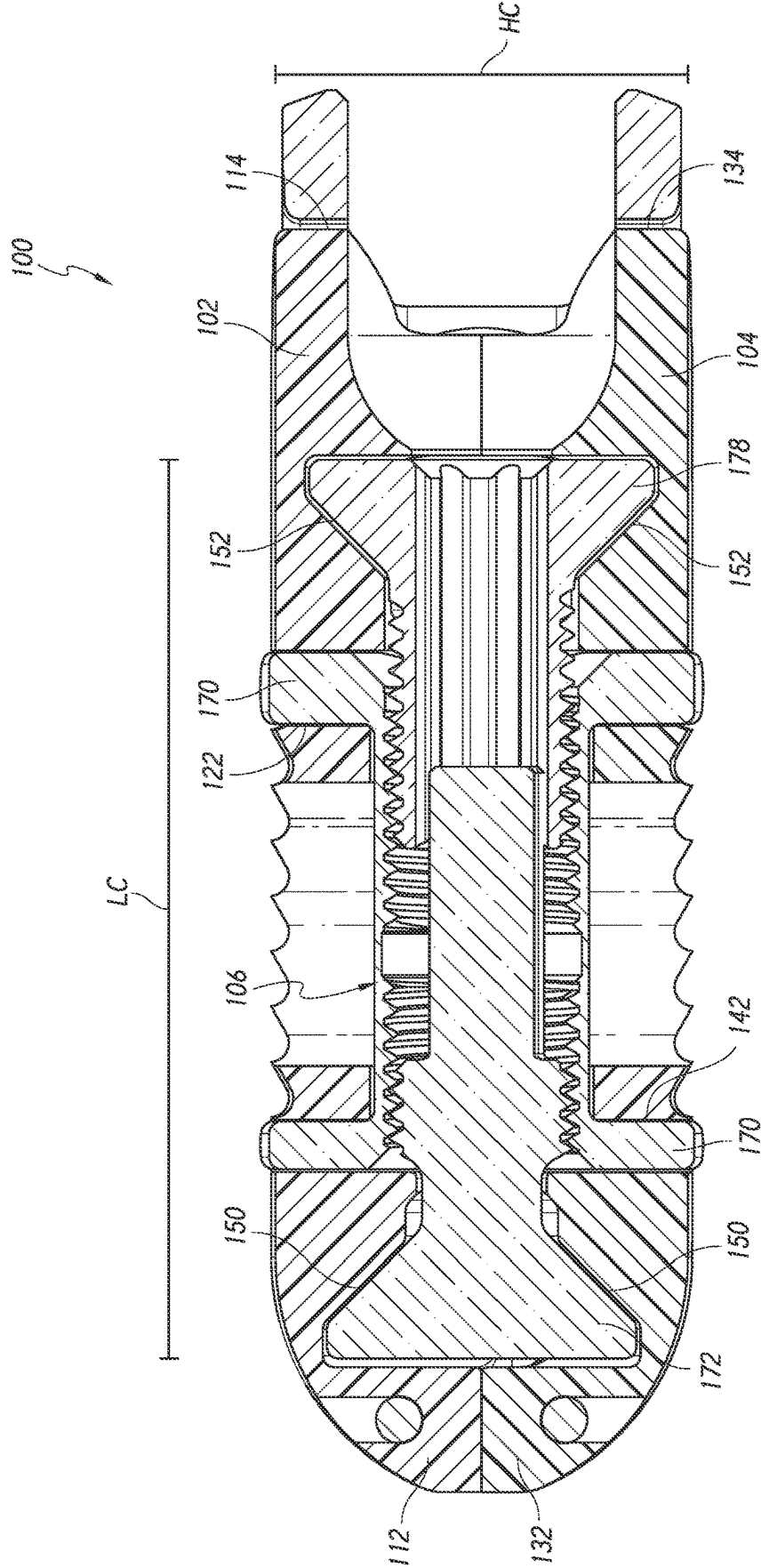


FIG. 11

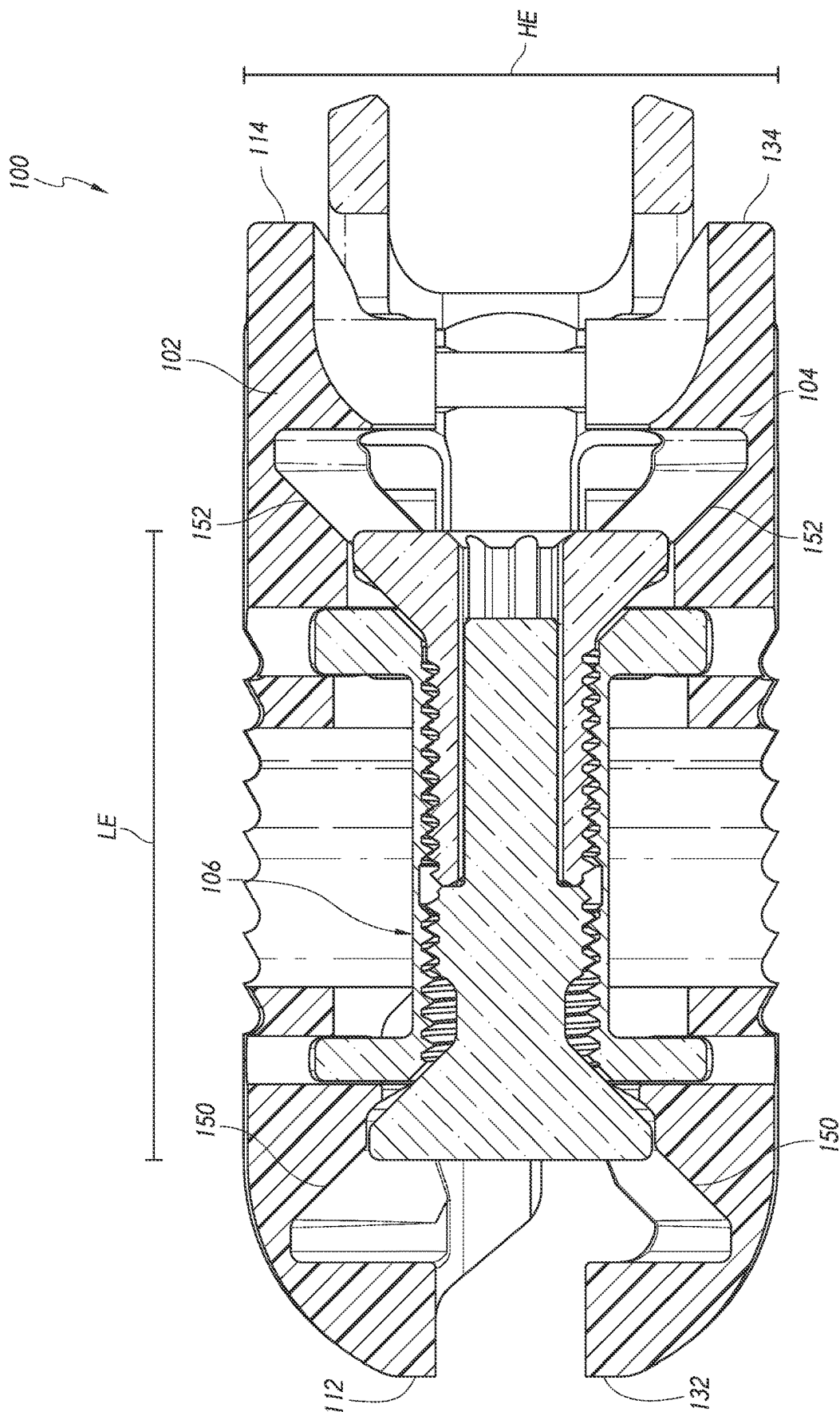
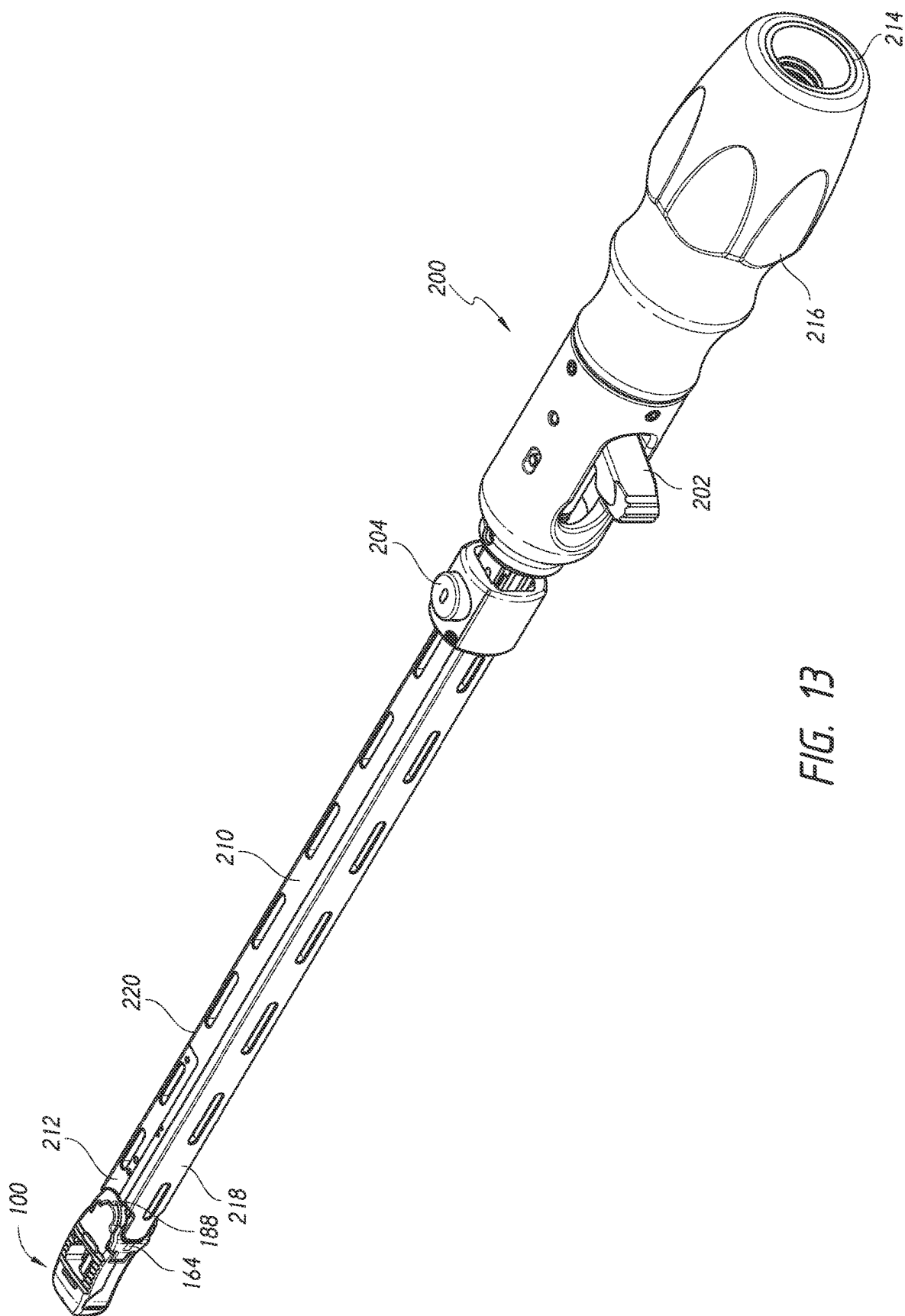


FIG. 12



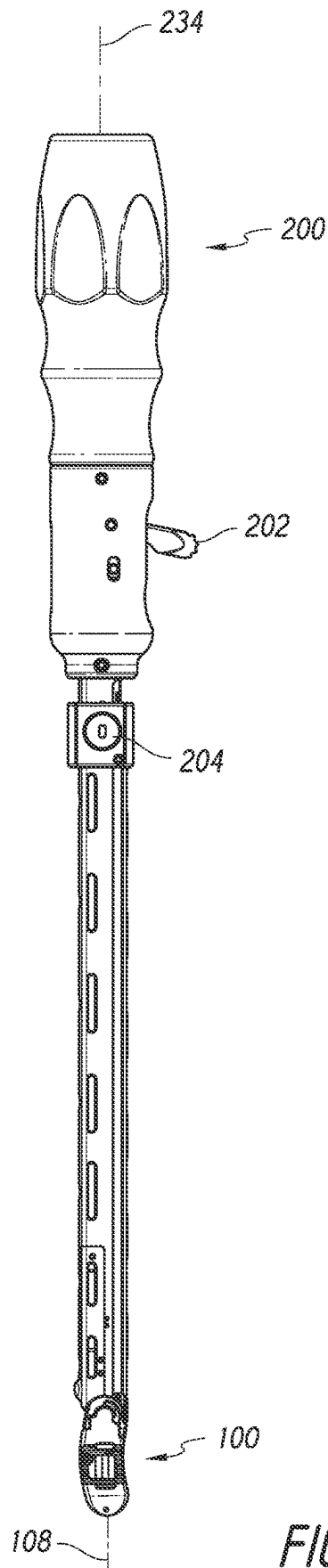


FIG. 14

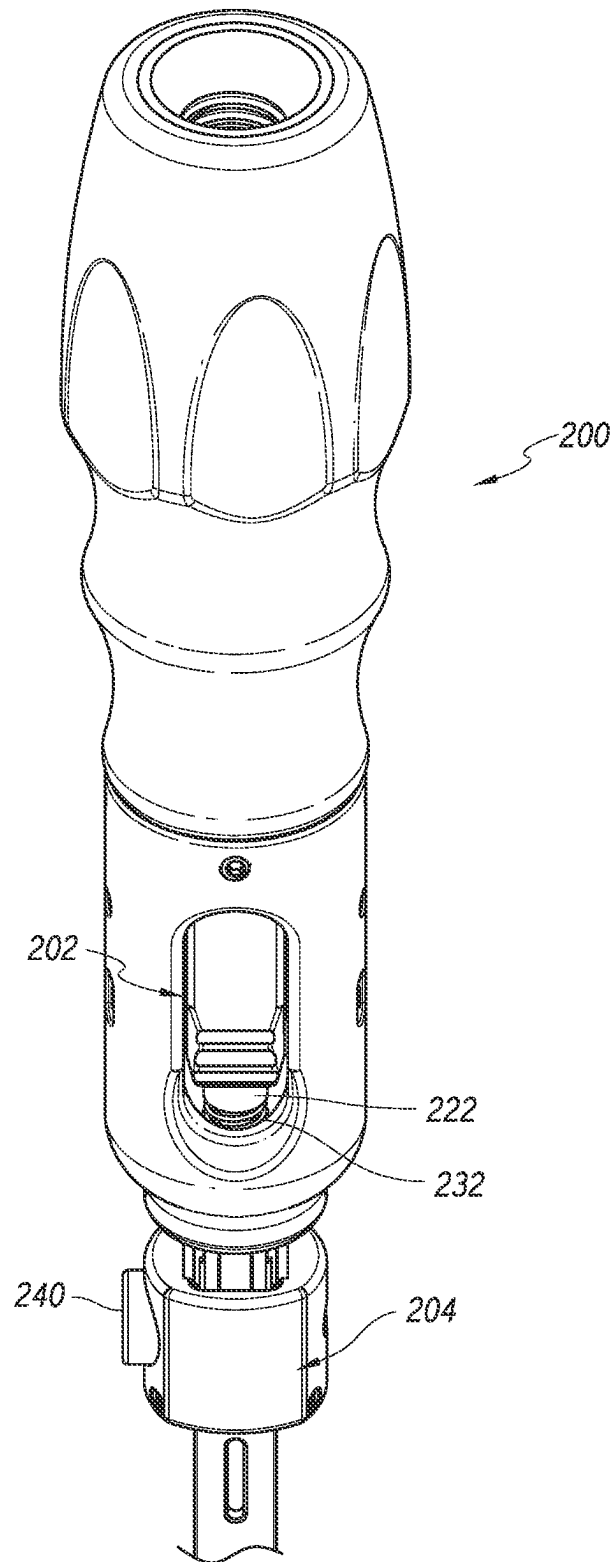


FIG. 15

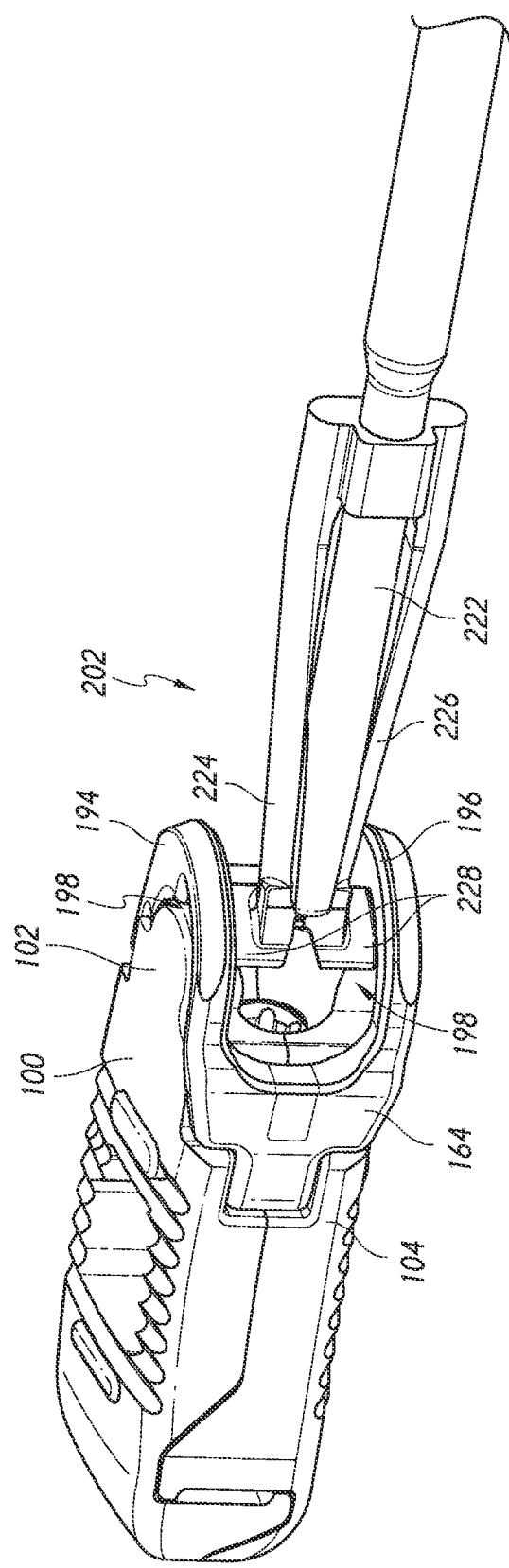


FIG. 16

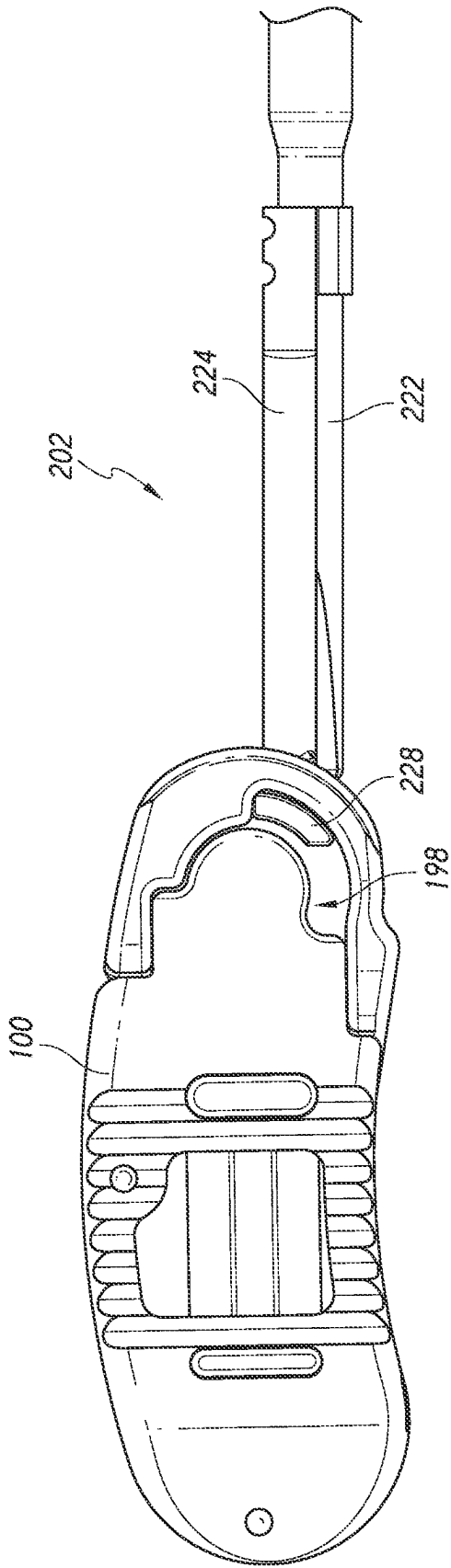


FIG. 17

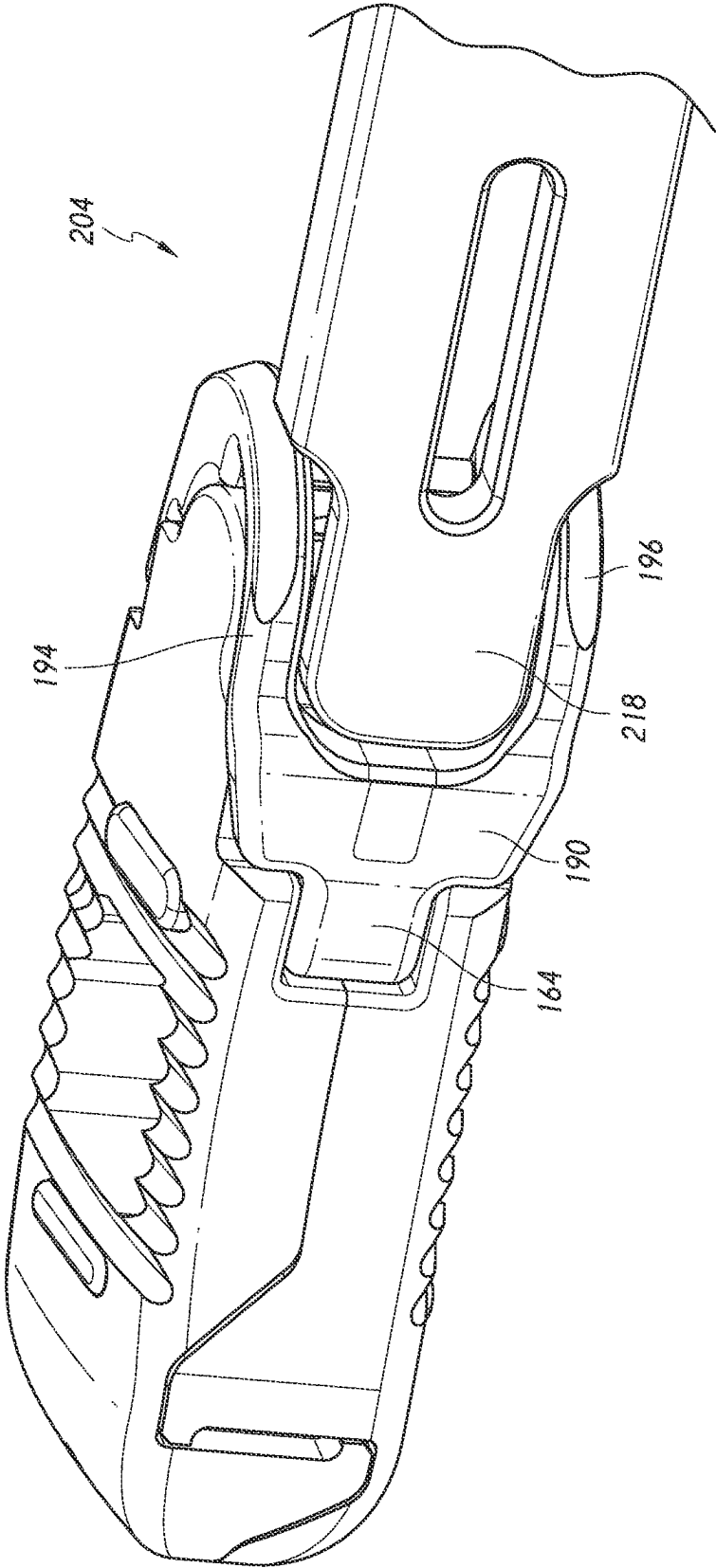


FIG. 18

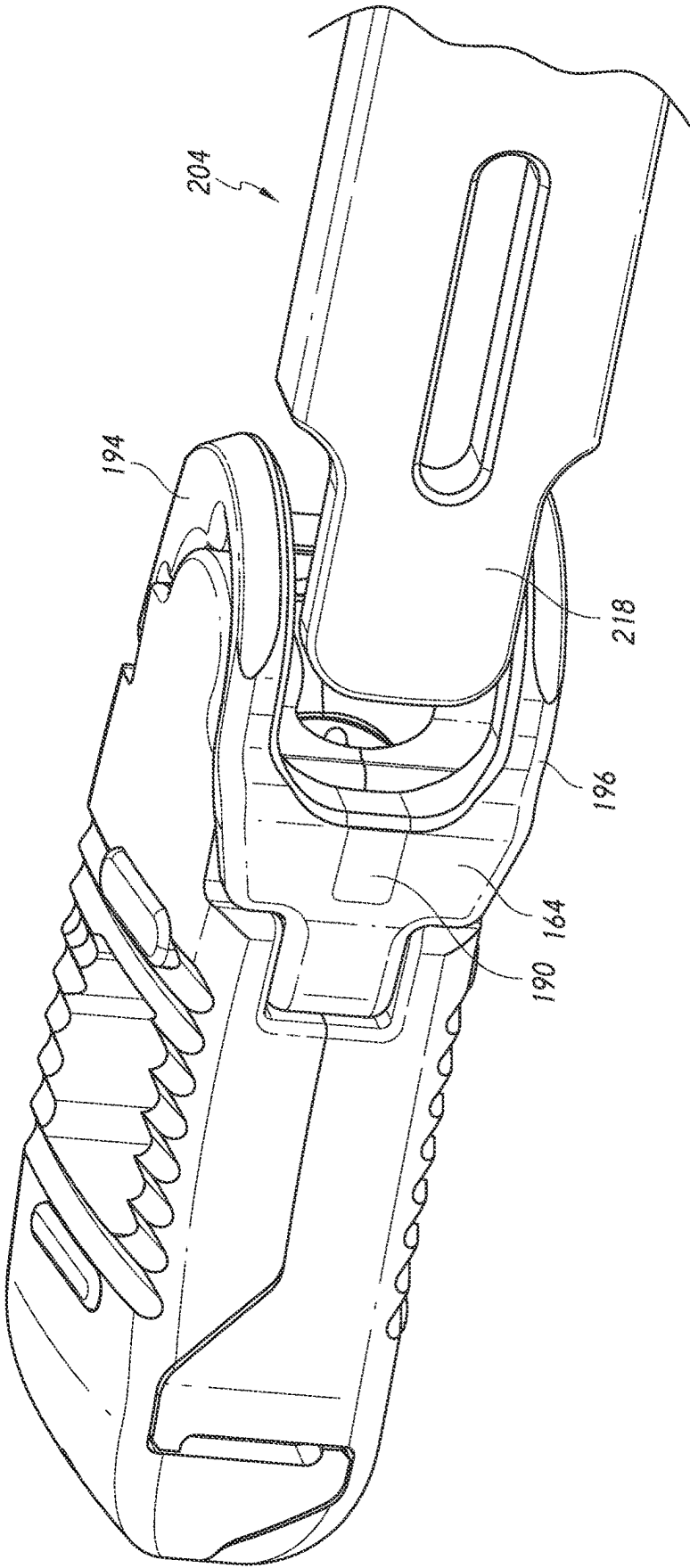


FIG. 19

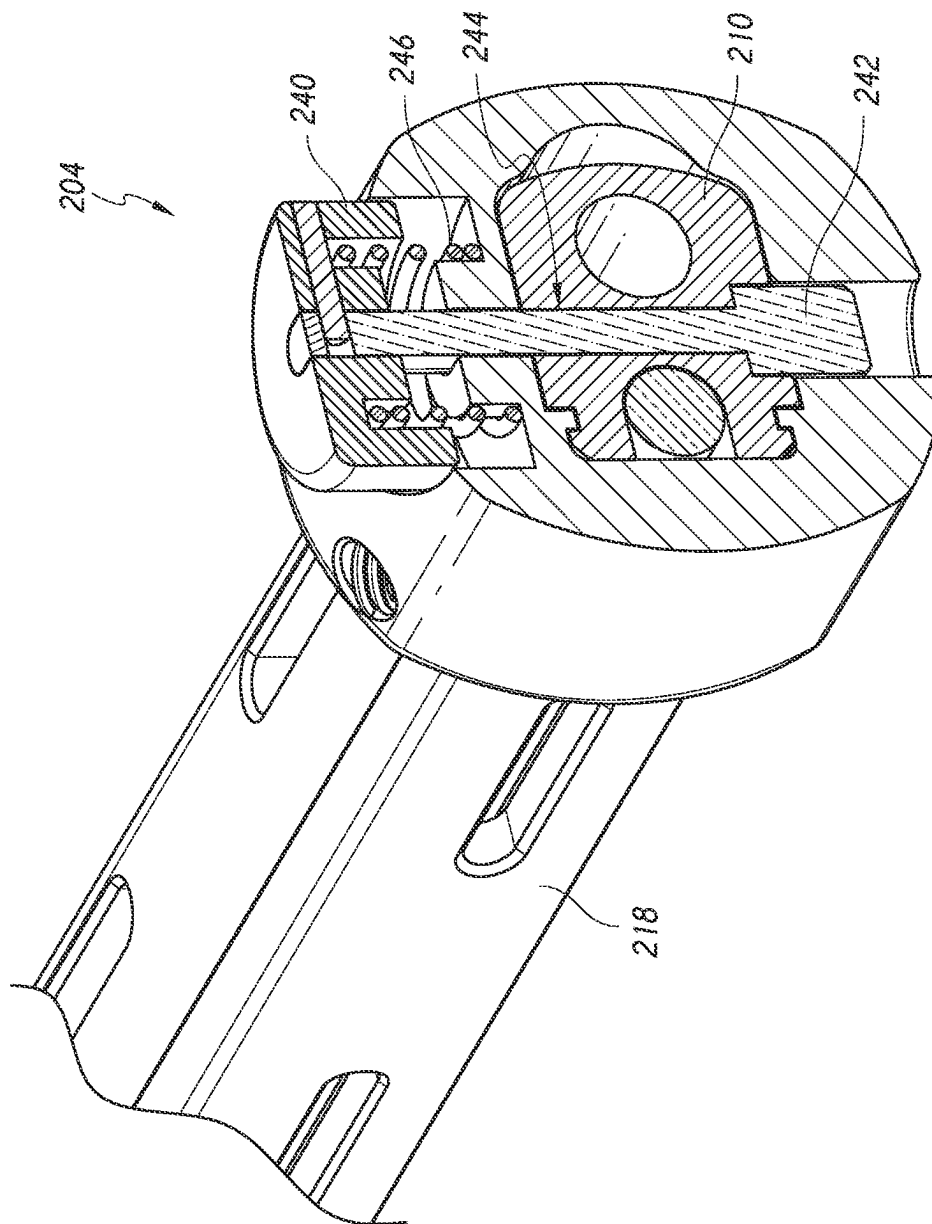


FIG. 20

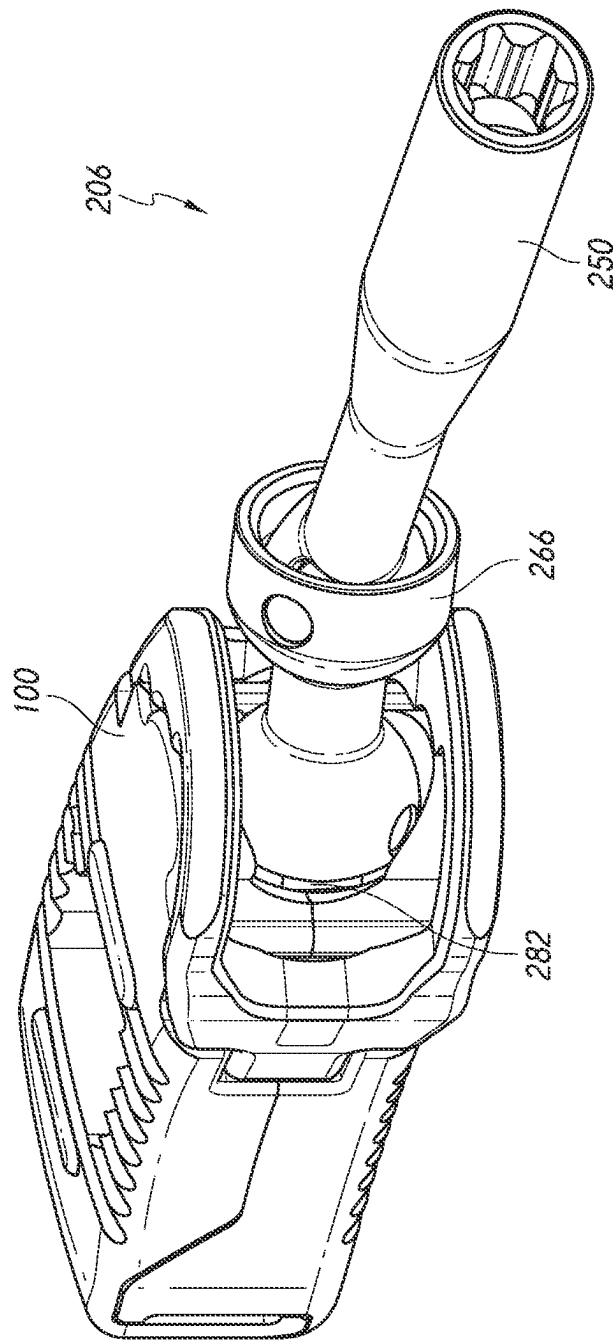


FIG. 21

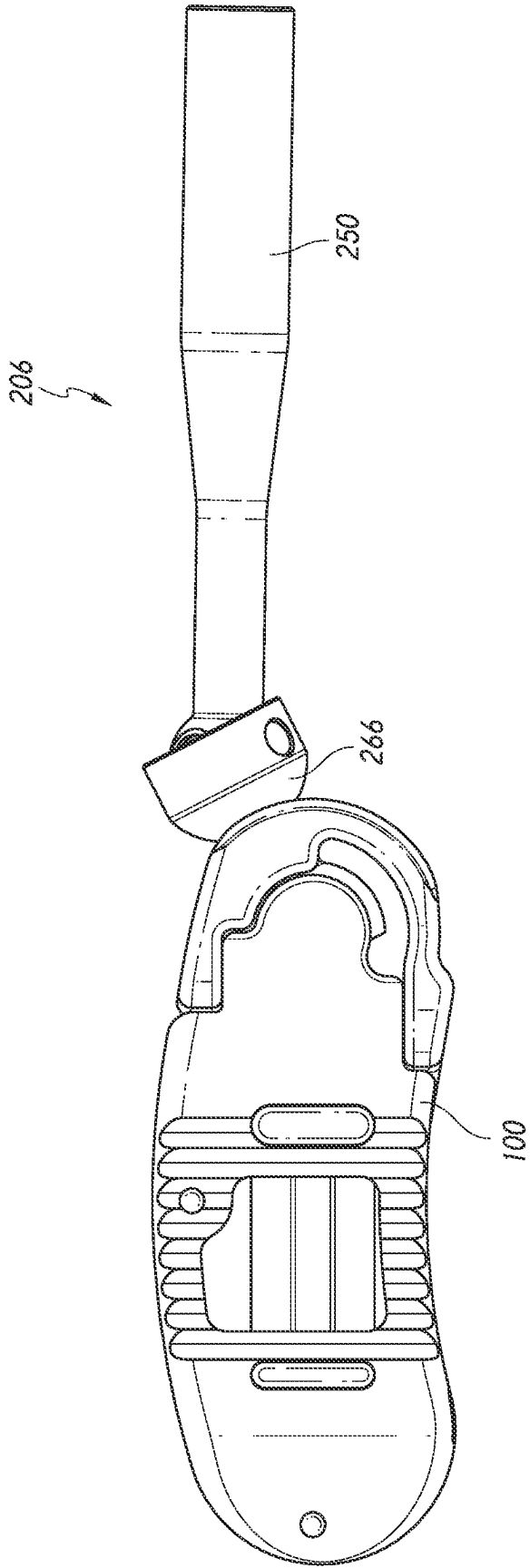


FIG. 22

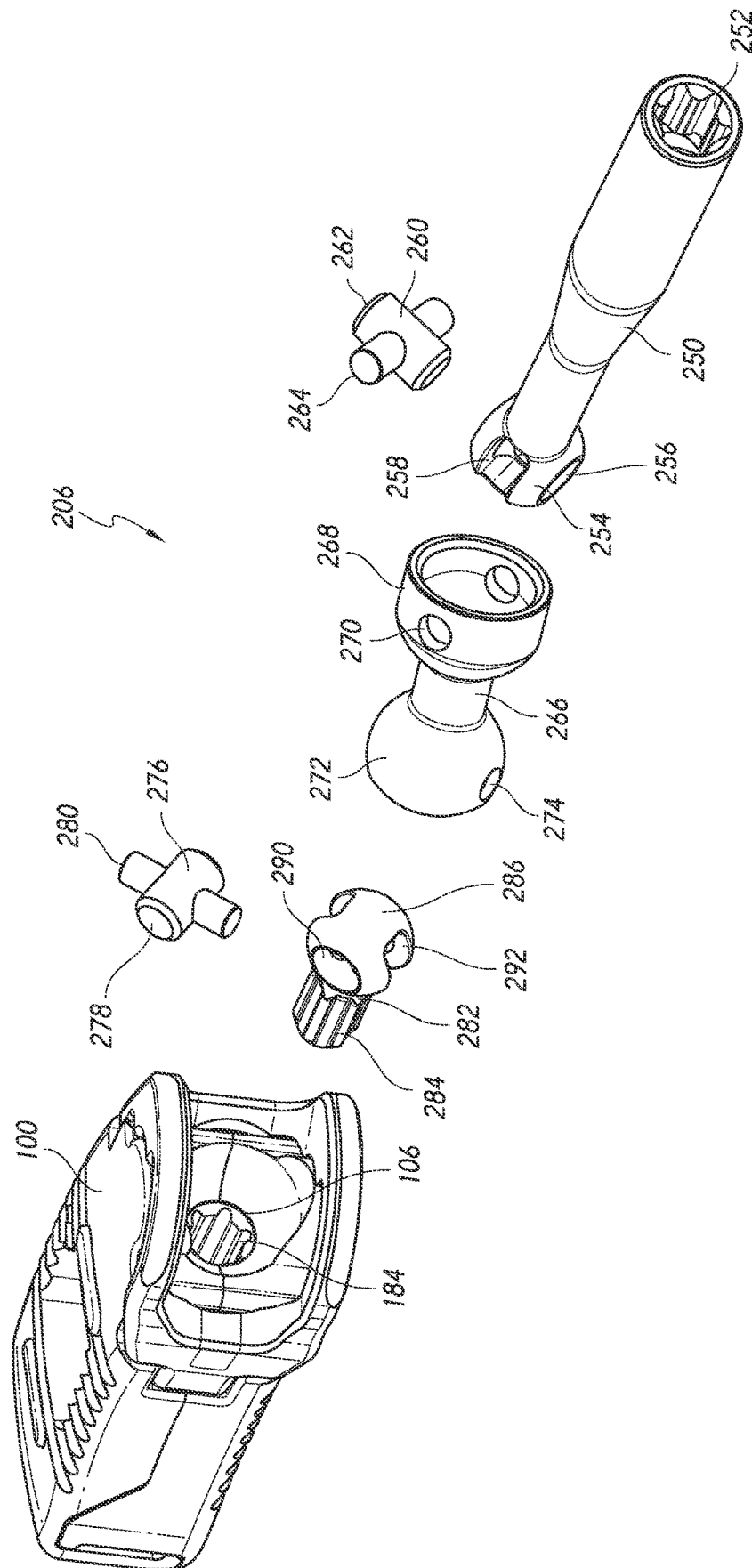


FIG. 23

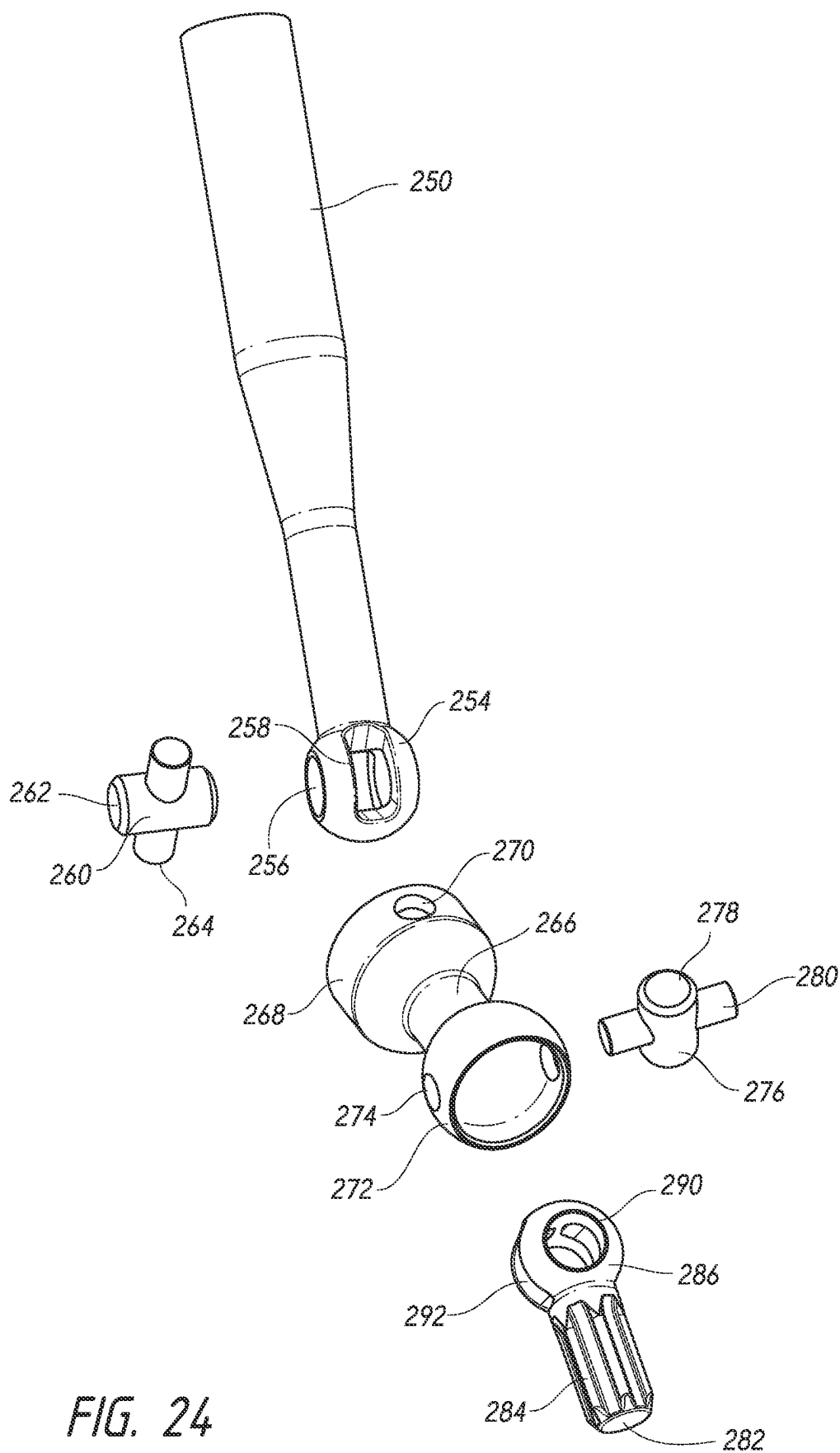


FIG. 24

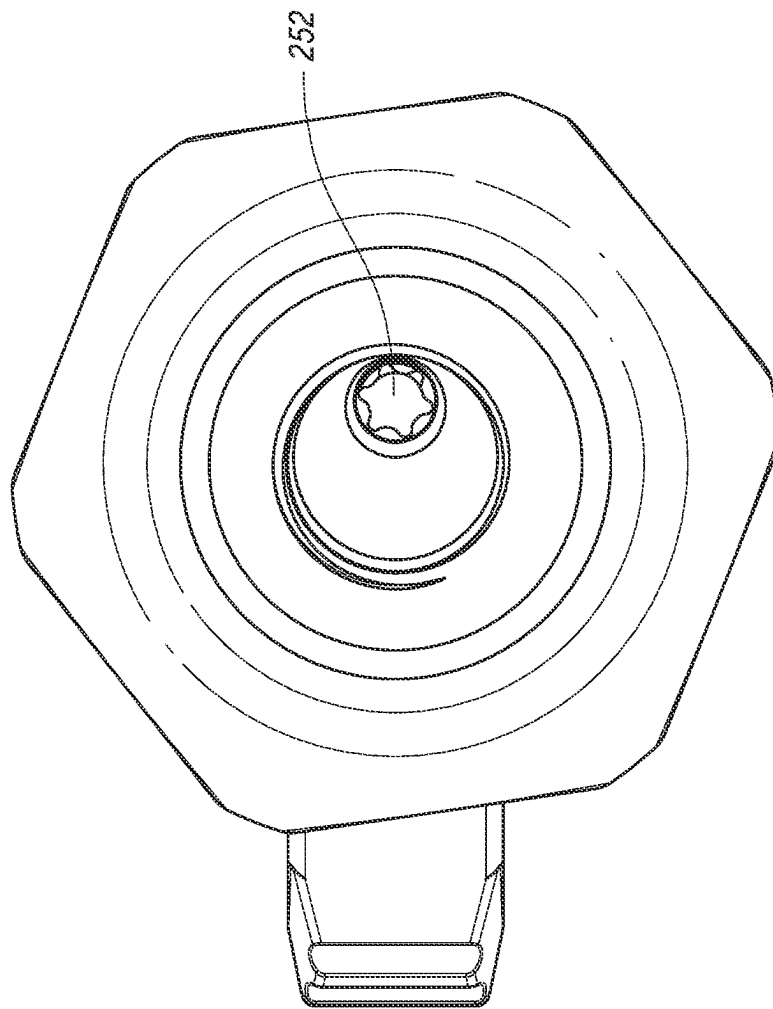


FIG. 25

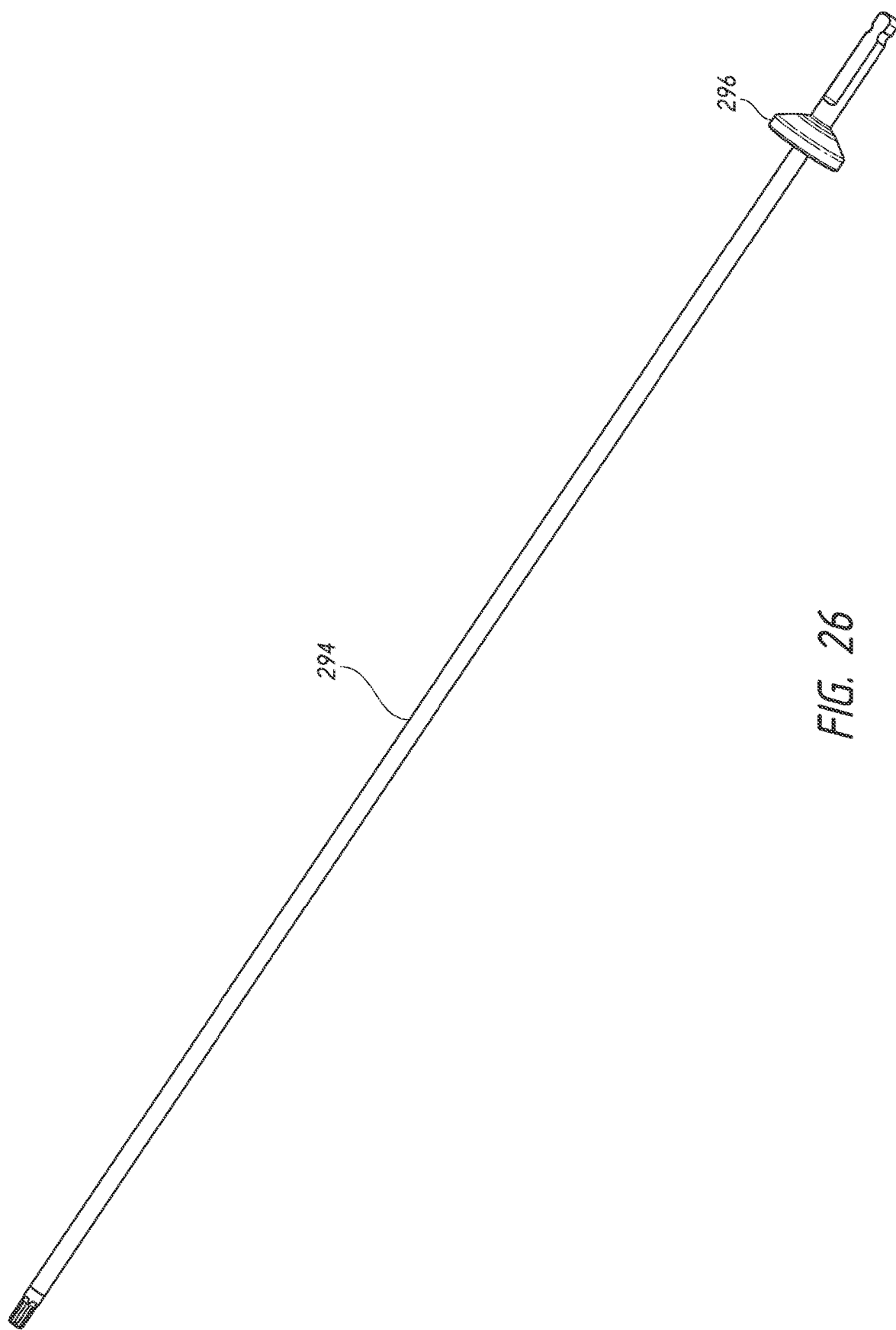


FIG. 26

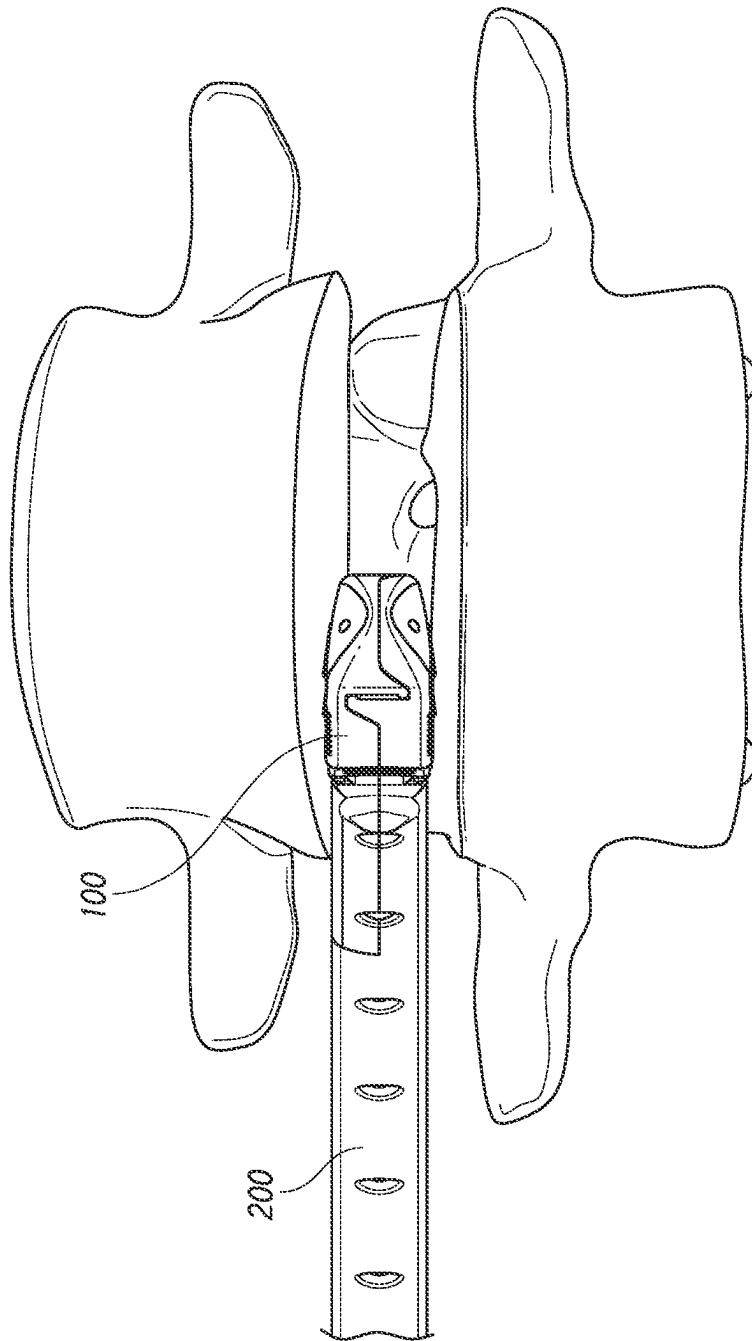


FIG. 27

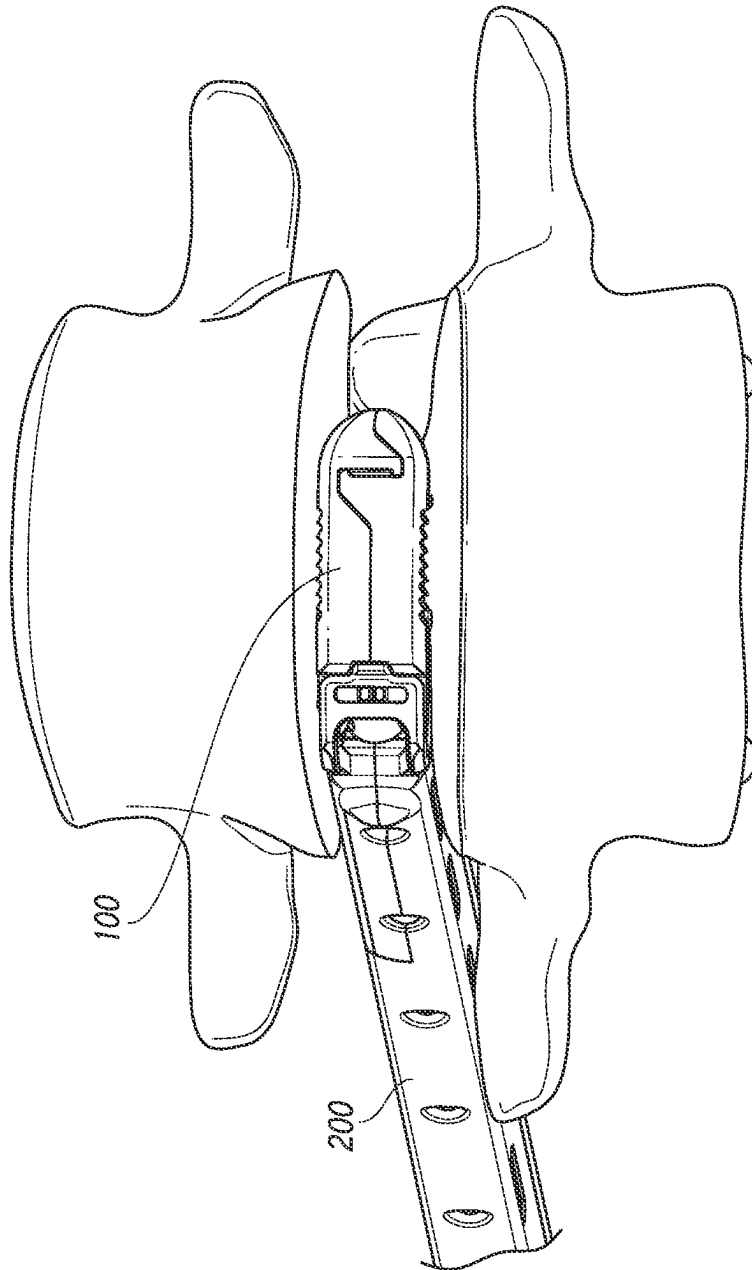


FIG. 28

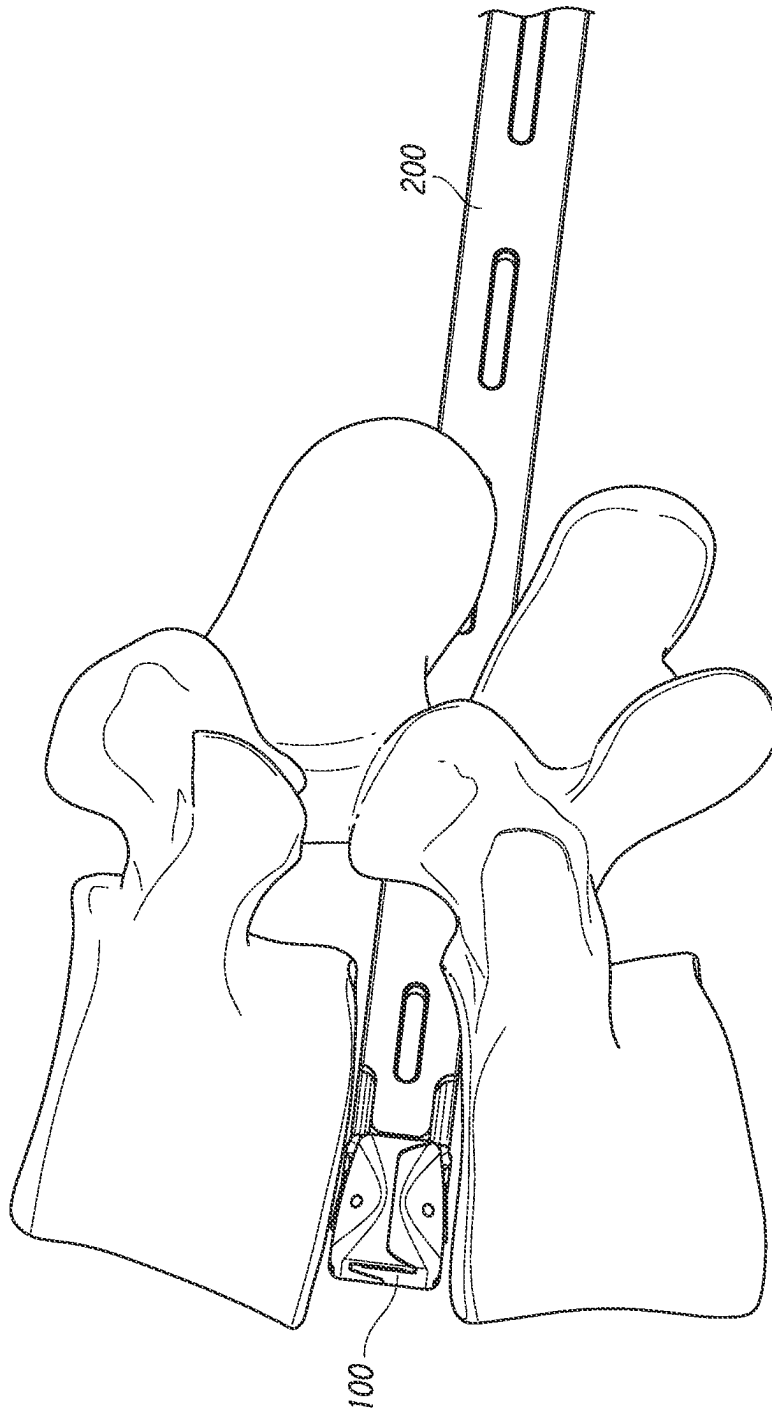
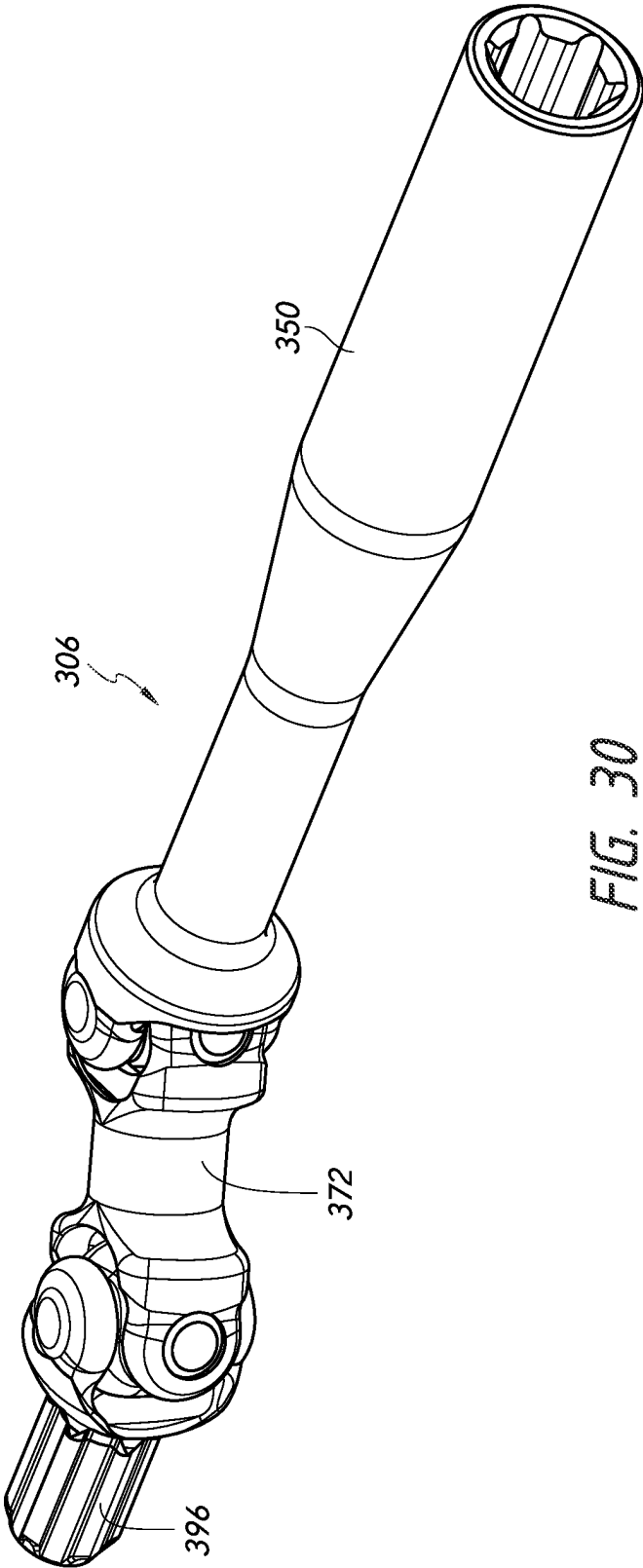


FIG. 29



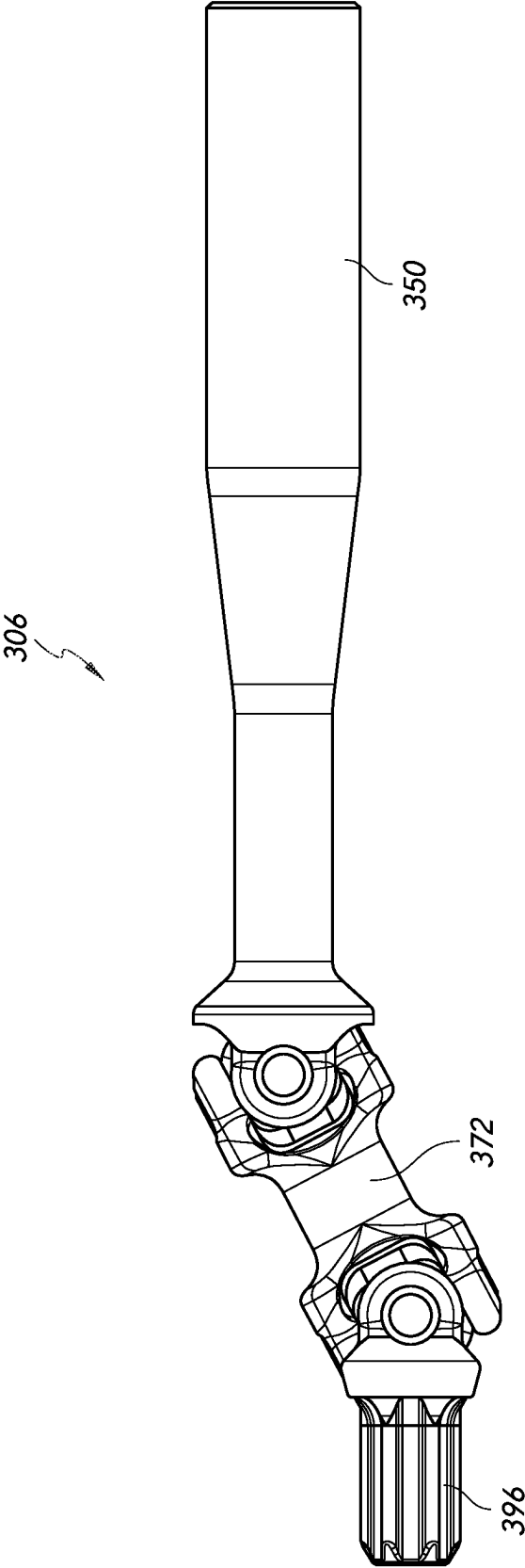


FIG. 31

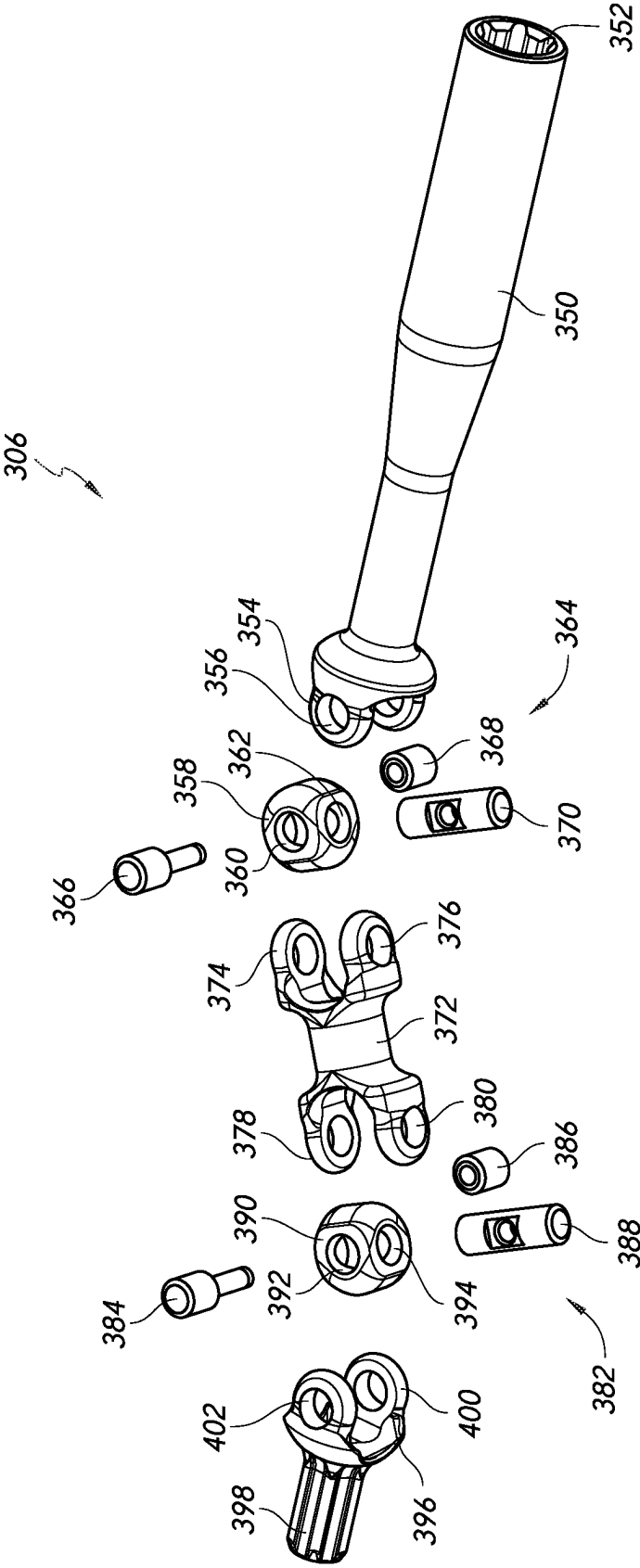


FIG. 32

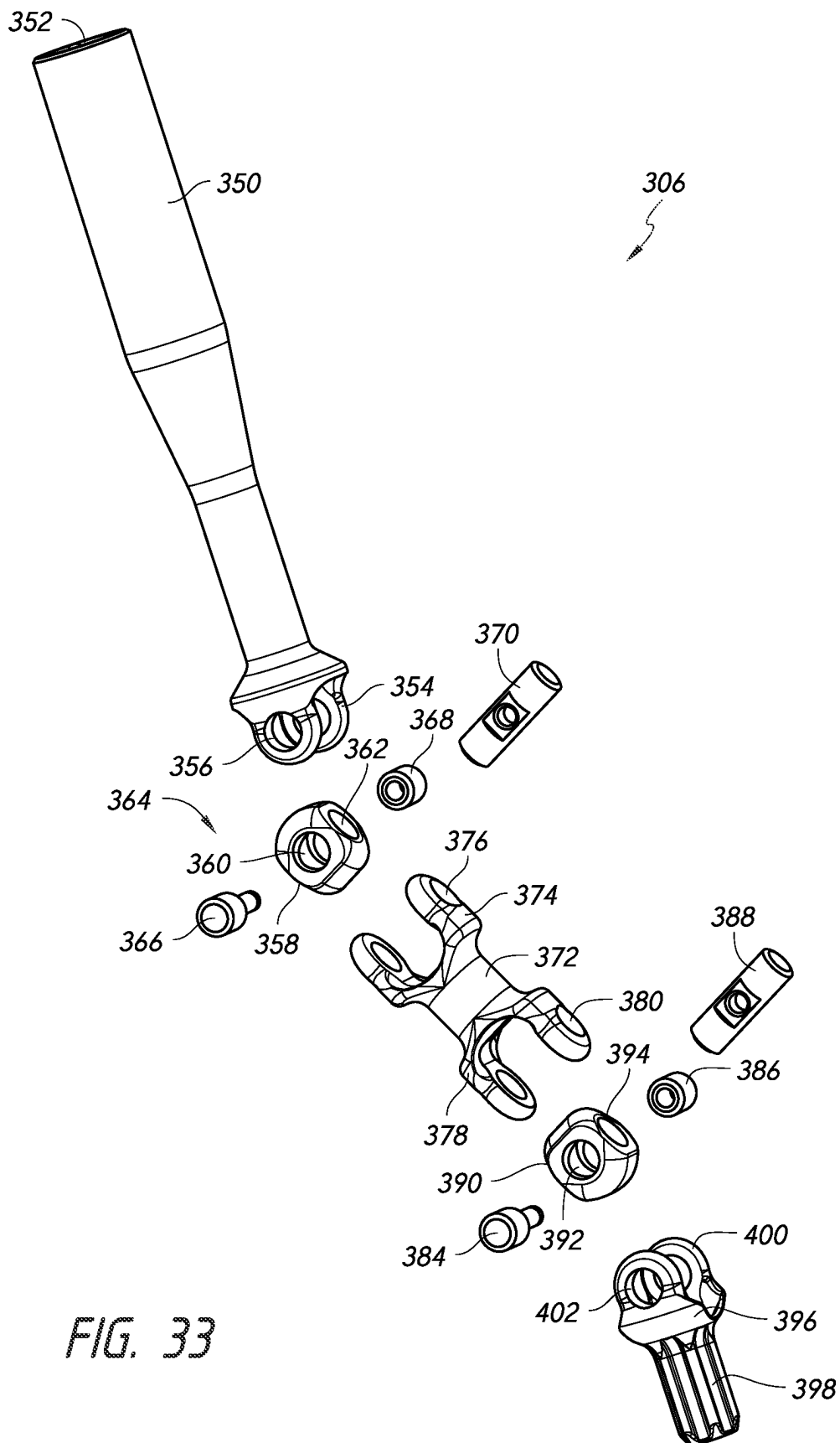
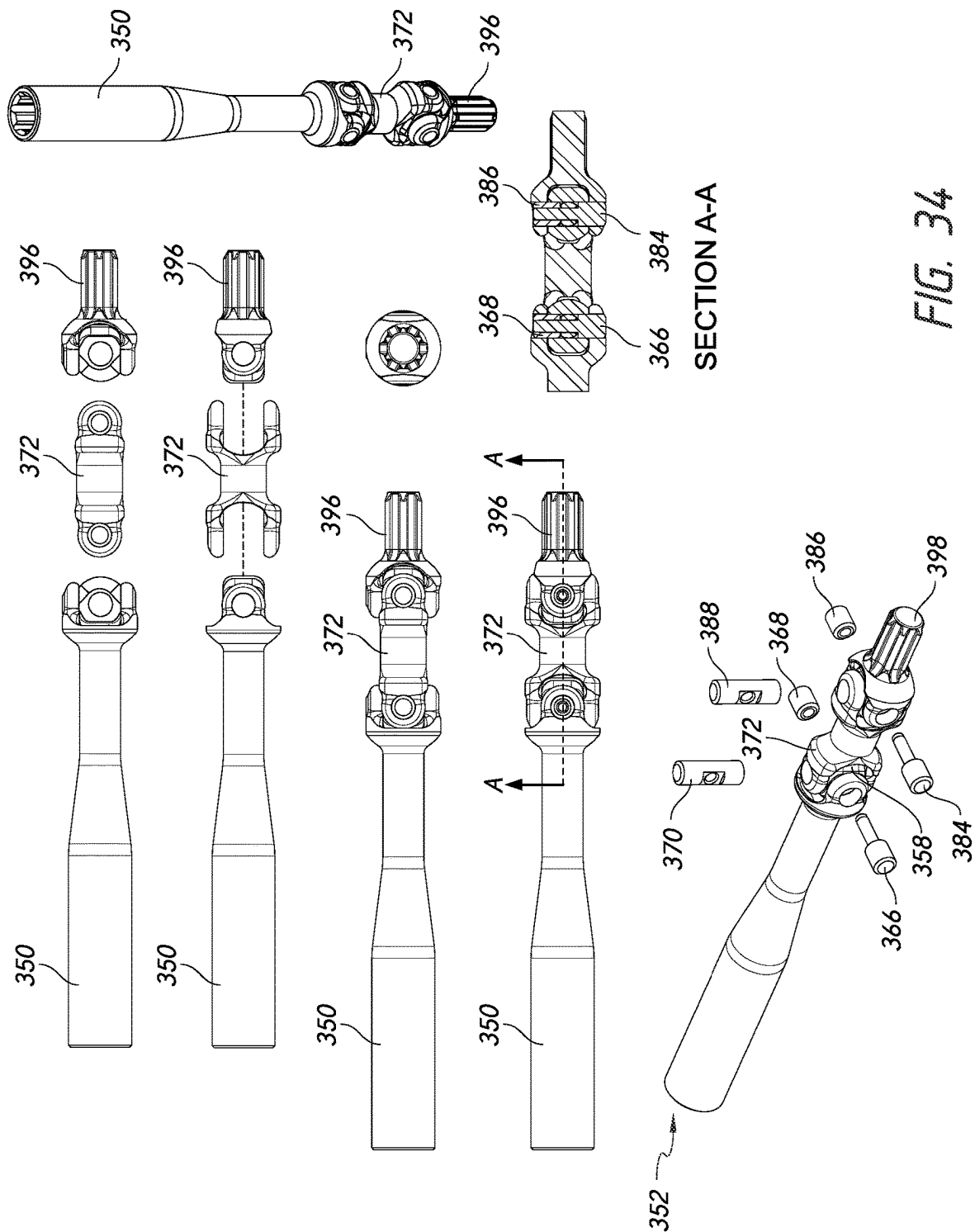


FIG. 33



1

CURVED EXPANDABLE INTERBODY DEVICES AND DEPLOYMENT TOOLS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/523,620, filed Nov. 10, 2021, which claims priority benefit to U.S. Provisional Patent Application No. 63/116,060, filed Nov. 19, 2020, and U.S. Provisional Patent Application No. 63/179,420, filed Apr. 25, 2021, each of which is incorporated herein by reference in its entirety for all purposes.

BACKGROUND

Field

The present disclosure generally relates to the field of spinal orthopedics, and more particularly to curved expandable spinal implants for placement in intervertebral spaces between adjacent vertebrae and deployment tools for the placement of curved expandable spinal implants.

Description of the Related Art

The spine is a flexible structure that extends from the base of the skull to the tailbone. The weight of the upper body is transferred through the spine to the hips and the legs. The spine contains a plurality of bones called vertebrae. The vertebrae are hollow and stacked one upon the other, forming a strong hollow column for support. The hollow core of the spine houses and protects the nerves of the spinal cord. The spine is held upright through the work of the back muscles, which are attached to the vertebrae. While the normal spine has no side-to-side curve, it does have a series of front-to-back curves, giving it a gentle "S" shape.

Each vertebra is separated from the vertebra above or below by a cushion-like, fibrocartilage called an intervertebral disc. The discs act as shock absorbers, cushioning the spine, and preventing individual bones from contacting each other. In addition, intervertebral discs act as a ligament that holds vertebrae together. Intervertebral discs also work with the facet joint to allow for slight movement of the spine. Together, these structures allow the spine to bend, rotate and/or twist.

The spinal structure can become damaged as a result of degeneration, dysfunction, disease and/or trauma. More specifically, the spine may exhibit disc collapse, abnormal curvature, asymmetrical disc space collapse, abnormal alignment of the vertebrae and/or general deformity, which may lead to imbalance and tilt in the vertebrae. This may result in nerve compression, disability and overall instability and pain. If the proper shaping and/or curvature are not present due to scoliosis, neuromuscular disease, cerebral palsy, or other disorder, it may be necessary to straighten or adjust the spine into a proper curvature with surgery to correct these spinal disorders.

Surgical treatments may involve manipulation of the spinal column by attaching a corrective device, such as rods, wires, hooks or screws, to straighten abnormal curvatures, appropriately align vertebrae of the spinal column and/or reduce further rotation of the spinal column. The correct curvature is obtained by manipulating the vertebrae into their proper position and securing that position with a rigid system of screws and rods. The screws may be inserted into the pedicles of the vertebrae to act as bone anchors, and the

2

rods may be inserted into heads of the screws. Two rods may run substantially parallel to the spine and secure the spine in the desired shape and curvature. The rods, which are shaped to mimic the correct spinal curvature, force the spine into proper alignment. Bone grafts are then placed between the vertebrae and aid in fusion of the individual vertebrae together to form a correctly aligned spine.

Other ailments of the spine result in degeneration of the spinal disc in the intervertebral space between adjacent vertebrae. Disc degeneration can cause pain and other complications. Conservative treatment can include non-operative treatment requiring patients to adjust their lifestyles and submit to pain relievers and a level of underlying pain. Operative treatment options include disc removal. This can relieve pain in the short term, but also often increases the risk of long-term problems and can result in motor and sensory deficiencies resulting from the surgery. Disc removal and more generally disc degeneration disease are likely to lead to a need for surgical treatment in subsequent years. The fusion or fixation will minimize or substantially eliminate relative motion between the fixed or fused vertebrae. In surgical treatments, interbody implants may be used to correct disc space collapse between adjacent vertebra, resulting in spinal fusion of the adjacent vertebra.

A fusion is a surgical method wherein two or more vertebrae are joined together (fused) by way of interbody implants, sometimes with bone grafting, to form a single bone. The current standard of care for interbody fusion requires surgical removal of all or a portion of the intervertebral disc. After removal of the intervertebral disc, the interbody implant is implanted in the interspace. In many cases, the fusion is augmented by a process called fixation. Fixation refers to the placement of screws, rods, plates, or cages to stabilize the vertebrae so that fusion can be achieved.

Interbody implants must be inserted into the intervertebral space in the same dimensions as desired to occupy the intervertebral space after the disc is removed. This requires that an opening sufficient to allow the interbody implant must be created through surrounding tissue to permit the interbody implant to be inserted into the intervertebral space. In some cases, the intervertebral space may collapse prior to insertion of the interbody implant. In these cases, additional hardware may be required to increase the intervertebral space prior to insertion of the implant.

In addition, minimally invasive surgical techniques have been used on the spine. Under minimally invasive techniques, access to the intervertebral space is taken to reach the spine through small incisions. Through these incisions, discs are removed and an interbody implant is placed in the intervertebral disc space to restore normal disc height. Minimally invasive spine surgery offers multiple advantages as compared to open surgery. Advantages include: minimal tissue damage, minimal blood loss, smaller incisions and scars, minimal post-operative discomfort, and relative quick recovery time and return to normal function.

SUMMARY

It would be desirable to insert a curved interbody device with a first smaller dimension into an intervertebral space and once in place, deploy to a second, relatively larger dimension to occupy the intervertebral space. This first smaller dimension can permit the use of minimally invasive surgical techniques for easy access to the intervertebral space, which can cause less disruption of soft and bony tissue in order to get to the intervertebral space. The curved

expandable interbody device may be implanted with or without the need of additional hardware.

Disclosed are curved expandable interbody devices that are configured to have an initial collapsed configuration having a first height suitable for being inserted into an intervertebral space between a pair of adjacent vertebrae, and an expanded configuration having a second height that is greater than the first height. The curved expandable interbody devices can be expanded from the initial collapsed configuration to the expanded configuration in-situ. The expanded configuration can provide support to the adjacent vertebrae while bone fusion occurs and can also provide rigid support between the adjacent vertebrae that withstands compressive forces. In some configurations, the curved expandable interbody device can help increase the distance or height between the adjacent vertebrae. By inserting the curved expandable interbody device in the initial collapsed configuration into the intervertebral space, it is possible to perform the surgery percutaneously with minimal disruption to tissues surrounding the surgical site and intervening soft tissue structures. The curved expandable interbody device can be implanted through a minimally invasive or an open wound procedure.

In some embodiments, a curved expandable interbody device for placement between vertebrae is provided. The curved expandable interbody device can include an upper structure configured to abut a superior vertebra. The curved expandable interbody device can include a lower structure configured to abut an inferior vertebra. The curved expandable interbody device can include a screw mechanism between the upper structure and the lower structure. In some embodiments, the screw mechanism comprises a proximal portion, a distal portion, and a coupler. In some embodiments, the coupler comprises a central portion at least partially between the proximal portion and the distal portion. In some embodiments, the coupler further comprises a proximal interface comprising an upper connector and a lower connector. In some embodiments, the proximal portion and the distal portion are configured to rotate as a unit to change a distance between the proximal portion and the distal portion from a first length to a second length. In some embodiments, the upper structure and the upper connector form an upper curved slot therebetween and the lower structure and the lower connector form a lower curved slot therebetween.

In some embodiments, the proximal portion comprises a frustoconical surface configured to wedge between the upper structure and the lower structure, and wherein the distal portion comprises a frustoconical surface configured to wedge between the upper structure and the lower structure to change a distance between the upper structure and the lower structure from a first height to a second height. In some embodiments, the proximal portion comprises first threads wound in a first direction configured to engage a proximal threaded hole in the coupler, and wherein the distal portion comprises second threads wound in a second direction, opposite the first direction, configured to engage a distal threaded hole in the coupler. In some embodiments, the first threads and the second threads have an equal pitch, such that when the screw mechanism is actuated, a proximal end of the curved expandable interbody device changes height at a same rate as a distal end of the interbody device. In some embodiments, the curved expandable interbody device is configured to expand 3 mm from an initial collapsed height. In some embodiments, one or more of the upper structure and the lower structure further comprise a plurality of protrusions. In some embodiments, one or more

of the upper structure and the lower structure comprise vertebrae engagement surfaces with a porous or roughened surface. In some embodiments, the vertebrae engagement surfaces comprise a titanium coating. In some embodiments, the proximal portion comprises a drive interface configured to be engaged by a deployment tool to rotate the proximal portion and the distal portion. In some embodiments, the distal portion comprises a keyed shaft configured to slideably engage with a matching keyed bore on the proximal portion. In some embodiments, the upper structure and the lower structure comprise a right side and a left side when viewed from the top, wherein the left side is concave. In some embodiments, the upper structure and the lower structure comprise a right side and a left side when viewed from the top, wherein the right side is convex. In some embodiments, the coupler further comprises at least one anti-rotational feature configured to engage the upper structure or the lower structure to prevent the coupler from rotating when the proximal portion and the distal portion are rotated.

In some embodiments, a system is provided comprising a curved expandable interbody device and a deployment tool.

In some embodiments, the upper curved slot and the lower curved slot are configured to be engaged by the deployment tool to lock the deployment tool to the curved expandable interbody device. In some embodiments, the upper and lower curved slot are configured to allow pivoting of the deployment tool relative to the curved expandable interbody device. In some embodiments, the upper and lower curved slot are configured to allow pivoting of the deployment tool in the direction of the concave surface of the curved expandable interbody device.

In some embodiments, a method is provided. The method can include coupling a deployment tool to a curved expandable interbody device. The method can include positioning the curved expandable interbody device between the superior vertebra and the inferior vertebra. The method can include pivoting the deployment tool and the curved expandable interbody device relative to each other to further position the curved expandable interbody device between the superior vertebra and the inferior vertebra. The method can include rotating the screw mechanism to expand the curved expandable interbody device.

In some embodiments, coupling the deployment tool to the curved expandable interbody device comprises coupling the deployment tool in a locked, straight position relative to the curved expandable interbody device. In some embodiments pivoting the deployment tool relative to the curved expandable interbody device comprises pivoting the deployment tool to a pivoted position of a plurality of pivoted positions.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view showing an embodiment of a curved expandable interbody device in a collapsed configuration.

FIG. 2 is a top view of the curved expandable interbody device of FIG. 1.

FIG. 3 is a bottom view of the curved expandable interbody device of FIG. 1.

FIG. 4 is a left side view of the curved expandable interbody device of FIG. 1.

FIG. 5 is a right side view of the curved expandable interbody device of FIG. 1.

FIG. 6 is a front view of the curved expandable interbody device of FIG. 1.

5

FIG. 7 is a rear view of the curved expandable interbody device of FIG. 1.

FIG. 8 is a perspective view showing the curved expandable interbody device of FIG. 1 in an expanded configuration.

FIG. 9 is a perspective exploded view showing the curved expandable interbody device of FIG. 1.

FIG. 10 is another perspective exploded view showing the curved expandable interbody device of FIG. 1.

FIG. 11 is a cross-sectional view of the curved expandable interbody device of FIG. 1 in a collapsed configuration.

FIG. 12 is a cross-sectional view of the curved expandable interbody device of FIG. 1 in an expanded configuration.

FIG. 13 is a perspective view of the curved expandable interbody device of FIG. 1 coupled to a deployment tool.

FIG. 14 is a top view of the curved expandable interbody device and the deployment tool of FIG. 13.

FIG. 15 is a close-up side view of the deployment tool of FIG. 13.

FIG. 16 is a close-up side view of a locking mechanism of the deployment tool of FIG. 13 in an unlocked configuration.

FIG. 17 is a close-up top view of the locking mechanism of the deployment tool of FIG. 13 in an unlocked configuration.

FIG. 18 is a close-up side view of a pivoting mechanism of the deployment tool of FIG. 13 in a distal position.

FIG. 19 is a close-up side view of the pivoting mechanism of the deployment tool of FIG. 13 in a proximal position.

FIG. 20 is a close-up cross-sectional view of the pivoting mechanism of the deployment tool of FIG. 13.

FIG. 21 is a close-up side view of a driving mechanism of the deployment tool of FIG. 13.

FIG. 22 is a close-up top view of the driving mechanism of the deployment tool of FIG. 13.

FIG. 23 is an exploded view of the driving mechanism of the deployment tool of FIG. 13.

FIG. 24 is another exploded view of the driving mechanism of the deployment tool of FIG. 13.

FIG. 25 is a rear view of the deployment tool of FIG. 13.

FIG. 26 is a perspective view of a keyed shaft.

FIG. 27 is a view of the curved expandable interbody device of FIG. 1 coupled to a deployment tool in a straight position.

FIGS. 28 and 29 are views of the curved expandable interbody device of FIG. 1 coupled to a deployment tool in a pivoted position.

FIG. 30 is a close-up side view of a driving mechanism of a deployment tool of FIG. 13.

FIG. 31 is a close-up top view of the driving mechanism of the deployment tool of FIG. 13.

FIG. 32 is an exploded view of the driving mechanism of the deployment tool of FIG. 13.

FIG. 33 is another exploded view of the driving mechanism of the deployment tool of FIG. 13.

FIG. 34 is additional views of the driving mechanism of the deployment tool of FIG. 13.

DETAILED DESCRIPTION

Curved expandable interbody devices can be configured to have an initial collapsed configuration having a first height suitable for being inserted into an intervertebral space between a pair of adjacent vertebrae, and an expanded configuration having a second height that is greater than the first height. The implant can be expanded from the initial collapsed configuration to the expanded configuration in-

6

situ. The use of a small curved interbody implant which may be expanded in-situ allows the possibility of performing the surgery percutaneously with minimal disruption to tissues surrounding the surgical site and intervening soft tissue structures, through a minimally invasive or open procedure. The curved expandable interbody device can include features that reduce displacement of soft tissue and structures during placement of the curved expandable interbody device while providing support after placement to the adjacent vertebrae while bone fusion occurs. The curved expandable interbody device includes a collapsed configuration with dimensions that can allow insertion of the curved expandable interbody device between the vertebrae. Once the curved expandable interbody device is positioned in a desired location between the vertebrae, the curved expandable interbody device may be expanded to an expanded configuration. The curved expanded configuration can increase the distance or height between the adjacent vertebrae and provide support to the adjacent vertebrae while bone fusion occurs. The expanded configuration can also provide rigid support between the adjacent vertebrae that withstands compressive forces. The curved expandable interbody device may sometimes be referred to as a device, an implant, an expandable implant, a curved expandable interbody spacer or a curved expandable corpectomy device, all of which are envisioned for the present disclosure.

Several non-limiting embodiments will now be described with reference to the figures, wherein like numerals reflect like elements throughout. The terminology used in the description presented herein is not intended to be interpreted in any limited or restrictive way, simply because it is being utilized in conjunction with a detailed description of certain specific embodiments. Furthermore, some embodiments may include several novel features, no single one of which is solely responsible for its desirable attributes or which is essential to the devices and methods described herein.

The words proximal and distal are applied herein to denote specific ends of components of the instrument described herein. A proximal end refers to the end of a component nearer to an operator of the instrument when the instrument is being used. A distal end refers to the end of a component further from the operator and extending towards the surgical area of a patient and/or the implant. The words top, bottom, left, right, upper and lower are used herein to refer to sides of the device from the described point of view. These reference descriptions are not intended to limit the orientation of the implanted curved expandable interbody device. The curved expandable interbody device can be positioned in any functional orientation. For example, in some configurations, the curved interbody device can be used in an upside-down orientation from the specific orientation described herein.

FIGS. 1-12 illustrate a curved expandable interbody device 100. FIG. 1 is a perspective view. FIG. 2 is a top view. FIG. 3 is a bottom view. FIG. 4 is a left side view. FIG. 5 is a right side view. FIG. 6 is a front view. FIG. 7 is a rear view. FIG. 8 is a perspective view showing the curved expandable interbody device 100 in an expanded configuration. FIGS. 9-10 are perspective exploded views showing the curved expandable interbody device 100. FIG. 11 is a cross-sectional view of the curved expandable interbody device 100 in a collapsed configuration. FIG. 12 is a cross-sectional view of the curved expandable interbody device 100 in an expanded configuration. The expandable interbody device 100 can be configured to couple to a deployment tool 200 described in FIGS. 11-25.

Referring to FIG. 1, the curved expandable interbody device 100 can include an upper structure 102, a lower structure 104, and a screw mechanism 106. The upper structure 102 can include any of the features of the lower structure 104. The lower structure 104 can include any of the features of the upper structure 102. The curved expandable interbody device 100 can include a longitudinal axis 108. The longitudinal axis 108 can extend along the screw mechanism 106. The expandable interbody device 100 can be changeable between a collapsed configuration, as shown in FIG. 1, to an expanded configuration, as shown in FIG. 8.

Referring to FIG. 2, the upper structure 102 can include a top surface 110, a distal surface 112, a proximal surface 114, a left surface 116, and a right surface 118. The right and left orientation are taken when the curved expandable interbody device 100 is viewed from the top and from the proximal end. In some embodiments, the curved expandable interbody device 100 curves to the left. The left surface 116 can be concave. The right surface 118 can be convex. In some embodiments, the curved expandable interbody device 100 curves to the right. The right surface 118 can be concave. The left surface 116 can be convex.

In some embodiments, one or more openings 120 can extend through the upper structure 102. The upper structure 102 can have one central opening 120. The upper structure 102 can have a plurality of openings 120. The curved expandable interbody device 100 can facilitate fusion when the one or more openings 120 are packed with material, as described herein. The one or more openings 120 can be configured to receive fluids, medication, bone graft material, or other material to help in the integration of the interbody device with the vertebrae, such as with allograft and/or Demineralized Bone Matrix ("DBM") packing. In some embodiments, the upper structure 102 may not have any openings and the top surface 110 can be closed.

The upper structure 102 can include one or more alignment slots 122. The upper structure 102 can include two alignment slots 122. The upper structure 102 can include a distal alignment slot 122 positioned between the distal surface 112 and the opening 120. The upper structure 102 can include a proximal alignment slot 122 positioned between the proximal surface 114 and the opening 120. The alignment slots 122 can facilitate alignment between the screw mechanism 106 and the upper structure 102.

The upper structure 102 can have any size configured to be placed between vertebrae. The distal surface 112, the proximal surface 114, the left surface 116, and the right surface 118 may have a varying height. The distal surface 112, the proximal surface 114, the left surface 116, and the right surface 118 may have a varying length. The distal surface 112, the proximal surface 114, the left surface 116, and the right surface 118 may have a varying width. The distal surface 112, the proximal surface 114, the left surface 116, and the right surface 118 may have a varying wall thickness. The distal surface 112, the proximal surface 114, the left surface 116, and the right surface 118 may have a varying curvature radius.

In some embodiments, the upper structure 102 can have one or more markers 126 to help visualization using radiation during the implantation procedure. The one or more markers 126 can be made of a radiopaque material, such as titanium. The one or more markers 126 can be spherical. The one or more markers 126 can be inserted into lumens in the upper structure 102. The upper structure 102 can include a marker 126 near the distal surface 112. The upper structure 102 can include a marker near the right surface 118 when viewed from the top.

The top surface 110 of the upper structure 102 can have a roughened surface, such as a plurality of protrusions 128. The protrusions 128 can be configured to be spaced throughout the top surface 110 or a portion thereof. The protrusions 128 can be located near the opening 120 and the alignment slots 122. The protrusions 128 can be configured to have variable thickness, height, and width as well as angled surfaces. In some embodiments, the top surface 110 can have protrusions 128 that are angled toward the proximal surface 114. The distal facing side of the protrusions 128 can be less steep than the proximal facing side of the protrusions 128.

Referring to FIG. 3, the lower structure 104 can include a bottom surface 130, a distal surface 132, a proximal surface 134, a left surface 136, and a right surface 138. In some embodiments, the curved expandable interbody device 100 curves to the left. The left surface 136 can be concave. The right surface 138 can be convex. In some embodiments, the curved expandable interbody device 100 curves to the right. The right surface 138 can be concave. The left surface 136 can be convex.

In some embodiments, one or more openings 140 can extend through the lower structure 104. The lower structure 104 can have one central opening 140. The lower structure 104 can have a plurality of openings 140. The lower structure 104 can facilitate fusion when the one or more openings 140 are packed with material, as described herein. The one or more openings 120 of the upper structure 102 and the one or more openings 140 of the lower structure 104 can be aligned. The one or more openings 120 of the upper structure 102 and the one or more openings 140 of the lower structure 104 can form a lumen through the curved expandable interbody device 100. In some embodiments, the one or more openings 120 and the one or more openings 140 can extend through to the curved expandable interbody device 100 and can be used as an access location for delivering fluids, medication or other material, as disclosed herein. In some embodiments, the lower structure 104 may not have any openings and the bottom surface 130 can be closed.

The lower structure 104 can include one or more alignment slots 142. The lower structure 104 can include two alignment slots 142. The lower structure 104 can include a distal alignment slot 142 positioned between the distal surface 132 and the opening 140. The lower structure 104 can include a proximal alignment slot 142 positioned between the proximal surface 134 and the opening 140. The alignment slots 122 and the alignment slots 142 can be aligned. The alignment slots 142 can facilitate alignment between the screw mechanism 106 and the lower structure 104.

The lower structure 104 can have any size configured to be placed between vertebrae. The distal surface 132, the proximal surface 134, the left surface 136, and the right surface 138 may have a varying height. The distal surface 132, the proximal surface 134, the left surface 136, and the right surface 138 may have a varying length. The distal surface 132, the proximal surface 134, the left surface 136, and the right surface 138 may have a varying width. The distal surface 132, the proximal surface 134, the left surface 136, and the right surface 138 may have a varying wall thickness. The distal surface 132, the proximal surface 134, the left surface 136, and the right surface 138 may have a varying curvature radius.

In some embodiments, the lower structure 104 can have one or more markers 146 to help visualization using radiation during the implantation procedure. The one or more markers 146 can be made of a radiopaque material, such as

titanium. The lower structure **104** can include a marker **146** near the distal surface **132**. The lower structure **104** can include a marker near the right surface **138**. The upper structure **102** and the lower structure **104** can include markers **126** that are aligned near the distal surfaces **112**, **132**. The upper structure **102** and the lower structure **104** can include markers **126** that are aligned near the right surfaces **118**, **138**.

The bottom surface **130** of the lower structure **104** can have a roughened surface, such as a plurality of protrusions **148**. The protrusions **128** can be configured to be spaced throughout the bottom surface **130** or a portion thereof. The protrusions **148** can be located near the opening **140** and the alignments slots **142**. The protrusions **148** can be configured to have variable thickness, height, and width as well as angled surfaces. In some embodiments, the bottom surface **130** can have protrusions **148** that are angled toward the proximal surface **134**. The distal facing side of the protrusions **148** can be less steep than the proximal facing side of the protrusions **148**.

The shape of the protrusions **128**, **148** can allow for easy insertion of the curved expandable interbody device **100** and help prevent backing out of the device from the intervertebral space. The protrusions **128**, **148** can be configured to provide additional support after the curved expandable interbody device **100** is implanted in the intervertebral space of the patient. For example, the friction between the vertebrae and the structures **102**, **104** provided at least in part by the protrusions **128**, **148** can help reduce movement of the curved expandable interbody device **100** in the intervertebral space.

Referring to FIG. 4, in some embodiments, the left surfaces **116**, **136** align. The left surfaces **116**, **136** can form a smooth concave side. The left surfaces **116**, **136** can have the same radius of curvature. Referring to FIG. 5, in some embodiments, the right surfaces **118**, **138** can align. The right surfaces **118**, **138** can form a smooth convex side. The right surfaces **118**, **138** can have the same radius of curvature. In some embodiments, the upper structure **102** and the lower structure **104** can have the same or similar perimeter. In some embodiments, the top surface **110** and the bottom surface **130** can be skewed relative to each other. The top surface **110** and the bottom surface **130** can be angled. In some embodiments, the top surface **110** and the bottom surface **130** can be parallel. The top surface **110** and the bottom surface **130** can be mirror images. The top surface **110** and the bottom surface **130** can be symmetrical.

Referring to FIG. 6, the distal surfaces **112**, **132** can align. In some embodiments, the distal surfaces **112**, **132** can form a smooth bullet shape. Referring to FIG. 7, the proximal surfaces **114**, **134** can align. In some embodiments, the proximal surfaces **114**, **134** can form a keyed shape as shown in FIG. 3. The keyed shape can be configured to interlock with a coupler of the screw mechanism **106** as described herein.

The upper structure **102** and lower structure **104**, or portions thereof, can be made of any of a variety of materials known in the art, including but not limited to a polymer such as polyetheretherketone (PEEK), polyetherketoneketone (PEKK), polyethylene, fluoropolymer, hydrogel, or elastomer; a ceramic such as zirconia, alumina, or silicon nitride; a metal such as titanium, titanium alloy, cobalt chromium or stainless steel; or any combination of the above materials. The curved expandable interbody device **100** may be made of multiple materials in combination. The upper structure **102** and the lower structure **104** can be made of the same material. The upper structure **102** and the lower structure

104 can be made of different materials. For example, the upper structure **102** can comprise a first material such as a polymer, such as PEEK or polyethylene, and the lower structure **104** can comprise a second, different material such as a metal or ceramic.

In some embodiments, the upper structure **102** and/or the lower structure **104** may be formed of a porous material or have a roughened surface. The surfaces may be formed of a porous material, coated with a porous material, or chemically etched to form a porous or roughened surface with pores, which may help participate in the growth of bone with the adjacent vertebra. In some embodiments, only portions of the curved expandable interbody device **100** may be formed of a porous material, coated with a porous material, or chemically etched to form a porous surface. For example, at least some portions of the top surface **110** and/or the bottom surface **130** can be coated with a porous material, such as a titanium coating. FIG. 8 illustrates the coating. In some embodiments, the surface porosity may be at least approximately 50 microns and less than or equal to approximately 300 microns.

Referring to FIG. 8, the curved expandable interbody device **100** can expand. The upper structure **102** can be configured to slideably fit with the lower structure **104**. The upper structure **102** and the lower structure **104** can have smooth surfaces at the interface between the structures. The upper structure **102** and the lower structure **104** can have smooth surfaces to form a slide bearing. In other embodiments, the upper structure **102** and lower structure **104** can have any of a plurality of different types of functional couplers to form a slideable connection.

Referring to FIGS. 9-10, the screw mechanism **106** is configured to expand the curved expandable interbody device **100**. The upper structure **102** and the lower structure **104** can have one or more grooves **150**, **152** that accepts the screw mechanism **106**. In the illustrated embodiments, the upper structure **102** can include the groove **150** near the distal surface **112** and the groove **152** near the proximal surface **114**. In the illustrated embodiments, the lower structure **104** can include the groove **150** near the distal surface **132** and the groove **152** near the proximal surface **134**. The grooves **150** near the distal surfaces **112**, **132** can form a cavity to accept a distal portion of the screw mechanism **106**. The grooves **152** near the proximal surface **114**, **134** can form a cavity to accept a proximal portion of the screw mechanism **106**.

Referring to FIGS. 11-12, the grooves **150**, **152** can have an angled surface. The grooves **150**, **152** can be conical. The grooves **150** can flare outward toward the distal surfaces **112**, **132**. The grooves **150** can be larger near the distal surfaces **112**, **132** than toward the middle. The grooves **152** can flare outward toward the proximal surfaces **114**, **134**. The grooves **152** can be larger near the proximal surface **114**, **134** than toward the middle. When the upper structure **102** and the lower structure **104** are in the collapsed configuration as shown in FIG. 11, the grooves **150**, **152** can form a frustoconical shape. The upper structure **102** can have approximately half of the cone and the lower structure **104** can have approximately half of the cone. The grooves **150**, **152** can interface with the screw mechanism **106** to transition the curved expandable interbody device **100** from the collapsed to expanded configuration, as disclosed herein.

The upper structure **102** and the lower structure **104** can have one or more cavities **154** to receive the screw mechanism **106**. The cavities **154** can be continuous with one or more grooves **150**, **152**. The cavities **154** can be continuous with one or more openings **120**, **140**. In some embodiments,

11

the cavities **154** can be packed with material. In some embodiments, only the openings **120**, **140** are packed with material. The cavities **154** can be larger than the corresponding components of the screw mechanism **106**. In some embodiments, only the grooves **150**, **152** contact movable components of the screw mechanism **106**. In some embodiments, the cavities **154** do not contact movable components of the screw mechanism **106**.

The cavities **154** can include one or more proximal cavities **156**. The one or more proximal cavities **156** can receive the deployment tool **200** as described herein. The one or more proximal cavities **156** can be open on the proximal surfaces **114**, **134**. The one or more proximal cavities can be concave between the upper structure **102** and the lower structure **104**. The one or more proximal cavities **156** can be semi-spherical. The one or more proximal cavities can be rounded inwards toward the middle. When the upper structure **102** and the lower structure **104** are in the collapsed configuration, the proximal cavities **156** can form a concave, semi-spherical shape. The upper structure **102** can have approximately half of the semi-sphere and the lower structure **104** can have approximately half of the semi-sphere. The one or more proximal cavities **156** can receive the deployment tool **200** to transition the curved expandable interbody device **100** from the collapsed to expanded configuration, as disclosed herein.

Referring back to FIG. **8**, the upper structure **102** and the lower structure **104** can include a catch **158**. The upper structure **102** and the lower structure **104** can be configured to expand vertically. The upper structure **102** and the lower structure **104** can move linearly apart from one another. The catch **158** can prevent additional expansion. In some embodiments, the upper structure **102** and the lower structure **104** can be configured to expand along a single axis. In some embodiments, the upper structure **102** and the lower structure **104** can be configured to expand without translation. In some embodiments, both the upper structure **102** and the lower structure **104** expand.

In some embodiments, the curved expandable interbody device **100** can have a slight inclination, called a lordosis angle. In some embodiments, the lordosis angle decreases the height of the curved expandable interbody device **100**. In some embodiments, the height of the curved expandable interbody device **100** decreases near the left surface **116**, **136** compared to the right surface **118**, **138**. In some embodiments, the lordosis angle is approximately 5 degrees. Other configurations are contemplated, for example 1 degree, 2 degrees, 3 degrees, 4 degrees, 5 degrees, 6 degrees, 7 degrees, 8 degrees, 9 degrees, 10 degrees, 11 degrees, 12 degrees, 13 degrees, 14 degrees, 15 degrees, 16 degrees, 17 degrees, 18 degrees, 19 degrees, 20 degrees, between 4 degrees and 6 degrees, between 0 degrees and 5 degrees, between 3 degrees and 5 degrees, between 8 degrees and 12 degrees, between 14 degrees and 16 degrees, or any range of two of the foregoing values. In some embodiments, the curved expandable interbody device **100** can be slightly enlarged at the convex surface. The right surface **118**, **138** can be the maximum height of the curved expandable interbody device **100**. The curved expandable interbody device **100** can taper slightly by the lordosis angle toward the concave surface. In some embodiments, the top surface **110** of the upper structure **102** can be tapered by the lordosis angle. In some embodiments, the bottom surface **130** of the lower structure **104** can be tapered by the lordosis angle. In some embodiments, both the top surface **110** and the bottom surface **130** taper toward the left. In some embodiments, both the top surface **110** and the bottom surface **130** taper

12

toward the concave surface. In some embodiments, the curved expandable interbody device **100** can have a width measured from the maximum left surface to the maximum right surface. This width can be the total tangent envelop of the device. In some embodiments, the curved expandable interbody device **100** can have a width measured from the convex surface to the concave surface. In some embodiments, the curved expandable interbody device **100** can have a width of 6 mm, 8 mm, 10 mm, 12 mm, 13 mm, 14 mm, 15 mm, 16 mm, or any range of two of the foregoing values. In some embodiments, the curved expandable interbody device **100** can have a length measured from the maximum proximal surfaces to the maximum distal surfaces of the upper and lower structures **102**, **104**. In some embodiments, the length accounts for the upper and lower structures **102**, **104** without the screw mechanism **106**. In some embodiments, the length accounts for the upper and lower structures **102**, **104** without the coupler **164**. In some embodiments, the curved expandable interbody device **100** can have a length measured from the maximum distal surfaces of the upper and lower structures **102**, **104** to the maximum proximal surfaces of the coupler **164**. In some embodiments, the curved expandable interbody device **100** can have a length measured from the maximum distal surfaces of the upper and lower structures **102**, **104** to the maximum proximal surfaces of the upper and lower connectors **194**, **196**. In some embodiments, the length of the curved expandable interbody device **100** includes the length of the coupler **164**. In some embodiments, the curved expandable interbody device **100** can have a length of 20 mm, 21 mm, 22 mm, 23 mm, 24 mm, 25 mm, 26 mm, 27 mm, 28 mm, 29 mm, 30 mm, 31 mm, 32 mm, 33 mm, 34 mm, 35 mm, 36 mm, 37 mm, 38 mm, 39 mm, 40 mm, 41 mm, 42 mm, or any range of the foregoing values. In some embodiments, the curved expandable interbody device **100** can have a height of 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, 10 mm, 11 mm, 12 mm, 13 mm, 14 mm, 15 mm, 16 mm, 17 mm, 18 mm, 19 mm, 20 mm, or any range of the foregoing values. In some embodiments, the curved expandable interbody device **100** can have a width of 12 mm to 13 mm, a length of 27 mm to 37 mm, a height of 8 mm to 12 mm, an expansion of up to 4 mm, and a lordosis of 5 degrees to 15 degrees. Other configurations are contemplated.

Referring back to FIGS. **9-12**, the screw mechanism **106** can include a distal section **160**, a proximal section **162** and a coupler **164**. The coupler **164** can have a distal opening **166** configured to engage the distal section **160**. The coupler **164** can have a proximal opening **168** configured to engage the proximal section **162**. The distal opening **166** and the proximal opening **168** can be threaded. The openings **166**, **168** can have threads in opposite directions. In some embodiments, the distal opening **166** is left handed threads and the proximal opening **168** is right handed threads.

The coupler **164** can include alignment protrusions **170** configured to engage with alignment slots **122**, **142** in the upper structure **102** and lower structure **104**, respectively. The alignment protrusions **170** can prevent the coupler **164** from rotating as the proximal section **162** of the screw mechanism **106** is rotated with the deployment tool **200**. The alignment protrusions **170** can facilitate smooth expansion along a single plane. The alignment protrusions **170** can prevent translation of the upper structure **102** relative to the lower structure **104** during expansion.

In the illustrated embodiment, the coupler **164** includes four alignment protrusions **170**. The coupler **164** can include any number of alignment protrusions **170**. The coupler **164** can include two alignment protrusions **170** configured to

13

engage the upper structure **102** and two alignment protrusions **170** configured to engage the lower structure **104**. The alignment protrusions **170** can be located to maximize the surface area of the openings **120**, **140**. The alignment protrusions **170** can have any shape. In the illustrated embodiment, the alignment protrusions **170** can have an elongated or oval shape. In some embodiments, the protrusions can have any of a variety of shapes, such as cylindrical or rectangular extensions.

The distal section **160** can include a head **172**. The head **172** can be configured to be positioned distally within the upper section **102** and the lower section **104**. The head **172** can be conical. The head **172** can flare outward toward the distal surfaces **112**, **132**. The head **172** can be larger near the distal surfaces **112**, **132** than toward the middle. The head **172** can form a frustoconical shape. The head **172** can have an angled surface configured to slide against the angled surfaces of the grooves **150**. The head **172** can be configured to slide and press against the grooves **150** of the upper structure **102** and lower structure **104**. The head **172** can have sufficient smoothness to functionally slide and press against the grooves **150**.

The distal section **160** can include a threaded portion **174**. The threaded portion **174** can extend proximally from the head **172**. The threaded portion **174** can be configured to couple with the distal opening **166** of the coupler **164**. The threaded portion **174** can be integrally or monolithically formed with the head **172**. The threaded portion **174** can be separately formed from the head **172** and coupled thereto.

The distal section **160** can also have a keyed shaft **176**. The keyed shaft **176** can extend proximally from the threaded portion **174**. The keyed shaft **176** can extend proximally along a longitudinal axis of the screw mechanism **106**. The keyed shaft **176** is configured to slideably couple with the proximal section **162**. As described below, the keyed shaft **176** can be keyed, such that when the proximal section **162** is rotated, the distal section **160** also rotates as a unit. The keyed shaft **176** can be integrally or monolithically formed with the head **172** and/or the threaded portion **174**. The keyed shaft **176** can be separately formed from the head **172** and/or the threaded portion **174** and coupled thereto.

The proximal section **162** can include a head **178**. The head **178** can be configured to be positioned proximally within the upper section **102** and the lower section **104**. The head **178** can be conical. The head **178** can flare outward toward the proximal surfaces **114**, **134**. The head **178** can be larger near the proximal surfaces **114**, **134** than toward the middle. The head **178** can form a frustoconical shape. The head **178** can have an angled surface configured to slide against the angled surfaces of the grooves **152**. The head **178** can be configured to slide and press against the grooves **152** of the upper structure **102** and lower structure **104**. The head **178** can have sufficient smoothness to functionally slide and press against the grooves **152**.

The proximal section **162** can include a threaded portion **180**. The threaded portion **180** can extend distally from the head **178**. The threaded portion **180** can be configured to couple with the proximal opening **168** of the coupler **164**. The threaded portion **180** can be integrally or monolithically formed with the head **178**. The threaded portion **180** can be separately formed from the head **178** and coupled thereto.

The proximal section **162** can also have a keyed bore **182**. The keyed shaft **176** of the distal section **160** can be configured to extend into the keyed bore **182** from the distal end of the proximal section **162**. The keyed shaft **176** of the distal section **160** can be inward from the threaded portion

14

180 of the proximal section **162** when disposed therein. The keyed bore **182** can extend along a longitudinal axis of the screw mechanism **106**. The keyed bore **182** is configured to slideably couple with the keyed shaft **176** of the distal portion **160**. The keyed bore **182** and the keyed shaft **176** transmit torque. As the proximal section **162** is rotated, the distal section **160** also rotates due to the interaction between the keyed bore **182** and the keyed shaft **176**. While the proximal portion **162** includes the keyed bore **182** and the distal portion **160** includes the keyed shaft **176**, in other embodiments, the proximal portion **162** includes the keyed shaft and the distal portion **160** includes the keyed bore.

The head **178** can include a drive interface **184** configured to receive the deployment tool **200** for rotating or driving the proximal section **162**. In the illustrated embodiment, the head **178** has a fluted socket with six flutes. In other embodiments, the head **178** can have any of a variety of drive interfaces, such as slotted, cross and polygonal heads. In some embodiments, the drive interface **184** and the keyed bore **182** are continuous. In some embodiments, the drive interface **184** and the keyed bore **182** have the same keyed shape.

The distal section **160** and the proximal section **162** can be threaded. The distal section **160** and the proximal section **162** can have threads in opposite directions. In some embodiments, the distal section **160** is left handed threads and the proximal section **162** is right handed threads. As disclosed herein, the threads in opposite directions enable the screw mechanism **106** to contract or extend when rotated. The distal section **160** can have threads that are configured to engage threads in the distal opening **166** of the coupler **164**. The proximal section **162** can have threads that are configured to engage threads in the proximal opening **168** of the coupler **164**. In the illustrated embodiment, the distal section **160** and the proximal section **162** have external threads while the coupler **164** has internal threads. In other embodiments, the coupler **164** can have external threads while the distal section **160** and the proximal section **162** can have internal threads.

The distal section **160** and the proximal section **162** can be rotatably linked with the keyed coupling, such that when the proximal section **162** is rotated, the distal section **160** also rotates as a unit. The keyed shaft **176** on the distal section **160** can have a keyed shape that slideably engages with the keyed bore **182** on the proximal section **162**. In the illustrated embodiment the keyed shaft **176** has six flutes that slideably engage the keyed bore **182** having a fluted socket with six flutes. Other suitable shapes or geometric configurations for a keyed connection between the distal section **160** and the proximal section **162** may be used in the screw mechanism **106** to achieve the desired results, such as triangular, hexagonal, oval, star-shaped, or other non-circular shape.

The drive interface **184** can be engaged by the deployment tool **200** to rotate the proximal portion **162**. The drive interface **184** can be rotated to compress the screw mechanism **106**. Rotation of the drive interface **184** can cause the distal section **160** and the proximal section **162** of the screw mechanism **106** to move toward each other. During movement, the heads **172**, **178** of the portions **160**, **162** slide along the grooves **150**, **152** in the upper structure **102** and lower structure **104** toward each other.

The movement of the distal portion **160** and the proximal portion **162** toward each other moves the two structures **102**, **104** away from each other to expand the curved expandable interbody device **100**. The movement of the distal portion **160** and the proximal portion **162** away from each other

15

moves the two structures **102**, **104** toward each other to collapse the curved expandable interbody device **100**. The distal portion **160** and the proximal portion **162** move toward and away from each other along the longitudinal axis of the screw mechanism **106**. The upper structure **102** and the lower structure **104** expand and collapse in a direction perpendicular to the longitudinal axis of the screw mechanism **106**. The alignment protrusions **170** prevent the translation of the upper structure **102** and the lower structure **104** during expansion.

In some embodiments, turning the drive interface **184** clockwise causes the expansion of the curved expandable interbody device **100**. In some methods, the drive interface **184** may be actuated in the opposite direction to change the curved expandable interbody device **100** from the expanded configuration back to the collapsed configuration. In some embodiments, turning the drive interface counter-clockwise causes the collapsing of the curved expandable interbody device **100**. This ability to reversibly expand and collapse the curved expandable interbody device **100** can allow the curved expandable interbody device **100** to be moved to another location, repositioned, or removed from the patient.

The screw mechanism **106** can vary in length to change the curved expandable interbody device **100** from the collapsed configuration to the expanded configuration. Initially, the length of the screw mechanism **106** between the heads **172**, **178** can be LC in the fully collapsed configuration as shown in FIG. 11. As the drive interface **184** of the proximal section **162** is rotated in a first direction, the proximal section **162** is screwed into the coupler **164** and the distal section **160** is screwed into the coupler **164**. The distal section **160** rotates with the proximal section **162** due to the keyed interface. The distal section **160** and the proximal section **162** can have opposite handed threads so as the proximal section **162** is screwed into the coupler **164**, the distal section **160** is also screwed into the coupler **164**. The length of the screw mechanism **106** between the heads **172**, **178** contracts to LE in the fully expanded configuration as shown in FIG. 12. By reversing rotation of the drive interface **184** in a second direction, opposite the first, the screw mechanism **106** can be extended in length from LE back to LC or any length therebetween. The curved expandable interbody device **100** can be expanded by any incremental height between the fully collapsed configuration and the fully expanded configuration. In some embodiments, the expansion of the curved expandable interbody device **100** is the expansion from the fully collapsed configuration to the fully expanded configuration. In some embodiments, the expansion of the curved expandable interbody device **100** is 2.5 mm. In some embodiments, the expansion of the curved expandable interbody device **100** is 3 mm. In some embodiments, the expansion of the curved expandable interbody device **100** is 3.5 mm. In some embodiments, the expansion of the curved expandable interbody device **100** is up to 4 mm. In some embodiments, the curved expandable interbody device **100** can be expanded by any height greater than 0 mm and up to 3 mm. In some embodiments, the curved expandable interbody device **100** can be expanded by any height greater than 0 mm and up to 4 mm. In some embodiments, the curved expandable interbody device **100** can be designed to be expanded by any height, including greater than 3 mm.

The alignment protrusions **170** on the coupler **164** are constrained in the alignment slots **122**, **142** on the upper structure **102** and lower structure **104** to prevent the coupler **164** from rotating with the distal section **160** and the proximal section **162** as the drive interface **184** is rotated.

16

The coupler **164** remains stationary as the distal section **160** and the proximal section **162** rotate relative to the coupler **164**. The coupler **164** remains stationary as the upper structure **102** and the lower structure **104** expand relative to the coupler **164**. The alignment protrusions **170** on the coupler **164** remain within the alignment slots **122**, **142** on the upper structure **102** and lower structure **104** in the fully collapsed configuration and the fully expanded configuration.

The rotation of the proximal portion **162** can cause the rotation of the distal portion **160**. The rotation of the proximal portion **162** can cause the distal portion **160** and the proximal portion **162** to move toward each other. The rotation of the proximal portion **162** can cause the distal portion **160** and the proximal portion **162** to move away from each other. The opposite threads of the distal portion **160** and the proximal portion **162** causes this relative movement. The keyed shaft **176** and keyed bore **182** transmit rotational forces from the drive interface **184** to the distal portion **160**.

In some embodiments, in the collapsed configuration the curved expandable interbody device **100** has a height of HC. The head **172** of the distal section **160** can engage the grooves **150** of the upper structure **102** and lower structure **104**. The head **178** of the proximal section **162** can contact the grooves **152** of the upper structure **102** and lower structure **104**. When the drive interface **184** is rotated in a first direction, the distal section **160** and the proximal section **162** can move toward each other from LC to LE, or any length therebetween. When this happens, the heads **172**, **178** can push against the grooves **150**, **152** in the upper structure **102** and lower structure **104**, causing the upper structure **102** and lower structure **104** to separate. The height between the upper structure **102** and the lower structure **104** can increase from HC (collapsed configuration) to HE (expanded configuration), or any height therebetween. The curved expandable interbody device **100** does not have to be completely expanded to HE and may only be expanded to a partial height between HC and HE, depending on the expansion needed between the adjacent vertebrae. In some embodiments, the grooves **150**, **152** can have features that increase resistance to turning of the screw mechanism **106**, so that increased actuating forces are required during select portions of the expansion procedure. This variation of actuating forces can provide tactile feedback to the user that the curved expandable interbody device **100** is nearing the limits of its expansion.

The threaded portion **174** of the distal section **160** can engage with the distal opening **166** of the coupler **164** and the threaded portion **180** of the proximal section **162** can engage with the proximal opening **168** of the coupler **164**. The distal opening **166** and the proximal opening **168** can have thread patterns in opposite directions. The threaded portions **174**, **180** can have thread patterns in opposite directions. In some embodiments, the openings **166**, **168** and the threaded portions **174**, **180** may have equal pitch, such that during expansion, the proximal portion and the distal portion of the upper structure **102** and the lower structure **104** translate or move at the same rate. In other embodiments, the openings **166**, **168** and the threaded portions **174**, **180** may have different pitches, such that during expansion, the proximal portion and the distal portion of the upper structure **102** and the lower structure **104** translate or move at different rates. For example, the proximal portion of the upper structure **102** and the lower structure **104** may translate or move at a first rate of speed and the distal portion of the upper structure **102** and the lower structure **104** may

17

translate or move at a second rate of speed. The first rate of speed may be faster or slower than the second rate of speed. This allows for some angularity between the upper structure 102 and lower structure 104 during expansion. The difference between the first and second rates of speed allows the user to select a curved expandable interbody device that has some angulation after expansion, for example to account for the lordotic curvature of the spine.

The screw mechanism 106 or portions of the screw mechanism 106 can be fabricated from any biocompatible material suitable for implantation in the human spine, such as metals including, but not limited to, stainless steel, titanium and titanium alloys, as well as surgical grade plastics, plastic composites, ceramics, bone, and other suitable materials. In some embodiments, the distal portion 160 and the proximal portion 162 may be formed of a porous material that participates in the growth of bone with the adjacent vertebral bodies. In some embodiments, the screw mechanism 106 can include a roughened surface that is coated with a porous material, such as a titanium coating, or the material may be chemically etched to form pores that participate in the growth of bone with the adjacent vertebra. In some embodiments, only portions of the screw mechanism 106 may be formed of a porous material, coated with a porous material, or chemically etched to form a porous surface, such as the heads 172, 178, which may be exposed to the native anatomy after implantation. In some embodiments, the surface porosity may be between 50 and 300 microns.

In some embodiments, the screw mechanism may be a compression screw having a proximal section threadably coupled to a distal section, the proximal section having a threaded shaft and the distal section having a threaded bore, or vice-versa, such that when the proximal section is rotated, the threaded shaft engages the threaded bore to shorten or lengthen the distance between the distal head 172 and the proximal head 178. The distal section can have anti-rotational features, such as for example an oblong head shape, to prevent it from rotating as the proximal section is engaged with distal section.

The coupler 164 can include features to facilitate coupling with the deployment tool 200 as described herein. The coupler 164 can include a central portion 186. The central portion 186 can be disposed between the distal portion 160 and the proximal portion 162 in use. The central portion 186 can include the distal opening 166 and the proximal opening 168. The central portion 186 can include the alignment protrusions 170 extending through the upper structure 102 and the lower structure 104. The central portion 186 can be viewed through the openings 120, 140. The central portion 186 can prevent material from encroaching the threads of the screw mechanism during use. The central portion 186 can protect the threads as the openings 120, 140 are packed with material.

The coupler 164 can include a proximal interface 188. The proximal interface 188 can include a left arm 190 and a right arm 192. In some embodiments, the left arm 190 and the right arm 192 extend laterally from the proximal opening 168. In some embodiments, the left arm 190 and the right arm 192 are aligned with the proximal alignment protrusions 170. The left arm 190 can have a shaped surface configured to complement a shaped surface of the left surface 116 of the upper structure 102 and the left surface 136 of the lower structure 104. The right arm 192 can have a shaped surface configured to complement a shaped surface of the right surface 118 of the upper structure 102 and the right surface 138 of the lower structure 104.

18

The coupler 164 can include an upper connector 194 and a lower connector 196. The upper connector 194 can span between the left arm 190 and the right arm 192. The lower connector 194 can span between the left arm 190 and the right arm 192. The upper connector 194 and the lower connector 196 can form the proximal end of the curved expandable interbody device 100. The upper connector 194 can be shaped to receive the proximal surface 114 of the upper structure 102. The lower connector 196 can be shaped to receive the proximal surface 134 of the lower structure 104. The upper connector 194 can form a portion of a ring. The lower connector 196 can form a portion of a ring.

In some embodiments, the upper connector 194 and the lower connector 196 can have the same or similar perimeter. In some embodiments, the upper connector 194 and the lower connector 196 can be parallel. In some embodiments, the upper connector 194 and the lower connector 196 can be mirror images. In some embodiments, the upper connector 194 and the lower connector 196 can be symmetrical. The upper connector 194 and the lower connector 196 can receive the deployment tool 200 therebetween.

Referring back to FIGS. 2-3, the curved expandable interbody device 100 can include one or more curved slots 198. The upper structure 102 and the upper connector 194 can form a curved slot 198 therebetween. The lower structure 104 and the lower connector 196 can form a curved slot 198 therebetween. The curved slots 198 accept a portion of the deployment tool 200 as described herein. The curved slots 198 can facilitate locking the deployment tool 200 to the curved expandable interbody device 100 while allowing the deployment tool 200 to pivot relative to the curved expandable interbody device 100, as described herein.

With reference to FIGS. 13-26, the deployment tool 200 can be used to implant the curved expandable interbody device 100 into the patient. FIG. 13 is a perspective view of the curved expandable interbody device 100 coupled to the deployment tool 200. FIG. 14 is a top view. FIG. 15 is a close-up side view. In use, an incision can be made on the patient to allow access to the implant site in the intervertebral space. The incision can be made for implanting the curved expandable interbody device 100 from any approach, including from the posterior, lateral or anterior directions. The incision can be small for a minimally invasive procedure or a larger incision can be used for an open surgery. Once the implant site is accessed, the two adjacent vertebrae can be distracted in some situations to open up the intervertebral space. In some situations, the curved expandable interbody device 100 can be used to at least partially distract the vertebrae during the implant procedure. In some situations, the intervertebral space may include a degenerated disc or other disorder that may require a partial or complete discectomy prior to insertion of the curved expandable interbody device 100.

In some configurations, more than one curved expandable interbody device 100 can be implanted between the adjacent vertebrae of the patient. In such embodiments, multiple curved expandable interbody device 100 can be placed in a side-by-side configuration or any other suitable configuration, thereby creating additional support. In some methods, only one curved expandable interbody device 100 is utilized in the intervertebral space. In some methods, two or more curved expandable interbody device 100 are utilized in the intervertebral space.

The deployment tool 200 can perform multiple functions. The deployment tool 200 can reversibly lock onto the curved expandable interbody device 100. The deployment tool 200 can include a locking mechanism 202. The deployment tool

200 can pivot relative to the curved expandable interbody device 100. The deployment tool 200 can include pivoting mechanism 204. The pivoting mechanism 204 can hold the deployment tool 200 in a straight position or the pivoting mechanism 204 can allow the deployment tool 200 to pivot. The deployment tool 200 can rotate the drive interface 184 of the screw mechanism 106. The deployment tool 200 can include a driving mechanism 206. The driving mechanism 206 can allow driving of the screw mechanism 106 in any straight or pivoted position of the deployment tool 200 relative to the curved expandable interbody device 100. The deployment tool 200 is configured to attach securely to the curved expandable interbody device 100 to provide control during insertion.

FIG. 16 is a close-up side view of the locking mechanism 202. FIG. 17 is a close-up top view of the locking mechanism 202. FIGS. 18-19 are close-up side views of the pivoting mechanism 204. FIG. 20 is a close-up cross-sectional view of the pivoting mechanism 204. FIG. 21 is a close-up side view of the driving mechanism 206. FIG. 22 is a close-up top view of the driving mechanism 206. FIGS. 23-24 are exploded views of the driving mechanism 206. FIG. 25 is a rear view of the deployment tool 200. FIG. 26 is a perspective view of a keyed shaft of the driving mechanism 206.

Referring back to FIG. 13, the deployment tool 200 can include an elongated shaft 210. The deployment tool 200 can include a distal end 212 that is configured to interact with the curved expandable interbody device 100. The deployment tool 200 can include a proximal end 214. The proximal end 214 of the deployment tool 200 can include a handle 216. The handle 216 can be configured to be gripped by the user. The handle 216 can include features to facilitate the grip of the user. The handle 216 can be made of a material to facilitate gripping. In some embodiments, the handle 216 comprises a silicone rubber.

In methods of use, the deployment tool 200 is brought into proximity of the curved expandable interbody device 100. The distal end 212 can have a complementary shape to the curved expandable interbody device 100. The distal end 212 can mate with the proximal interface 188 of the coupler 164 shown in greater detail in FIG. 9. The distal end 212 facilitates alignment between the deployment tool 200 and the curved expandable interbody device 100 before the deployment tool is locked with the locking mechanism 202.

FIGS. 16 and 17 illustrate the locking mechanism 202 with portions of the deployment tool 200 removed. The locking mechanism 202 can be configured to interact with one or more curved slots 198 as described herein. The one or more curved slots 198 can be between the structures 102, 104 and the coupler 164. In some embodiments, the one or more curved slots 198 can be configured to allow pivoting in a certain direction. In some embodiments, the one or more curved slots 198 can extend toward the left of the curved expandable interbody device 100. In some embodiments, the one or more curved slots 198 can extend toward the concave surface of the curved expandable interbody device 100. In some embodiments, the one or more curved slots 198 can be configured to allow clockwise pivoting of the deployment tool 200 relative to the curved expandable interbody device 100 from the straight position. In some embodiments, the one or more curved slots 198 can extend toward the right of the curved expandable interbody device 100. In some embodiments, the one or more curved slots 198 can extend toward the convex surface of the curved expandable interbody device 100. In some embodiments, the one or more curved slots 198 can be configured to allow counter-clock-

wise pivoting of the deployment tool 200 relative to the curved expandable interbody device 100 from the straight position.

In some embodiments, the locking mechanism 202 can extend along the left side of the deployment tool 200. The locking mechanism 202 can include an inner shaft 222. The locking mechanism 202 can include an upper deflection arm 224 and a lower deflection arm 226. The upper deflection arm 224 and the lower deflection arm 226 can be coupled near a proximal end of the deflection arms 224, 226. The deflection arms 224, 226 can include one or more tines 228. The upper deflection arm 224 can include a tine 228 extending upward. The lower deflection arm 226 can include a tine 228 extending downward. The inner shaft 222 can extend between the upper deflection arm 224 and the lower deflection arm 226. The inner shaft 222 can move proximally and distally relative to the upper deflection arm 224 and the lower deflection arm 226.

The locking mechanism 202 can include a locking actuator 230 as shown in FIG. 15. The locking actuator 230 can control the proximal and distal movement of the inner shaft 222 relative to the upper deflection arm 224 and the lower deflection arm 226. The locking actuator 230 can be a switch. The locking actuator 230 can be near the proximal end of the deployment tool 200.

The locking actuator 230 can include a neutral position as shown in FIG. 15. In some embodiments, the locking actuator 230 is perpendicular to the deployment tool 200 in the neutral position. In some embodiments, the locking actuator 230 extends outward from the deployment tool 200 in the neutral position. In some embodiments, the locking actuator 230 can be biased toward the neutral position. The locking mechanism 202 can include a spring 232. The spring 232 can bias the inner shaft 222 in the proximal direction. The spring 232 can bias the locking actuator 230 outward from the deployment tool 200.

In the neutral position, the deployment tool 200 is not locked to the curved expandable interbody device 100. The deployment tool 200 can be in the neutral position as the user positions the deployment tool 200 relative to the curved expandable interbody device 100. The deployment tool 200 is properly positioned relative to the curved expandable interbody device 100 when the distal end 212 of the deployment tool 200 abuts the upper connector 194 and the lower connector 196 of the coupler 164 of the curved expandable interbody device 100. The distal end 212 of the deployment tool 200 and the proximal end of the coupler 164 can have a corresponding shape to facilitate alignment between the deployment tool 200 and the curved expandable interbody device 100. The distal end 212 can have a shaped surface that complements the curved expandable interbody device 100 to facilitate the proper positioning of the components. The distal end 212 can include a left arm 218 and a right arm 220. The locking mechanism 202 can constrain at least one degree of freedom such that the deployment tool 200 is secured to the curved expandable interbody device 100.

Once the deployment tool 200 is properly positioned relative to the curved expandable interbody device 100, the locking mechanism 202 can be activated. The locking actuator 230 can include a retention position. In some embodiments, the locking actuator 230 is parallel to a longitudinal axis 234 of the deployment tool 200 in the retention position. In some embodiments, the locking actuator 230 sits in-line with the longitudinal axis 234 of the deployment tool 200 in the retention position. In some embodiments, the locking actuator 230 is pushed inward. The locking actuator 230 is coupled to the inner shaft 222 such that inward movement of

21

the locking actuator **230** causes distal movement of inner shaft **222**. The inner shaft **222** moves distally between the upper deflection arm **224** and the lower deflection arm **226**, referring to FIG. 16. The inner shaft **222** can play the upper deflection arm **224** and the lower deflection arm **226**. The upper deflection arm **224** and the lower deflection arm **226** move radially outward from the inner shaft **222**. The tines **228** of the upper deflection arm **224** and the lower deflection arm **226** move into the curved slots **198**. The upper structure **102** and the upper connector **194** can form a curved slot **198** that receives the tine **228** of the upper deflection arm **224**. The lower structure **104** and the lower connector **196** can form a curved slot **198** that receives the tine **228** of the lower deflection arm **226**. The tines **228** are retained within the curved slots **198**. The tines **228** constrain an additional degree of freedom. The tines **228** limit proximal movement of the deployment tool **200** relative to the curved expandable interbody device **100**. The tines **228** function as a stop. The locking mechanism **202** can be considered a retention mechanism. The locking mechanism **202** retains the curved expandable interbody device **100** relative to the deployment tool **200**. As described herein, the deployment tool **200** can pivot relative to the curved expandable interbody device **100** when the curved expandable interbody device **100** is retained.

FIGS. 18 and 19 illustrate the pivoting mechanism **204** with portions of the deployment tool **200** removed. The deployment tool **200** can allow pivoting of the deployment tool **200** relative to the curved expandable interbody device **100** in some configurations and prevent pivoting of the deployment tool **200** relative to the curved expandable interbody device **100** in other configurations. The deployment tool **200** can include the pivoting mechanism **204**. Referring back to FIG. 14, the curved expandable interbody device **100** can include the longitudinal axis **108**. The deployment tool **200** can include the longitudinal axis **234**. The deployment tool **200** can have a straight position as shown in FIG. 14. The longitudinal axis **234** of the deployment tool **200** can be coaxial with the longitudinal axis **108** of the curved expandable interbody device **100** in the straight position. The longitudinal axis **234** of the deployment tool **200** can be aligned with the longitudinal axis **108** of the curved expandable interbody device **100** in the straight position. The longitudinal axis **234** of the deployment tool **200** can be parallel to the longitudinal axis **108** of the curved expandable interbody device **100** in the straight position.

The deployment tool **200** can have one or more pivoted position. The longitudinal axis **234** of the deployment tool **200** can be angled with respect to the longitudinal axis **108** of the curved expandable interbody device **100** in a pivoted position. The longitudinal axis **234** of the deployment tool **200** can be skewed relative to the longitudinal axis **108** of the curved expandable interbody device **100** in a pivoted position. The longitudinal axis **234** of the deployment tool **200** is not aligned with the longitudinal axis **108** of the curved expandable interbody device **100** in a pivoted position. The deployment tool **200** can have a plurality of pivoted positions. The deployment tool **200** can smoothly transition between pivoted positions as the tines **228** slide within the curved slots **198**. The deployment tool **200** can include an infinite number of pivoted positions. The maximum angle of the longitudinal axis **234** of the deployment tool **200** with respect to the longitudinal axis **108** of the curved expandable interbody device **100** can be determined by the curved slots **198**. In some embodiment, the deployment tool **200** can pivot relative to the curved expandable

22

interbody device **100** by 0 degrees, 10 degrees, 20 degrees, 30 degrees, 40 degrees, 50 degrees, 60 degrees, 70 degrees, 80 degrees, 90 degrees, 100 degrees, 110 degrees, 120 degrees, 130 degrees, 140 degrees, 150 degrees, 160 degrees, 170 degrees, 180 degrees, between 70 degrees and 90 degrees, between 50 degrees and 70 degrees, or any range of two of the foregoing values. In the illustrated embodiment, the angle of the deployment tool **200** relative to the curved expandable interbody device **100** can be between 0 degrees and 50 degrees. In some embodiments, the nominal pivot range of the deployment tool **200** relative to the curved expandable interbody device **100** is approximately in the range of 0° to 60°. In some embodiments, the nominal pivot range of the deployment tool **200** and curved expandable interbody device **100** relative to each other is approximately in the range of 4° to 55°. In some embodiments, the deployment tool **200** and the curved expandable interbody device **100** can pivot relative to each other by 0 degrees, 5 degrees, 10 degrees, 15 degrees, 20 degrees, 25 degrees, 30 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, or any range of two of the foregoing values. The range can account for manufacturing tolerances on both the curved expandable interbody device **100** and the deployment tool **200**. In some embodiments, by the nature of the double-joint driving mechanism **206** of the deployment tool **200** as described herein, the designed operating range is from 0 degrees to 60 degrees. In some embodiments, other ranges are contemplated.

In some embodiments, the pivoting mechanism **204** can extend along the left side of the deployment tool **200**. The pivoting mechanism **204** can include the left arm **218**. The left arm **218** can extend along a portion of the length of the deployment tool **200**. The left arm **218** can align with the left arm **190** of the coupler **164**. The left arm **218** can fit within a space between the upper connector **194** and the lower connector **196** of the coupler **164**. The left arm **218** can generally compliment the shape of the upper connector **194** and the lower connector **196**.

Referring to FIGS. 15 and 20, the pivoting mechanism **204** can include a pivoting actuator **240**. The pivoting actuator **240** can control the proximal and distal movement of the left arm **218** relative to the curved expandable interbody device **100**. The pivoting actuator **240** can control the proximal and distal movement of the left arm **218** relative to the upper connector **194** and the lower connector **196**. The pivoting actuator **240** can be a button. The pivoting actuator **240** can be near the proximal end of the deployment tool **200**.

The pivoting actuator **240** can include two discrete position. The pivoting actuator **240** can slide between these two discrete positions. The pivoting actuator **240** can be biased to be in one of two discrete positions. The pivoting actuator **240** can include a straight position. The pivoting actuator **240** can include a pivoting position. The pivoting actuator **240** can be depressed to slide between the two positions. The pivoting actuator **240** can include a shaft **242**. The upper portion of the shaft **242** can include a smaller profile. The lower portion of the shaft **242** can include a larger profile. The elongated shaft **210** can have a shaped slot **244** to receive the shaft **242**. The lower portion of the shaft **242** with the larger profile fits within a shaped slot **244** at the straight position and the pivoting position. The shaft **242** must be depressed to lower the larger profile from the shaped slot **244** to allow the smaller profile of the shaft to be within the shaped slot **244**. Depressing the pivoting actuator **240** lowers the shaft **242** such that the shaft **242** can move between

23

the straight position and the pivoting position. The pivoting actuator **240** can include a spring **246**. The spring **246** can bias the button outward from the deployment tool **200**. The spring **246** can bias the lower portion of the shaft **242** into the shaped slot **244**. The spring **246** can bias the upper portion of the shaft **242** out of the shaped slot **244**.

The pivoting actuator **240** can include the straight position as shown in FIG. **18**. The left arm **218** can be interlocked with the coupler **164** of the curved expandable interbody device **100** when the pivoting actuator **240** is in the straight position. The left arm **218** can be between the upper connector **194** and the lower connector **196**. The pivoting actuator **240** can prevent or limit pivoting when the pivoting actuator **240** is in the straight position. The pivoting actuator **240** can prevent or limit pivoting by positioning the left arm **218** relative to the curved expandable interbody device **100**. The pivoting actuator **240** can provide tactile feedback to the user that the pivoting actuator **240** is in the straight position. In some embodiments, the spring **246** can push the button outward when the pivoting actuator **240** is in the straight position. In some embodiments, the straight position can correspond to a distal position of the pivoting actuator **240** on the deployment tool **200** as shown in FIG. **15**.

The pivoting actuator **240** can include the pivoting position as shown in FIG. **19**. The left arm **218** can be at least partially withdrawn relative to the curved expandable interbody device **100** when the pivoting actuator **240** is in the pivoting position. The left arm **218** can be at least partially removed from between the upper connector **194** and the lower connector **196**. The pivoting actuator **240** can allow pivoting when the pivoting actuator **240** is in the pivoting position. The pivoting actuator **240** can allow pivoting by retracting the left arm **218** proximally relative to the curved expandable interbody device **100**. The pivoting actuator **240** can provide tactile feedback to the user that the pivoting actuator **240** is in the pivoting position. In some embodiments, the spring **246** can push the button outward when the pivoting actuator **240** is in the pivoting position. In some embodiments, the pivoting position of the pivoting actuator **240** can allow the deployment tool **200** to assume one or more pivoted position upon a torque being applied by the user to pivot the deployment tool **200**. In some embodiments, the pivoting position can correspond to a proximal position of the pivoting actuator **240** on the deployment tool **200**.

The locking mechanism **202** and the pivoting mechanism **204** can be utilized in any order. In some embodiments, the locking mechanism **202** is actuated before the pivoting actuator **240** is actuated. The locking mechanism **202** is actuated to lock the deployment tool **200** first. In some embodiments, the locking mechanism **202** is actuated to lock the deployment tool **200** to the curved expandable interbody device **100** when the pivoting actuator **240** is in the pivoting position. In some embodiments, the locking mechanism **202** is actuated to lock the deployment tool **200** to the curved expandable interbody device **100** when the pivoting actuator **240** is actuated to the straight position. In some embodiments, pivoting actuator **240** is actuated to the straight position after the locking mechanism **202** is locked. Other configurations are possible. In some embodiments, the locking mechanism **202** is actuated to lock the deployment tool **200** to the curved expandable interbody device **100** when the pivoting actuator **240** is in the straight position. In some embodiments, the locking mechanism **202** is actuated after the pivoting actuator **240** is actuated.

24

In some methods, the deployment tool **200** can be locked in the straight position relative to the curved expandable interbody device **100** by pressing the pivoting actuator **240** and moving the pivoting actuator **240** distally into the straight position. In some methods, the deployment tool **200** can allow pivoting motion relative to the curved expandable interbody device **100** by pressing the pivoting actuator **240** and moving the pivoting actuator **240** proximally into the pivoting position. The deployment tool **200** can retain the curved expandable interbody device **100** via the locking mechanism **202** regardless of whether the deployment tool **200** is in a straight position or a pivoted position.

FIGS. **21-24** illustrate the driving mechanism **206** with portions of the deployment tool **200** removed. In some embodiments, the driving mechanism **206** can extend along the left side of the deployment tool **200**. The driving mechanism **206** can include a first shaft **250**. The first shaft **250** can be generally straight within the deployment tool **200**. The first shaft **250** can include a keyed bore **252**. The keyed bore **252** can extend from the proximal end of the first shaft **250**. The keyed bore **252** can extend a portion of the length of the first shaft **250**. The keyed bore **252** can extend along a longitudinal axis of the first shaft **250**.

The first shaft **250** can include a ball end **254**. The ball end **254** can include a first bore **256**. The ball end **254** can include a first slot **258**. The first bore **256** and the first slot **258** can be perpendicular.

The driving mechanism **206** can include a first connector **260**. The first connector can include a first bore pin **262** and a first slot pin **264**. The first bore pin **262** can be configured to be received in the first bore **256**. The first bore pin **262** can be configured to couple the first connector **260** and the first shaft **250**. The first slot pin **264** can be configured to be received in the first slot **258**. In some methods of use, the first slot pin **264** can slide within the first slot **258**.

The driving mechanism **206** can include a second shaft **266**. The second shaft **266** can be generally skewed within the deployment tool **200**. The second shaft **266** can include a proximal socket end **268**. The proximal socket end **268** can extend from the proximal end of the second shaft **266**. The proximal socket end **268** can receive the ball end **254** of the first shaft **250**. The proximal socket end **268** can allow pivoting motion of the ball end **254** of the first shaft **250**. The proximal socket end **268** can extend a portion of the length of the second shaft **266**. The proximal socket end **268** can extend along a longitudinal axis of the second shaft **266**. The proximal socket end **268** can include a second proximal bore **270**. The first slot pin **264** can be configured to be received in the second proximal bore **270**. The first slot pin **264** can couple the first shaft **250** and the second shaft **266**. The first slot pin **264** can allow the first shaft **250** and the second shaft **266** to pivot relative to each other over a very limited range. In some embodiments, the first shaft **250** and the second shaft **266** do not pivot. In some embodiments, the first shaft **250** and the second shaft **266** are held at a fixed pivot angle. This fixed pivot angle can be any angle. In some embodiments, the fixed pivot angle is between 27 degrees and 30 degrees. In some embodiments, the first slot **258** is designed to accommodate the fixed pivot angle. The first shaft **250** and the second shaft **266** can be held by an internal pocket geometry in the deployment tool **200**. The first shaft **250** and the second shaft **266** can be held by an internal pocket geometry in the right arm **220** near the distal end **212** of the deployment tool **200**. This joint design can allow rotation and/or torque to be transmitted from the axis of the first shaft **250** to the axis of the second shaft **266**. The first shaft **250**

25

can function as the input drive shaft. The second shaft 266 can function as a double joint coupling, as described herein.

The second shaft 266 can include a distal socket end 272. The distal socket end 272 can extend from the distal end of the second shaft 266. The distal socket end 272 can extend a portion of the length of the second shaft 266. The distal socket end 272 can extend along a longitudinal axis of the second shaft 266. The distal socket end 272 can include a second distal bore 274. The second proximal bore 270 and the second distal bore 274 can be perpendicular.

The driving mechanism 206 can include a second connector 276. The second connector 276 can include a second bore pin 278 and second slot pin 280. The second slot pin 280 can be configured to be received in the second distal bore 274.

The driving mechanism 206 can include a third shaft 282. The third shaft 282 can be generally straight within the deployment tool 200. The third shaft 282 can include a keyed shaft 284. The keyed shaft 284 can extend from the distal end of the third shaft 282. The keyed shaft 284 can extend a portion of the length of the third shaft 282. The keyed shaft 284 can extend along a longitudinal axis of the third shaft 282. The keyed shaft 284 can be configured to engage the drive interface 184. The screw mechanism 106 can include the drive interface 184 configured to receive the keyed shaft 284 of the driving mechanism 206 for rotating the screw mechanism 106.

The third shaft 282 can include a ball end 286. The ball end 286 can include a third bore 290. The ball end 286 can include a third slot 292. The third bore 290 and the third slot 292 can be perpendicular. The distal socket end 272 can receive the ball end 286 of the third shaft 282. The distal socket end 272 can allow pivoting motion of the ball end 286 of the third shaft 282.

The second bore pin 278 can be configured to be received in the third bore 290. The second bore pin 278 can be configured to couple the second connector 276 and the third shaft 282. The second slot pin 280 can be configured to be received in the third slot 292. In some methods of use, the second slot pin 280 can slide within the third slot 292.

The second slot pin 280 can be configured to be received in the second distal bore 274. The second slot pin 280 can couple the second shaft 266 and the third shaft 282. The second slot pin 280 can allow the second shaft 266 and the third shaft 282 to pivot. In some embodiments, the first shaft 250 and the second shaft 266 do not pivot. In some embodiments, only the second shaft 266 and the third shaft 282 pivot. In some embodiments, the second shaft 266 and the third shaft 282 allow for pivoting for the total range of the deployment tool 200 relative to the curved expandable interbody device 100.

The first shaft 250, the first connector 260, the second shaft 266, the second connector 276, and the third shaft 282 couple together to form an assembly. In methods of use, the rotation of the first shaft 250 cause rotation of the second shaft 266 and the third shaft 282. The torque is transmitted by the first slot pin 264 which couples the first shaft 250 and the second shaft 266. The torque is transmitted by the second slot pin 280 which couples the second shaft 266 and the third shaft 282. The third shaft 282 can be coupled to the screw mechanism 106. The rotation of the first shaft 250 cause rotation of the screw mechanism 106 to expand the curved expandable interbody device 100.

The first shaft 250, the first connector 260, the second shaft 266, the second connector 276, and the third shaft 282 allow the deployment tool 200 to be in any pivoted position while allowing the driving mechanism 206 to rotate the

26

screw mechanism 106. As the deployment tool 200 pivots, the first slot pin 264 can slide within the first slot 258 of the first shaft 250. As the deployment tool 200 pivots, the second slot pin 280 can slide within the third slot 292 of the third shaft 282. The slots 258, 292 accommodate the range of pivoted positions of the deployment tool 200. The slots 258, 292 accommodate pivoting up to 30 degrees, 40 degrees, 50 degrees, 60 degrees, 70 degrees, 80 degrees, 90 degrees, or any range of two of the foregoing values. In some methods of use, the first shaft 250 and the third shaft 282 can remain stationary as the deployment tool 200 pivots. In some methods of use, the third shaft 282 remains coaxial with the longitudinal axis 108 of the curved expandable interbody device 100 regardless of the position of the deployment tool 200. The first shaft 250 and the second shaft 266 can pivot relative to the third shaft 282 as the deployment tool 200 pivots.

As described herein, the deployment tool 200 can have a straight position. In the straight position of the deployment tool 200, the first shaft 250 and the second shaft 266 can be at the fixed pivot angle relative to each other. In the straight position, the first slot pin 264 transmits torque between the first shaft 250 and the second shaft 266. In the straight position, the first slot pin 264 slides within the first slot 258 as the first shaft 250 rotates. The first slot 258 enables a sweep of the first slot pin 264 that is necessitated by the fixed pivot angle between the first shaft 250 and the second shaft 266. The first slot pin 264 slides along the length of the first slot 258 during a full rotation of the first shaft 250. In some embodiments, the fixed pivot angle between the first shaft 250 and the second shaft 266 is between 27 degrees and 30 degrees. The first slot pin 264 can transmit torque when the first shaft 250 and the second shaft 266 are at this fixed pivot angle. The first slot pin 264 can slide twice the fixed pivot angle. In some embodiments, the first slot pin 264 needs to slide about twice the relative angle between the first shaft 250 and second shaft 266. Since the first shaft 250 and second shaft 266 are at the fixed pivot angle of 27 degrees to 30 degrees, the first slot pin 264 will slide or sweep from about negative 27 degrees to 30 degrees to positive 27 degrees to 30 degrees during a single rotation of the joint between the first shaft 250 and the second shaft 266. The sliding of the first slot pin 264 can allow the deployment tool 200 to transmit torque in the straight position.

In the straight position of the deployment tool 200, the second shaft 266 and the third shaft 282 can be skewed. In the straight position, the second slot pin 280 transmits torque between the second shaft 266 and the third shaft 282. In the straight position, the second slot pin 280 slides within the third slot 292 as the second shaft 266 rotates. In some embodiments, the second slot pin 280 can slide twice the fixed pivot angle in the straight position. In some embodiments, the second shaft 266 and the third shaft 282 are at an angle of 27 degrees to 30 degrees when the deployment tool 200 is in the straight position. In some embodiments, the second slot pin 280 needs to slide about twice the relative angle between the second shaft 266 and the third shaft 282. The second slot pin 280 will slide or sweep from about negative 27 degrees to 30 degrees to positive 27 degrees to 30 degrees during a single rotation of the joint between the second shaft 266 and the third shaft 282. The sliding of the second slot pin 280 can allow the deployment tool 200 to transmit torque in the straight position.

As described herein, the deployment tool 200 can have one or more pivoted positions. In the pivoted positions of the deployment tool 200, the first shaft 250 and the second shaft 266 remain at the fixed pivot angle relative to each other. In

27

the pivoted positions, the first slot pin **264** transmits torque between the first shaft **250** and the second shaft **266**. The first slot pin **264** slides within the first slot **258** as the first shaft **250** rotates. The motion of the first slot pin **264** accommodates the fixed pivot angle between the first shaft **250** and the second shaft **266**. The first slot pin **264** slides along the length of the first slot **258** during a full rotation of the first shaft **250**. The first slot pin **264** can slide up to twice the fixed pivot angle between the first shaft **250** and the second shaft **266**. In some embodiments, the first slot pin **264** continues to slide twice the relative angle between the first shaft **250** and second shaft **266** in the pivoted positions of the deployment tool **200**. Since the first shaft **250** and second shaft **266** are at the fixed pivot angle of 27 degrees to 30 degrees, the first slot pin **264** can slide or sweep from about negative 27 degrees to 30 degrees to positive 27 degrees to 30 degrees during a single rotation of the joint between the first shaft **250** and the second shaft **266** regardless of the angle of the deployment tool **200**.

In a pivoted position of the deployment tool **200**, the second shaft **266** and the third shaft **282** can be pivoted relative to each other. The second shaft **266** and the third shaft **282** can pivot relative to each other approximately negative 30 degrees to positive 30 degrees. This pivoting of the second shaft **266** and the third shaft **282** combined with the fixed pivot angle of 27 to 30 degrees between the first shaft **250** and the second shaft **266** allows the deployment tool **200** to achieve cumulative total deployment tool pivot angles of 0 degrees to 60 degrees. In the pivoted positions, the second slot pin **280** transmits torque between the second shaft **266** and the third shaft **282**. In at least some pivoted positions, the second slot pin **280** slides within the third slot **292** as the second shaft **266** rotates. The third slot **292** accommodates the motion of the second slot pin **280** at the various pivoted positions between the second shaft **266** and the third shaft **282**. The second slot pin **280** can slide along the length of the third slot **292** or a portion thereof during a full rotation of the second shaft **266**. The second slot pin **280** transmits torque when the second shaft **266** and the third shaft **282** are angulated. In some embodiments, the second slot pin **280** needs to slide about twice the relative angle between the second shaft **266** and the third shaft **282**. Since the second shaft **266** and the third shaft **282** can have a variable angle, the second slot pin **280** can slide or sweep from any range between negative 30 degrees and positive 30 degrees during a single rotation of the joint between the second shaft **266** and the third shaft **282**. Unlike the first shaft **250** and the second shaft **266** that have a fixed angle, the second shaft **266** and the third shaft **282** are free to pivot relative to each other.

In one pivoted position, wherein the deployment tool **200** is approximately 30 degrees relative to the curved expandable interbody device **100**, the second shaft **266** and the third shaft **282** can be coaxial. In this position, the second slot pin **280** transmits torque between the second shaft **266** and the third shaft **282**. In this position, the longitudinal axis of the second shaft **266** can be aligned with the longitudinal axis of the third shaft **282**. In some embodiments, the second slot pin **280** does not slide within the third slot **292** in this position. In some embodiments, the second slot pin **280** rotates in a plane in this position. If the second shaft **266** and third shaft **282** have a relative angle of 0 degrees, the second slot pin **280** does not slide within the third slot **292**. This coupling between the second shaft **266** and the third shaft **282** will act as a straight joint in this position, and torque does not need to be transmitted at an angle.

28

If the second shaft **266** and the third shaft **282** have a relative angle of negative 30 degrees or positive 30 degrees, then similar with the first slot pin **264**, the second slot pin **280** needs to sweep from negative 30 degrees to positive 30 degrees. If the second shaft **266** and the third shaft **282** have a relative angle of negative 15 degrees, then the second slot pin **280** needs to sweep from negative 15 degrees to positive 15 degrees. If the second shaft **266** and the third shaft **282** have a relative angle of positive 15 degrees, then the second slot pin **280** also needs to sweep from negative 15 degrees to positive 15 degrees. If the second shaft **266** and the third shaft **282** have a relative angle of 0 degrees, then the second slot pin **280** needs to sweep 0 degrees.

The combined sliding of the first slot pin **264** and the second slot pin **280** can allow the deployment tool **200** to transmit torque in the straight position and one or more pivoted positions. The second shaft **266** and the third shaft **282** can pivot relative to each other approximately negative 30 degrees to positive 30 degrees. This pivoting of the second shaft **266** and the third shaft **282**, combined with the fixed angle of 27 degrees to 30 degrees between the first shaft **250** and second shaft **266** can achieve a cumulative total pivot angle of the deployment tool **200** of between 0 degrees to 60 degrees.

FIG. **25** is a rear view of the deployment tool **200**. The deployment tool **200** can include a passageway toward the keyed bore **252** of the first shaft **250**. The keyed bore **252** is configured to receive a keyed shaft **294** inserted into the deployment tool **200**. FIG. **26** is a perspective view of the keyed shaft **294**. In some embodiments, the keyed shaft **294** is removable. The keyed shaft **294** can be inserted into the deployment tool **200** to expand the curved expandable interbody device **100**. The keyed shaft **294** can be a separate component from the deployment tool **200**. In other embodiments, the keyed shaft **294** is located within the deployment tool **200**.

The keyed shaft **294** is configured to couple with the keyed bore **252** to transmit torque. As the keyed shaft **294** is rotated, the first shaft **250** also rotates. While the first shaft **250** includes the keyed bore **252**, in other embodiments, the first shaft **250** includes the keyed shaft and a keyed socket is inserted into the deployment tool **200**. The keyed shaft **294** can be keyed along the entire length or a distal portion thereof. In the illustrated embodiment, the keyed shaft **294** has six flutes that slideably engages the keyed bore **252** having a fluted socket with six flutes. Other suitable shapes or geometric configurations for a keyed connection between the keyed bore **252** and the keyed shaft **294** may be used to achieve the desired results, such as triangular, hexagonal, oval, star-shaped, or other non-circular shape. The keyed shaft **294** can include a flange **296**. The flange **296** can act as a depth stop of the keyed shaft **294**. The keyed shaft **294** can include unitary structure. The keyed shaft **294** can have a two-piece or multiple piece construction. The keyed shaft **294** can be manufactured from a single piece of metal stock or from multiple pieces coupled together through any manufacturing technique.

In operation, the keyed shaft **294** can be placed through a passageway extending through the right side of the deployment tool **200** as shown in FIG. **25**. After the curved expandable interbody device **100** is inserted and positioned within the intervertebral space between two vertebrae, the keyed shaft **294** can be used to deploy and expand the curved expandable interbody device **100** by applying a rotational force to the first shaft **250**. By rotating the keyed shaft **294** at the proximal portion of the deployment tool **200**, the first shaft **250** is also rotated, which in turn rotates the second

shaft **266** and the third shaft **282**, and thus the drive interface **184** of the screw mechanism **106** to expand the curved expandable interbody device **100**.

As the driving mechanism **206** applies the rotational force, the curved expandable interbody device **100** gradually expands as described above. The curved expandable interbody device **100** can be expanded until it contacts the two adjacent vertebrae. In some embodiments, the curved expandable interbody device **100** can be used to distract the two adjacent vertebrae and open up the intervertebral space. The keyed shaft **294** can advantageously transmit sufficient torque to the screw mechanism **106** to enable distraction using the curved expandable interbody device **100**. In some embodiments, the keyed shaft **294** can have a torque-limiting feature to prevent over-tightening of the screw mechanism **106**. For example, the torque-limiting feature can include a spring-loaded clutch mechanism along the keyed shaft **294** that can only transmit a predetermined amount of torque before the clutch slips. The amount of torque that can be transmitted can depend on the stiffness of the clutch spring. In other embodiments, the torque-limiting feature can be a portion of the keyed shaft **294** that is configured to break at a predetermined torque. In other embodiments, the feature can be any functional torque-limiting device.

The deployment tool **200** can include any features to increase rigidity and functionality. The deployment tool **200** can include one or more sleeves disposed over the components of the deployment tool **200**. The deployment tool **200** can include a passageway to receive the keyed shaft **294**. The deployment tool **200** can include rails to facilitate the sliding of the left arm **218**. The deployment tool **200** can include an impaction surface. In some embodiments, the deployment tool **200** can include an impaction cap. In some embodiments, the impaction cap can be a separate piece that is threadably engageable with the deployment tool **200**. In some embodiments, a separate impaction cap can reduce or eliminate direct impaction force on the handle **216**. The deployment tool can be configured to be struck by a mallet to facilitate placement of the curved expandable interbody device **100** between vertebrae. The deployment tool **200** can be configured to be struck by a mallet to insert the curved expandable interbody device **100** between vertebrae in the straight position. The deployment tool can be configured to be struck by a mallet to insert the curved expandable interbody device **100** between vertebrae in one or more pivoted positions. In some methods, striking the deployment tool **200** can change the trajectory of the deployment tool **200** when the deployment tool **200** is able to pivot. In some methods, striking the deployment tool **200** can cause the deployment tool to pivot to a greater angle when the deployment tool **200** is able to pivot.

The deployment tool **200** can include the locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206**. The locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206** can extend through the length of the deployment tool **200**. The locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206** can be independently actuated.

The locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206** can be actuated in a preferred order. In some methods, the locking mechanism **202** is actuated first to lock the deployment tool **200** to the curved expandable interbody device **100**. In some methods, the pivoting mechanism **204** is actuated next to limit pivoting movement during positioning of the curved expandable interbody device **100**. The pivoting mechanism **204** is

actuated to the straight position. During positioning of the curved expandable interbody device **100**, the pivoting mechanism **204** can be actuated to allow pivoting. The pivoting mechanism **204** is actuated to allow pivoting to facilitate further positioning of the curved expandable interbody device **100** within the disc space. In some methods, the driving mechanism **206** is actuated last to expand the curved expandable interbody device **100**.

The locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206** can be located at any position along the length of the deployment tool. In some embodiments, the locking mechanism **202** and the pivoting mechanism **204** are generally on the left side of the deployment tool **200** and the driving mechanism **206** is generally in the center and right side of the deployment tool **200**. The actuators **230**, **240** can be located on the lateral and proximal portion on the deployment tool **200**. In some embodiments, the keyed shaft **294** is inserted through the proximal end of the deployment tool **200** to actuate the driving mechanism **206**. Other configurations and placements of the locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206** are contemplated.

In some embodiments, the deployment tool **200** can be coupled to the curved expandable interbody device **100** through other mechanisms. In some embodiments, the deployment tool **200** can have any mechanism configured to lock the deployment tool **200** to the curved expandable interbody device **100**. In some embodiments, the deployment tool **200** can have any mechanism configured to prevent or allow pivoting of the deployment tool **200** relative to the curved expandable interbody device **100**. In some embodiments, the deployment tool **200** can have any mechanism configured to rotate the screw mechanism **106**.

The deployment tool **200** can have a generally straight position while the deployment tool is initially inserted into the intervertebral space as shown in FIG. **27**. The deployment tool **200** can be pivoted within the intervertebral space as shown in FIGS. **28-29**. The deployment tool **200** can be pivoted by the user to navigate the anatomy. In some configurations, the deployment tool **200** can have a variable angle shaft such that the shape of the deployment tool **200** can be adjusted during use. For example, the deployment tool **200** can function as a hinge that adjusts the pivot angle of the deployment tool **200** for improved fitment of the deployment tool through the incision and to the target implant site. The deployment tool **200** can be stiff, bendable, or partially stiff and partially bendable. In some embodiments, a power source may be provided for hydraulic, pneumatic or other power-assisted manipulation of the deployment tool **200**. In some embodiments, the keyed shaft **294** can be coupled to a power source to rotate the keyed shaft **294**. In some embodiments, the locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206** are manually driven by the user.

The locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206** can have any actuation means. The locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206** can include a knob, a lever, a flat protrusion, a drive interface or other suitable mechanism for actuation. While a lever is shown for the actuator **230** and a button is shown for the actuator **240**, any combination of interfaces can be utilized.

In some embodiments, the curved expandable interbody device **100** can be used to deliver fluids, medication or other materials, especially materials that can help in the integration of the curved expandable interbody device **100** with the vertebrae, such as allograft, Demineralized Bone Matrix

31

("DBM") packing, and/or other bone graft material. The material can fill the openings **120**, **140**. The curved expandable interbody device **100** can be packed with material before implantation. The curved expandable interbody device **100** can be packed with material after positioning the implant in the intervertebral space. The curved expandable interbody device **100** can be packed with material after expanding the implant. The material can fill up the empty cavity created between the upper structure **102** and lower structure **104** upon expansion, helping to provide support to the vertebrae.

The deployment tool **200** can be made of any appropriate material for the particular part. Materials can include, but are not limited to, stainless steel, surgical steel, cutlery steel, tool steel, cobalt and its alloys, nickel and its alloys, chromium and its alloys, titanium and its alloys, zirconium and its alloys, aluminum and its alloys, magnesium and its alloys, polymers, elastomers, and ceramics. Ceramics may include, but are not limited to silicon carbide, silicon oxide (s), silicon nitride, aluminum oxide, alumina, zirconia, tungsten carbide, other carbides.

The sizes of the curved expandable interbody device **100** and the deployment tool **200** are appropriate for treating the particular bone. Smaller devices can be used for smaller vertebra and larger devices for larger vertebra. In addition, while described in relation to the spine, the implant and deployment tool can be used on bones other than the vertebra and on bones for humans and non-humans.

Methods of implanting the curved expandable interbody device **100** can include selecting the implant size according to the sizing and goals of the procedure. In some embodiments, the curved expandable interbody device **100** can expand up to approximately 3 mm from its initial collapsed height. The user can select the curved expandable interbody device **100** from a plurality of implants. In some embodiments, the internal chamber of the curved expandable interbody device **100** may be packed with grafting material. Material can be packed through the openings **120**, **140**. Material can be packed when the curved expandable interbody device **100** is in a collapsed configuration. In some methods, the curved expandable interbody device **100** can be packed with material before coupling to the deployment tool **200**. In some methods, the curved expandable interbody device **100** can be packed with material after coupling to the deployment tool **200**. In some methods, the curved expandable interbody device **100** can be packed with material after placement within the intervertebral space.

The method can include coupling the curved expandable interbody device **100** to the deployment tool **200**. The curved expandable interbody device **100** can have a dedicated deployment tool **200** that attaches securely to the curved expandable interbody device **100**. The deployment tool **200** can provide control during insertion of the curved expandable interbody device **100**. The method can include loading the curved expandable interbody device **100**. The curved expandable interbody device **100** can be placed onto the distal end of the deployment tool **200**. The deployment tool **200** can engage the curved expandable interbody device **100** by manipulating the locking mechanism **202**. The curved expandable interbody device **100** can be secured to the deployment tool **200** by pushing the actuator **230** forward. The inner shaft **222** can be advanced between the deflection arms **224**, **226** to position the tines **228** of the deflection arms **224** into the curved slots **198**. The curved expandable interbody device **100** can be retained by the deployment tool **200**.

32

The method can include preventing pivoting between the curved expandable interbody device **100** and deployment tool **200**. The deployment tool **200** can be locked in a straight position by manipulating the pivoting mechanism **204**. The position of the deployment tool **200** can be controlled by pressing the actuator **240** down and moving the actuator **240** forward to the straight position. The curved expandable interbody device **100** and the deployment tool **200** can be locked in a straight position by moving the pivoting mechanism **204** distally. The pivoting mechanism **204** moves the left arm **218** against the curved expandable interbody device **100**. The left arm **218** can prevent pivoting as the curved expandable interbody device **100** is impacted.

An incision can be made on the patient to allow access to the implant site in the intervertebral space. The incision can be made for implanting the curved expandable interbody device **100** from any approach, including the posterior, lateral or anterior directions. The incision can be small for a minimally invasive procedure or a larger incision can be used for an open surgery. In some situations, two adjacent vertebrae can be distracted to open up the intervertebral space. In some configurations, the curved expandable interbody device **100** can be used to at least partially distract the vertebrae during the implant procedure. The incision and retraction of the soft tissues as well as a discectomy and any required decompression should be performed in accordance with procedures used in a spinal procedure. In some methods, the interbody device **100** can be used in a Lumbar Interbody Fusion (LIF) procedure.

The curved expandable interbody device **100** can be inserted into the intervertebral space. The curved expandable interbody device **100** and the deployment tool **200** can be locked for initial impaction. The curved expandable interbody device **100** and the deployment tool **200** can be in a straight position for initial impaction as shown in FIG. **27**. The curved expandable interbody device **100** can be placed into the prepared site using gentle impaction, if necessary. A mallet can be included in the instrument set to aid in impacting the curved expandable interbody device **100** into the site. A user can manipulate the handle **216** of the deployment tool **200** to position the curved expandable interbody device **100** in the intervertebral space.

The deployment tool **200** can allow for pivoting motion to further assist in positioning the curved expandable interbody device **100** in the intervertebral space. The deployment tool **200** can be set into a pivoting mode by manipulating the pivoting mechanism **204**. The position of the deployment tool **200** can be controlled by pressing the actuator **240** and moving the actuator **240** backward to the pivoting position. The curved expandable interbody device **100** and the deployment tool can remain locked by the locking mechanism **202** while allowing pivoting. The pivoting mechanism **204** can retract the left arm **218** relative to the curved expandable interbody device **100**. The left arm **218** can be retracted to allow the deployment tool **200** to pivot relative to the curved expandable interbody device **100**. The user can continue impacting the deployment tool **200** until the curved expandable interbody device **100** is manipulated into desired position within the disc space. The user can verify anterior-posterior and lateral placement with fluoroscope or other appropriate means. In some embodiments, the structures **102**, **104** can have one or more markers **126**, **146** to help visualization using radiation during the implantation procedure. The deployment tool **200** can be pivoted within the intervertebral space as shown in FIGS. **28-29**.

The deployment tool **200** can expand the curved expandable interbody device **100**. Once the curved expandable

33

interbody device **100** is positioned between adjacent vertebrae, the driving mechanism **206** can be rotated to rotate the screw mechanism **106**. The screw mechanism **106** changes length from a first length to a second length when rotated. The screw mechanism **106** expands the upper structure **102** and the lower structure **104** from a first height to a second height when rotated.

With the deployment tool **200** still locked to the curved expandable interbody device **100**, the keyed shaft **294** may be placed inside the passageway of the deployment tool **200** and used to rotate the drive mechanism **206**. The user can turn the keyed shaft **294** clockwise to expand the curved expandable interbody device **100**. The user can monitor expansion via fluoroscopy or X-ray. The user can expand until the desired height is reached. The expansion may be stopped whenever desired between a collapsed configuration and the maximum expanded configuration. In some embodiments, the keyed shaft **294** can provide audible feedback such as a click when the limit of expansion is reached if expansion is not stopped prior to the maximum expanded configuration.

In some embodiments, materials such as fluids, medication, bone graft material, allograft and/or Demineralized Bone Matrix (DBM) can be delivered to the interior cavity of the curved expandable interbody device **100** after expansion. The material can be delivered through a delivery tube and into the proximal section of the curved expandable interbody device **100**. In some embodiments, the material can be delivered through other paths to reach the cavities of the curved expandable interbody device **100** after expansion.

The deployment tool **200** can release the curved expandable interbody device **100**. When the positioning and/or expansion are complete, the user can release the curved expandable interbody device **100** from the deployment tool **200** by actuating the locking mechanism **202**. The user can flip the actuator **230** back into released position. The inner shaft **222** can be retracted relative to the deflection arms **224**, **226** to retract the tines **228** of the deflection arms **224** from the curved slots **198**. The curved expandable interbody device **100** can be released. The deployment tool **200** can be removed from the patient.

In some configurations, more than one curved expandable interbody device **100** can be implanted between the adjacent vertebrae of the patient. In such embodiments, multiple curved expandable interbody device **100** can be placed in a side-by-side configuration or any other suitable configuration, thereby creating additional support. The deployment tool **200** can be reused during a procedure to position one or more additional curved expandable interbody devices **100**.

In some embodiments of the deployment tool **200**, the movement of the locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **206** can be effected by manual force applied by a person, such as by his or her hands, or alternatively it can be supplied or supplemented with a motor, pneumatics, hydraulics, springs, and/or magnetics. Some embodiments of the deployment tool **200** may comprise one or more actuators for actuating the mechanisms therein. Other embodiments of the deployment tool **200** can include mechanisms that include compound leverage, ratcheting, and/or multistep closing.

The method can include supplemental fixation. Supplemental fixation such as posterior pedicle screw and rod systems, anterior plate systems, and anterior screw and rod systems can be used in conjunction with the curved expandable interbody device **100**. The kit can include manufacturers' recommendations for the implantation of these systems

34

or other instructions for use. In some methods, systems that have been cleared for such uses can be implanted.

The method can include closure. The method can include post-operative care. Closing of the wound can be completed in line with normal practices for spinal procedures. In some methods, post-operative care and activity restriction can be administered and discussed with the patient in line with normal practices for spinal procedures.

In some methods, the method can include removal. In some methods, the method can include revision. In some methods, the user can determine the amount of bone removal necessary to free the curved expandable interbody device **100** if the curved expandable interbody device **100** needs to be removed. Once the curved expandable interbody device **100** is free from any surrounding bone, the deployment tool **200** may be used to re-engage and remove the curved expandable interbody device **100**. The deployment tool **200** can collapse the curved expandable interbody device **100** before removal. The deployment tool **200** can be pivoted to facilitate removal. In some embodiments, the deployment tool **200** can include one or more markers to help visualization using radiation during the implantation and/or removal procedure.

FIG. **30** is a close-up side view of a driving mechanism **306** of a deployment tool **200**. FIG. **31** is a close-up top view of the driving mechanism **306** of the deployment tool **200**. FIGS. **32-33** are exploded views of the driving mechanism **306** of the deployment tool **200**. FIG. **34** is additional views of the driving mechanism **306** of the deployment tool **200**. The driving mechanism **306** can include any features of the driving mechanism **206** described herein.

Referring back to FIG. **13**, the deployment tool **200** can include the elongated shaft **210**. The deployment tool **200** can include the distal end **212** that is configured to interact with the curved expandable interbody device **100**. The deployment tool **200** can include the proximal end **214** with the handle **216** configured to be gripped by the user. Referring back to FIGS. **16** and **17**, the deployment tool **200** can include the locking mechanism **202**. The locking mechanism **202** can be configured to interact with one or more curved slots **198** of the curved expandable interbody device **100**. Once the deployment tool **200** is properly positioned relative to the curved expandable interbody device **100**, the locking mechanism **202** can be activated to couple the deployment tool **200** to the curved expandable interbody device **100**. Referring back to FIGS. **18** and **19**, the deployment tool **200** can include the pivoting mechanism **204** to prevent pivoting of the deployment tool **200** relative to the curved expandable interbody device **100**. The deployment tool **200** can have a straight position as shown in FIG. **14**. The deployment tool **200** can have one or more pivoted positions.

FIGS. **30-34** illustrate the driving mechanism **306** with portions of the deployment tool **200** removed. The driving mechanism **306** can be an alternative embodiment to the driving mechanism **206**. In some embodiments, the driving mechanism **306** can extend along a side of the deployment tool **200**. The driving mechanism **306** can be along the right side of the deployment tool **200**. The locking mechanism **202** and the pivoting mechanism **204** can be along the left side of the deployment tool **200**. Other configurations of the mechanisms within the elongated shaft **210** are contemplated.

The driving mechanism **306** can include a first shaft **350**. The first shaft **350** can be generally straight within the deployment tool **200**. The first shaft **350** can include a driver interface. The first shaft **350** can include a keyed bore **352**. The keyed bore **352** can extend from the proximal end of the

35

first shaft 350. The keyed bore 352 can extend along a portion of the length of the first shaft 350. The keyed bore 352 can extend along a longitudinal axis of the first shaft 350. The first shaft 350 can include a yoke end 354. The yoke end 354 can include a pair of arms. The yoke end 354 can define a first bore 356.

The driving mechanism 306 can include a first block 358. The first block 358 can include a ball end. The first block 358 can fit into the yoke end 354. The first block 358 can fit between the pair of arms. The first block 358 can include a second bore 360. The first block 358 can include a third bore 362. The second bore 360 and the third bore 362 can be perpendicular. The first block 358 can form a portion of a universal joint. The first block 358 can be a universal joint block.

The driving mechanism 306 can include a first connector 364. The first connector 364 can include a first pin 366. The first connector 364 can include a first retainer sleeve 368. The first connector 364 can include a second pin 370. The first pin 366 and the first retainer sleeve 368 can couple to form a pivot. The first pin 366 and the first retainer sleeve 368 can be perpendicular to the second pin 370. The second pin 370 can receive a post of the first pin 366 therethrough. The post of the first pin 366 can be received by the first retainer sleeve 368. The first pin 366 and the first retainer sleeve 368 can intersect the second pin 370.

The first pin 366 and the first retainer sleeve 368 can be configured to be received in the first bore 356 of the first shaft 350. The first pin 366 and the first retainer sleeve 368 can be configured to be received in the second bore 360 of the first block 358. The first pin 366 and the first retainer sleeve 368 can be configured to couple the first block 358 and the first shaft 350. The yoke end 354 can be configured to receive the first block 358. The first block 358 can rotate relative to the yoke end 354 about the first pin 366 and the first retainer sleeve 368.

The driving mechanism 306 can include a second shaft 372. The second shaft 372 can be a linkage yoke. The second shaft 372 can be generally skewed within the deployment tool 200. The second shaft 372 can include a proximal yoke end 374. The proximal yoke end 374 can include a pair of arms. The first block 358 can fit into the proximal yoke end 374. The first block 358 can fit between the pair of arms. The proximal yoke end 374 can extend from the proximal end of the second shaft 372. The proximal yoke end 374 can receive the first block 358. The proximal yoke end 374 can extend along a portion of the length of the second shaft 372. The proximal yoke end 374 can extend along a longitudinal axis of the second shaft 372.

The proximal yoke end 374 can include fourth bore 376. The second pin 370 can be configured to be received in the fourth bore 376 of the second shaft 372. The second pin 370 can be configured to be received in the third bore 362 of the first block 358. The second pin 370 can couple the first block 358 and the second shaft 372. The second shaft 372 including the proximal yoke end 374, the first block 358, and the second pin 370 can rotate relative to the first shaft 350. In some embodiments, the first shaft 350 is fixed within the deployment tool 200. The yoke end 354 of the first shaft 350 is shaped to allow rotation of the second shaft 372 relative to the first shaft 350. The first block 358 has dual axis rotation. The first block 358 can rotate about the first pin 366 and the first retainer sleeve 368. The first block 358 can rotate about the about the axis of the second pin 370. The first connector 364 and the first block 358 can connect the rigid rods of the first shaft 350 and the second shaft 372. The axes of the first shaft 350 and the second shaft 372 can be

36

inclined relative to each other at a fixed pivot angle. The first shaft 350 and the second shaft 372 can transmit rotary motion. The first pin 366 and the first retainer sleeve 368 act as first hinge. The second pin 370 acts as a second hinge. The first connector 370 orients the rotational hinges perpendicularly.

The first connector 364 and the first block 358 can allow a fixed pivot angle between the first shaft 350 and the second shaft 372. In some embodiments, the fixed pivot angle is an angle that the first shaft 350 and the second shaft 372 are relative to each other regardless of the orientation of the deployment tool 200 to the curved expandable interbody device 100. In some embodiments, the fixed pivot angle between the first shaft 350 and the second shaft 372 is 10 degrees, 15 degrees, 20 degrees, 21 degrees, 22 degrees, 23 degrees, 24 degrees, 25 degrees, 26 degrees, 27 degrees, 28 degrees, 29 degrees, 30 degrees, 31 degrees, 32 degrees, 33 degrees, 34 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, 80 degrees, 85 degrees, 90 degrees, 95 degrees, 100 degrees, 105 degrees, 110 degrees, 115 degrees, 120 degrees, between 27 degrees and 30 degrees, between 20 degrees and 45 degrees, between 40 degrees and 90 degrees, or any range of two of the foregoing values. In some embodiments, the entire range of motion is between 5 degrees, 10 degrees, 15 degrees, 20 degrees, 25 degrees, 30 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, 80 degrees, 85 degrees, 90 degrees, 95 degrees, 100 degrees, 105 degrees, 110 degrees, 115 degrees, 120 degrees, between 40 degrees and 90 degrees, between 50 degrees and 70 degrees, or any range of two of the foregoing values.

In some embodiments, the first shaft 350 and the second shaft 372 have a fixed angle therebetween. In other embodiments, the first shaft 350 and the second shaft 372 can have a variable angle therebetween. The first connector 364 and the first block 358 can maintain an angle between the first shaft 350 and the second shaft 372 when the driving mechanism 306 is disposed within the elongated shaft 210. In some embodiments, the first block 358 and first connector 364 can allow the first shaft 350 and the second shaft 372 to pivot relative to each other over a very limited range while within the deployment tool 200. In some embodiments, the first shaft 350 and the second shaft 372 do not pivot within the deployment tool 200. In some embodiments, the first shaft 350 and the second shaft 372 are held at a fixed pivot angle relative to each other in the deployment tool 200. This fixed pivot angle can be any angle. In some embodiments, the fixed pivot angle is between 27 degrees and 30 degrees. In some embodiments, the first block 358 and the first connector 364 are designed to accommodate the fixed pivot angle.

In some embodiments, the first shaft 350 and the second shaft 372 can be held by an internal pocket geometry in the deployment tool 200. The first shaft 350 and the second shaft 372 can be held by an internal pocket geometry near the distal end 212 of the deployment tool 200. This joint design can allow rotation and/or torque to be transmitted from the axis of the first shaft 350 to the axis of the second shaft 372. The first shaft 350 can function as the input drive shaft. The second shaft 372 can function as a universal joint.

The second shaft 372 can include a distal yoke end 378. The distal yoke end 378 can include a pair of arms. The distal yoke end 378 can extend from the distal end of the second shaft 372. The distal yoke end 378 can extend along a portion of the length of the second shaft 372. The distal yoke end 378 can extend along a longitudinal axis of the second shaft 372. The distal yoke end 378 can include a fifth

37

bore 380. The fourth bore 376 and the fifth bore 380 can be parallel. The fourth bore 376 and the fifth bore 380 can be at a fixed angle relative to each other.

The driving mechanism 306 can include a second connector 382. The second connector 382 can include a third pin 384. The second connector 382 can include a third retainer sleeve 386. The second connector 382 can include a fourth pin 388. The third pin 384 and the third retainer sleeve 386 can couple to form a pivot. The third pin 384 and the third retainer sleeve 386 can be perpendicular to the fourth pin 388. The fourth pin 388 can receive a post of the third pin 384 therethrough. The post of the third pin 384 can be received by the third retainer sleeve 386. The third pin 384 and the third retainer sleeve 386 can intersect the fourth pin 388.

The driving mechanism 306 can include a second block 390. The second block 390 can include a ball end. The second block 390 can fit into the distal yoke end 378. The second block 390 can fit between the pair of arms. The second block 390 can include a sixth bore 392. The second block 390 can include a seventh bore 394. The sixth bore 392 and the seventh bore 394 can be perpendicular. The second block 390 can form a portion of a universal joint. The second block 390 can be a universal joint block.

The distal yoke end 378 can include the fifth bore 380. The fourth pin 388 can be configured to be received in the fifth bore 380 of the second shaft 372. The fourth pin 388 can be configured to be received in the seventh bore 394 of the second block 390. The fourth pin 388 can couple the second block 390 and the second shaft 372.

The driving mechanism 306 can include a third shaft 396. The third shaft 396 can be a driver tip. The third shaft 396 can be generally straight within the deployment tool 200. The third shaft 396 can include a keyed shaft 398. The keyed shaft 398 can extend from the distal end of the third shaft 396. The keyed shaft 398 can extend along a portion of the length of the third shaft 396. The keyed shaft 398 can extend along a longitudinal axis of the third shaft 396. The keyed shaft 398 can be configured to engage the drive interface 184. The screw mechanism 106 can include the drive interface 184 configured to receive the keyed shaft 396 of the driving mechanism 306 for rotating the screw mechanism 106.

The third shaft 396 can include a yoke end 400. The yoke end 400 can include a pair of arms. The yoke end 400 can define an eighth bore 402. The second block 390 can fit into the yoke end 400. The second block 390 can fit between the pair of arms. The yoke end 400 can allow pivoting motion of the second block 390.

The third pin 384 and the third retainer sleeve 386 can be configured to be received in the eighth bore 402 of the third shaft 396. The third pin 384 and the third retainer sleeve 386 can be configured to be received in the sixth bore 392 of the second block 390. The third pin 384 and the third retainer sleeve 386 can be configured to couple the second block 390 and the third shaft 396. The yoke end 400 can be configured to receive the second block 390. The second block 390 can pivot relative to the yoke end 400.

The second block 390 and the second connector 382 can couple the second shaft 372 and the third shaft 396. The second block 390 and the second connector 382 can allow the second shaft 372 and the third shaft 396 to pivot within the deployment tool 200. The second block 390 and the second connector 382 can allow the deployment tool 200 to change from a straight position to one or more pivoted positions. In some embodiments, the first shaft 350 and the second shaft 372 do not pivot within the deployment tool

38

200. In some embodiments, only the second shaft 372 and the third shaft 396 pivot within the deployment tool 200. The second shaft 372 and the third shaft 396 can allow for pivoting for the total range of the deployment tool 200 relative to the curved expandable interbody device 100. The second block 390 and the second connector 382 can be designed to allow for pivoting for the total range of the deployment tool 200 relative to the curved expandable interbody device 100.

The second shaft 372 including the distal yoke end 378, the second block 390, and the fourth pin 388 can rotate relative to the third shaft 396. In some embodiments, the third shaft 396 is fixed within the deployment tool 200. The yoke end 400 of the third shaft 396 is shaped to allow rotation of the second shaft 372 relative to the third shaft 396. The second block 390 has dual axis rotation. The second block 390 can rotate about the third pin 384 and the third retainer sleeve 386. The second block 390 can rotate about the axis of the fourth pin 388. The second connector 382 and the second block 390 can connect the rigid rods of the second shaft 372 and the third shaft 396. The axes of the second shaft 372 and the third shaft 396 can be inclined relative to each other. The second shaft 372 and the third shaft 396 can transmit rotary motion. The third pin 384 and the third retainer sleeve 386 act as a first hinge. The fourth pin 388 acts as a second hinge. The second connector 382 orients the rotational hinges perpendicularly.

In other embodiments, the first shaft 350 and the second shaft 372 pivot. The first shaft 350 and the second shaft 372 are not maintained at the fixed pivot angle. The first shaft 350, the second shaft 372, and the third shaft 396 can allow for pivoting for the total range of the deployment tool 200 relative to the curved expandable interbody device 100. The first block 358, the first connector 364, the second block 390, and the second connector 382 can be designed to allow for pivoting for the total range of the deployment tool 200 relative to the curved expandable interbody device 100.

The first shaft 350, the first block 358, the first connector 364, the second shaft 372, the second connector 382, the second block 390, and the third shaft 396 couple together to form an assembly. The first shaft 350, the first block 358, the first connector 364, the second shaft 372, the second connector 382, the second block 390, and the third shaft 396 can form a universal joint. The rotation of the first shaft 350 can cause rotation of the second shaft 372 which can cause rotation of the third shaft 396. The torque is transmitted by the first block 358 and the first connector 364 which couples the first shaft 350 and the second shaft 372. The torque is transmitted by the second block 390 and the second connector 382 which couples the second shaft 372 and the third shaft 396. The third shaft 396 can be coupled to the screw mechanism 106. The rotation of the first shaft 350 can cause rotation of the screw mechanism 106 to expand the curved expandable interbody device 100.

The first shaft 350, the first block 358, the first connector 364, the second shaft 372, the second connector 382, the second block 390, and the third shaft 396 allow the deployment tool 200 to be in any pivoted position while allowing the driving mechanism 306 to rotate the screw mechanism 106.

As the deployment tool 200 pivots, the first block 358 can rotate relative to the first shaft 350. The first block 358 rotates about the first pin 366 and the first retainer sleeve 368. The first block 358 rotates about the second pin 370. The first block 358 rotates about two perpendicular axes. During the transmission of torque and expansion of the curved expandable interbody device 100, the first shaft 350

can be fixed relative to the deployment tool. The dual axis pivoting allows torque to be transmitted from the first shaft 350 to the second shaft 372. As the deployment tool 200 pivots, the second block 390 can rotate relative to the third shaft 396. The second block 390 rotates about the third pin 384 and the third retainer sleeve 386. The second block 390 rotates about the fourth pin 388. The second block 390 rotates about two perpendicular axes. During the transmission of torque and expansion of the curved expandable interbody device 100, the third shaft 396 is coupled to the curved expandable interbody device 100. The dual axis pivoting allows torque to be transmitted from the second shaft 372 to the third shaft 396. As the deployment tool 200 pivots, the angle of the second shaft 372 changes relative to the fixed shaft 350, 396. The first shaft 350 can be fixed relative to the elongated shaft 210. The third shaft 396 can be fixed relative to the elongated shaft 210. The first block 358, the first connector 364, the second block 390, and the second connector 382 allow the second shaft 372 to reorient to accommodate the pivoted positions.

As the deployment tool 200 changes to one or more pivoted positions, the second shaft 372 must pivot relative to the third shaft 396. As the deployment tool 200 pivots, the second block 390 can rotate within the yoke end 400 of the third shaft 396. The second shaft 372 rotates relative to the third shaft 396 about two axes. The yoke end 400 of the third shaft 396 can accommodate the range of pivoted positions of the deployment tool 200 due to the dual axis rotation. The yoke end 400 of the third shaft 396 can accommodate rotation of the second block 390 up to 30 degrees, 40 degrees, 50 degrees, 60 degrees, 70 degrees, 80 degrees, 90 degrees, or any range of two of the foregoing values. In some methods of use, the first shaft 350 and the third shaft 396 can remain stationary as the deployment tool 200 assumes one or more pivoted positions. In some methods of use, the third shaft 396 remains coaxial with the longitudinal axis 108 of the curved expandable interbody device 100 regardless of the position of the deployment tool 200. In embodiments with a fixed pivot angle between the first shaft 350 and the second shaft 376, the first shaft 350 and the second shaft 376 can pivot relative to the third shaft 396 as the deployment tool 200 pivots.

The first block 358 and the second block 390 can accommodate the range of pivoted positions of the deployment tool 200. The first block 358 and the first connector 364 can maintain the fixed pivot angle between the first shaft 350 and the second shaft 372. The second block 390 and the second connector 382 can accommodate pivoting up to 30 degrees, 40 degrees, 50 degrees, 60 degrees, 70 degrees, 80 degrees, 90 degrees, or any range of two of the foregoing values. The pivoted positions can be accommodated due to the second block 390 and the second connector 382 allowing pivoting between the second shaft 372 and the third shaft 396.

As described herein, the deployment tool 200 can have a straight position. In the straight position of the deployment tool 200, the first shaft 350 and the second shaft 372 can be at the fixed pivot angle relative to each other. In the straight position, the first block 358 and the first connector 364 transmits torque between the first shaft 350 and the second shaft 372. In the straight position, the first shaft 350 rotates the first block 358 and the first connector 364 which rotates the second shaft 372. The first block 358 and the first connector 364 enables a sweep that is necessitated by the fixed pivot angle between the first shaft 350 and the second shaft 372. The first block 358 rotates relative to the first pin 366 and first retainer sleeve 368 forming a first hinge. The first block 358 rotates relative to second pin 370 forming a

second hinge. The first block 358 and the first connector 364 accommodates the fixed pivot angle between the first shaft 350 and the second shaft 372 during a full rotation of the first shaft 350. In some embodiments, the fixed pivot angle between the first shaft 350 and the second shaft 372 is between 27 degrees and 30 degrees. The first block 358 and the first connector 364 can rotate twice the fixed pivot angle. Since the first shaft 350 and second shaft 372 are at the fixed pivot angle of 27 degrees to 30 degrees, the first block 358 and the first connector 364 will rotate from about negative 27 degrees to 30 degrees to positive 27 degrees to 30 degrees during a single rotation of the joint between the first shaft 350 and the second shaft 372. The universal coupling of the first block 358 and the first connector 364 can allow the deployment tool 200 to transmit torque in the straight position between the first shaft 350 and the second shaft 372.

In the straight position of the deployment tool 200, the second shaft 372 and the third shaft 396 can be skewed. In some embodiments, the angle between the second shaft 372 and the third shaft 396 in the straight position is 10 degrees, 15 degrees, 20 degrees, 21 degrees, 22 degrees, 23 degrees, 24 degrees, 25 degrees, 26 degrees, 27 degrees, 28 degrees, 29 degrees, 30 degrees, 31 degrees, 32 degrees, 33 degrees, 34 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, 80 degrees, 85 degrees, 90 degrees, 95 degrees, 100 degrees, 105 degrees, 110 degrees, 115 degrees, 120 degrees, between 27 degrees and 30 degrees, between 20 degrees and 45 degrees, between 40 degrees and 90 degrees, or any range of two of the foregoing values. In some embodiments, the entire range of motion is between 5 degrees, 10 degrees, 15 degrees, 20 degrees, 25 degrees, 30 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, 80 degrees, 85 degrees, 90 degrees, 95 degrees, 100 degrees, 105 degrees, 110 degrees, 115 degrees, 120 degrees, between 40 degrees and 90 degrees, between 50 degrees and 70 degrees, or any range of two of the foregoing values. The second block 390 rotates relative to the third pin 384 and third retainer sleeve 386 forming a first hinge. The second block 390 rotates relative to the fourth pin 388 forming a second hinge. The second block 390 and the second connector 382 accommodates the pivot angle between the second shaft 372 and the third shaft 396 during a full rotation of the second shaft 372. In some embodiments, the second shaft 372 and the third shaft 396 are at an angle of 27 degrees to 30 degrees when the deployment tool 200 is in the straight position. In the straight position, the second block 390 and the second connector 382 transmit torque between the second shaft 372 and the third shaft 396. The second block 390 and the second connector 382 enable a sweep that is necessitated by the pivot angle between the second shaft 372 and the third shaft 396 when the deployment tool 200 is in the straight position. The second block 390 and the second connector 382 accommodates this pivot angle between the second shaft 372 and the third shaft 396 during a full rotation of the second shaft 372. In some embodiments, the pivot angle between the second shaft 372 and the third shaft 396 is between 27 degrees and 30 degrees when the deployment tool 200 is in the straight position. The second block 390 and the second connector 382 can rotate twice the relative angle between the second shaft 372 and the third shaft 396. Since the second shaft 372 and the third shaft 396 are at the relative angle of 27 degrees to 30 degrees, the second block 390 and the second connector 382 will rotate from about negative 27 degrees to 30 degrees to positive 27 degrees to 30 degrees during a single rotation of the joint between the second shaft 372 and the

41

third shaft 392 when the deployment tool 200 is in the straight position. The sliding of the second block 390 and the second connector 382 can allow the deployment tool 200 to transmit torque in the straight position between the second shaft 372 and the third shaft 396.

As described herein, the deployment tool 200 can have one or more pivoted positions. In the pivoted positions of the deployment tool 200, the first shaft 350 and the second shaft 372 remain at the fixed pivot angle relative to each other. In some embodiments, the fixed pivot angle between the first shaft 350 and the second shaft 372 is 10 degrees, 15 degrees, 20 degrees, 21 degrees, 22 degrees, 23 degrees, 24 degrees, 25 degrees, 26 degrees, 27 degrees, 28 degrees, 29 degrees, 30 degrees, 31 degrees, 32 degrees, 33 degrees, 34 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, 80 degrees, 85 degrees, 90 degrees, 95 degrees, 100 degrees, 105 degrees, 110 degrees, 115 degrees, 120 degrees, between 27 degrees and 30 degrees, between 20 degrees and 45 degrees, between 40 degrees and 90 degrees, or any range of two of the foregoing values. In the pivoted positions, the first block 358 and the first connector 364 transmits torque between the first shaft 350 and the second shaft 372. The first block 358 and the first connector 364 rotate as the first shaft 350 rotates. The motion of the first block 358 and the first connector 364 accommodates the fixed pivot angle between the first shaft 350 and the second shaft 372 in one or more pivoted positions. The first block 358 and the first connector 364 can rotate up to twice the fixed pivot angle between the first shaft 350 and the second shaft 366 in both the straight positions and in the one or more pivoted positions of the deployment tool 200. Since the first shaft 250 and second shaft 266 are at the fixed pivot angle of 27 degrees to 30 degrees, the first block 358 and the first connector 364 can rotate from about negative 27 degrees to 30 degrees to positive 27 degrees to 30 degrees during a single rotation of the joint between the first shaft 350 and the second shaft 372 regardless of the angle of the deployment tool 200.

In a pivoted position of the deployment tool 200, the second shaft 372 can be at a different pivot angle relative to the third shaft 396. In some embodiments, the angle between the second shaft 372 and the third shaft 396 in the pivoted position is negative 45 degrees, negative 40 degrees, negative 35 degrees, negative 30 degrees, negative 25 degrees, negative 20 degrees, negative 15 degrees, negative 10 degrees, negative 5 degrees, 0 degrees, 5 degrees, 10 degrees, 15 degrees, 20 degrees, 25 degrees, 30 degrees, 31 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, 80 degrees, 85 degrees, 90 degrees, 95 degrees, 100 degrees, 105 degrees, 110 degrees, 115 degrees, 120 degrees, between 27 degrees and 30 degrees, between 20 degrees and 45 degrees, between 40 degrees and 90 degrees, between negative 30 degrees and 30 degrees, between negative 45 degrees and 45 degrees, between negative 20 degrees and 20 degrees, between 0 degrees and 30 degrees, or any range of two of the foregoing values. The second shaft 372 and the third shaft 396 can pivot relative to each other approximately negative 30 degrees to positive 30 degrees.

This pivoting of the second shaft 372 and the third shaft 396 combined with the fixed pivot angle of 27 to 30 degrees between the first shaft 350 and the second shaft 372 allows the deployment tool 200 to achieve cumulative total deployment tool pivot angles of 0 degrees to 60 degrees. In the pivoted positions, the second block 390 and the second connector 382 transmits torque between the second shaft 372 and the third shaft 396. In at least some pivoted

42

positions, the second shaft 372 becomes skewed relative to the third shaft 396. The second block 390 and the second connector 382 transmits torque when the second shaft 372 and the third shaft 396 are angulated. In some embodiments, the second block 390 and the second connector 382 need to rotate about twice the relative angle between the second shaft 372 and the third shaft 396. Since the second shaft 372 and the third shaft 396 can have a variable angle, the second block 390 and the second connector 382 can rotate from any range between negative 30 degrees and positive 30 degrees during a single rotation of the joint between the second shaft 372 and the third shaft 396. Unlike the first shaft 350 and the second shaft 372 that have a fixed pivot angle, the second shaft 372 and the third shaft 396 do not have a fixed pivot angle. The second shaft 372 and the third shaft 396 can have a variable angle depending on the angulation of the deployment tool 200.

In one pivoted position, wherein the deployment tool 200 is approximately 30 degrees relative to the curved expandable interbody device 100, the second shaft 372 and the third shaft 396 can be coaxial. In this position, the longitudinal axis of the second shaft 372 can be aligned with the longitudinal axis of the third shaft 396.

If the second shaft 372 and the third shaft 396 have a relative angle of negative 30 degrees or positive 30 degrees, then the second block 390 and the second connector 382 need to rotate from negative 30 degrees to positive 30 degrees. If the second shaft 372 and the third shaft 396 have a relative angle of negative 15 degrees, then the second block 390 and the second connector 382 need to rotate from negative 15 degrees to positive 15 degrees. If the second shaft 372 and the third shaft 396 have a relative angle of positive 15 degrees, then the second block 390 and the second connector 382 also need to rotate from negative 15 degrees to positive 15 degrees. If the second shaft 372 and the third shaft 396 have a relative angle of 0 degrees, then the second block 390 and the second connector 382 need to rotate 0 degrees.

The combined rotating of the first block 358 and the second block 390 can allow the deployment tool 200 to transmit torque in the straight position and one or more pivoted positions. The second shaft 372 and the third shaft 396 can pivot relative to each other approximately negative 30 degrees to positive 30 degrees. This pivoting of the second shaft 372 and the third shaft 396, combined with the fixed angle of 27 degrees to 30 degrees between the first shaft 350 and second shaft 372 can achieve a cumulative total pivot angle of the deployment tool 200 of between 0 degrees to 60 degrees.

The combined rotating of the second block 390 can facilitate engagement of the third shaft 396 with the curved expandable interbody device 100. The second block 390 rotates relative to the third pin 384 and third retainer sleeve 386 forming a first hinge. The second block 390 rotates relative to fourth pin 388 forming a second hinge. The second block 390 and the second connector 382 can allow polyaxial movement of the third shaft 396. This polyaxial movement of the third shaft 396 can facilitate coupling the deployment tool 200 to the curved expandable interbody device 100.

Referring back to FIG. 25, the deployment tool 200 can include a passageway toward the keyed bore 352 of the first shaft 350. The keyed bore 352 is configured to receive the keyed shaft 294 inserted into the deployment tool 200. The keyed shaft 294 can be inserted into the deployment tool 200 to expand the curved expandable interbody device 100. As the keyed shaft 294 is rotated, the first shaft 350 also rotates.

43

By rotating the keyed shaft **294** at the proximal portion of the deployment tool **200**, the first shaft **350** is also rotated, which in turn rotates the second shaft **372** and the third shaft **396**, and thus the drive interface **184** of the screw mechanism **106** to expand the curved expandable interbody device **100**.

As the driving mechanism **306** applies the rotational force, the curved expandable interbody device **100** gradually expands as described herein. The keyed shaft **294** can advantageously transmit sufficient torque to the screw mechanism **106** to enable distraction using the curved expandable interbody device **100**.

The deployment tool **200** can include the locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **306**. The locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **306** can be independently actuated. The locking mechanism **202**, the pivoting mechanism **204**, and the driving mechanism **306** can be actuated in a preferred order. In some methods, the locking mechanism **202** is actuated first to lock the deployment tool **200** to the curved expandable interbody device **100**. In some methods, the pivoting mechanism **204** is actuated next to limit pivoting movement during positioning of the curved expandable interbody device **100**. In some methods, the driving mechanism **306** is actuated last to expand the curved expandable interbody device **100**. The driving mechanism **306** can include any features of the driving mechanism **306** described herein.

Although certain embodiments, features, and examples have been described herein, it will be understood by those skilled in the art that many aspects of the methods and devices illustrated and described in the present disclosure may be differently combined and/or modified to form still further embodiments. For example, any one component of the device illustrated and described above can be used alone or with other components without departing from the spirit of the present disclosure. Additionally, it will be recognized that the methods described herein may be practiced in different sequences, and/or with additional devices as desired. Such alternative embodiments and/or uses of the methods and devices described above and obvious modifications and equivalents thereof are intended to be included within the scope of the present disclosure. Thus, it is intended that the scope of the present disclosure should not be limited by the particular embodiments described above, but should be determined only by a fair reading of the claims that follow.

What is claimed is:

1. A system comprising:

- a curved expandable interbody device comprising:
 - an upper structure configured to abut a superior vertebra, wherein the upper structure comprises an upper structure proximal surface;
 - a lower structure configured to abut an inferior vertebra, wherein the lower structure comprises a lower structure proximal surface; and
 - a screw mechanism between the upper structure and the lower structure, the screw mechanism comprising a proximal portion, a distal portion, and a coupler, wherein the coupler comprises a central portion at least partially between the proximal portion and the distal portion, wherein the coupler further comprises a proximal interface comprising an upper connector and a lower connector,
- wherein the proximal portion and the distal portion are configured to rotate to change a distance between the proximal portion and the distal portion from a first

44

length to a second length thereby expanding the upper structure and the lower structure without translation of the upper structure and the lower structure, and

wherein the upper structure proximal surface and the upper connector form an upper curved slot therebetween, wherein the lower structure proximal surface and the lower connector form a lower curved slot therebetween; and

a deployment tool, wherein the upper connector and the lower connector are configured to receive an upper tine and a lower tine of the deployment tool therebetween, wherein the upper tine of the deployment tool is configured to be deflected upward into engagement with the upper curved slot and the lower tine of the deployment tool is configured to be deflected downward into engagement with the lower curved slot.

2. The system of claim 1, wherein the upper structure and the lower structure comprise a right side and a left side when viewed from the top, wherein the left side is concave and wherein the right side is convex.

3. The system of claim 1, wherein the coupler further comprises at least one anti-rotational feature configured to engage the upper structure or the lower structure to prevent the coupler from rotating when the proximal portion and the distal portion are rotated.

4. The system of claim 1, wherein the upper curved slot is open when viewed from the top.

5. The system of claim 1, wherein the curved expandable interbody device curves to the right.

6. The system of claim 1, wherein a right surface of the upper structure and a right surface of the lower structure comprise the same radius of curvature.

7. The system of claim 1, wherein the upper connector is shaped to receive the upper structure proximal surface.

8. The system of claim 1, wherein the upper tine is configured to be retained within the upper curved slot and the lower tine is configured to be retained within the lower curved slot.

9. The system of claim 1, wherein the upper tine and the lower tine are configured to limit proximal movement of the deployment tool relative to the curved expandable interbody device.

10. The system of claim 1, wherein the deployment tool is configured pivot relative to the curved expandable interbody device by sliding the upper tine within the upper curved slot and sliding the lower tine within the lower curved slot.

11. The system of claim 1, wherein the deployment tool is configured to rotate the proximal portion of the screw mechanism.

12. A method comprising:

coupling a deployment tool to a curved expandable interbody device, wherein the curved expandable interbody device comprises:

- an upper structure configured to abut a superior vertebra, wherein the upper structure comprises an upper structure proximal surface;

- a lower structure configured to abut an inferior vertebra, wherein the lower structure comprises a lower structure proximal surface; and

- a screw mechanism between the upper structure and the lower structure, the screw mechanism comprising a proximal portion, a distal portion, and a coupler, wherein the coupler comprises a central portion at least partially between the proximal portion and the

45

distal portion, wherein the coupler further comprises a proximal interface comprising an upper connector and a lower connector,

wherein the proximal portion and the distal portion are configured to rotate to change a distance between the proximal portion and the distal portion from a first length to a second length thereby expanding the upper structure and the lower structure without translation of the upper structure and the lower structure, wherein the upper structure proximal surface and the upper connector form an upper curved slot therebetween, wherein the lower structure proximal surface and the lower connector form a lower curved slot therebetween, and

wherein the upper connector and the lower connector receive an upper tine and a lower tine of the deployment tool therebetween, wherein the upper tine of the deployment tool is deflected upward into engagement with the upper curved slot and the lower tine of the deployment tool is deflected downward into engagement with the lower curved slot; and positioning the curved expandable interbody device between the superior vertebra and the inferior vertebra.

46

13. The method of claim 12, wherein the upper tine is retained within the upper curved slot and the lower tine is retained within the lower curved slot.

14. The method of claim 12, wherein the upper tine and the lower tine limit proximal movement of the deployment tool relative to the curved expandable interbody device.

15. The method of claim 12, wherein the deployment tool pivots relative to the curved expandable interbody device by sliding the upper tine within the upper curved slot and sliding the lower tine within the lower curved slot.

16. The method of claim 12, wherein the curved expandable interbody device expands from an initial collapsed configuration to an expanded configuration in situ.

17. The method of claim 12, wherein the deployment tool rotates the proximal portion of the screw mechanism.

18. The method of claim 12, further comprising rotating the screw mechanism to expand the curved expandable interbody device with the deployment tool.

19. The method of claim 12, further comprising pivoting the deployment tool relative to the curved expandable interbody device to further position the curved expandable interbody device between the superior vertebra and the inferior vertebra.

* * * * *